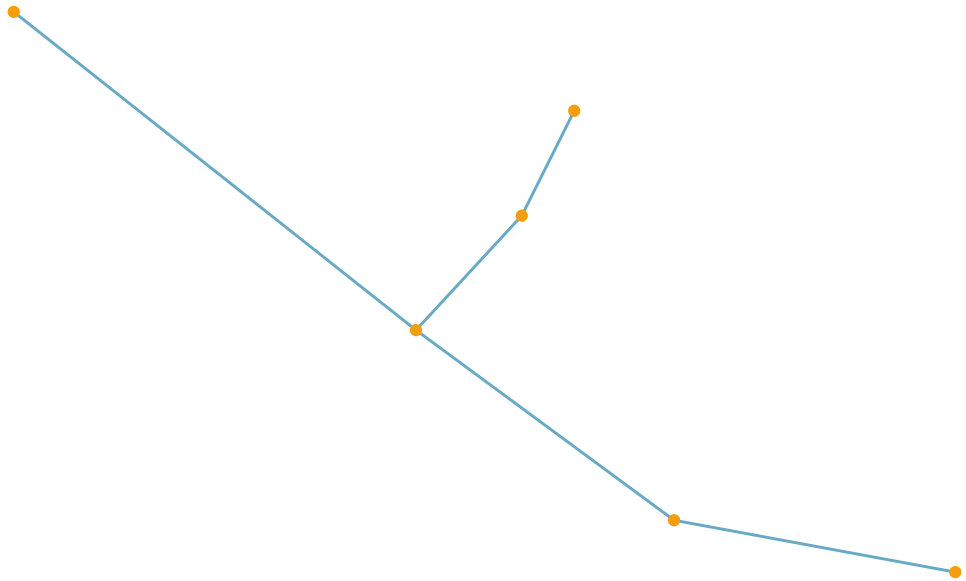
01/88 Andromeda (And) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

01/88 Andromeda (And) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 677	2.096542	+29.090833	HIP 3092
HIP 3092	9.831667	+30.861222	HIP 677, HIP 5447
HIP 5447	17.432500	+35.620833	HIP 3092, HIP 9640, HIP 4436
HIP 9640	30.974667	+42.329861	HIP 5447
HIP 4436	14.187917	+38.499250	HIP 5447, HIP 3881
HIP 3881	12.453458	+41.078944	HIP 4436

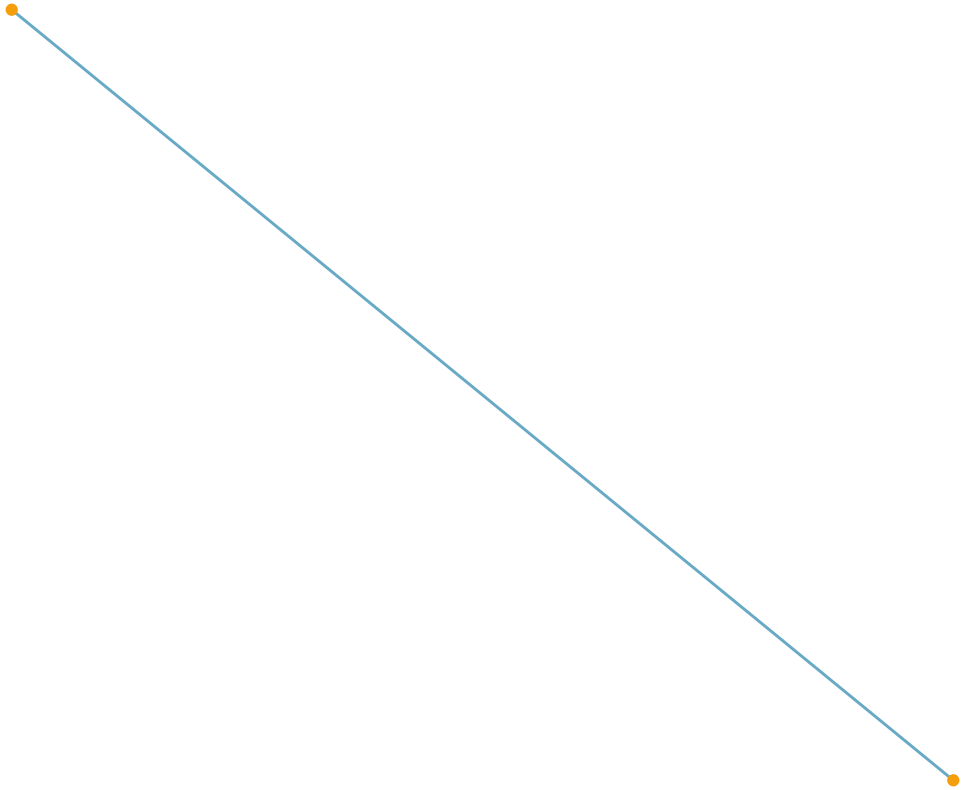
Effective release edges: 5 • unique plotted points: 6 • source: release_effective_edges.csv

02/88 Antlia (Ant) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

02/88 Antlia (Ant) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 51172	156.788167	-31.067806	HIP 48926
HIP 48926	149.718083	-35.890944	HIP 51172

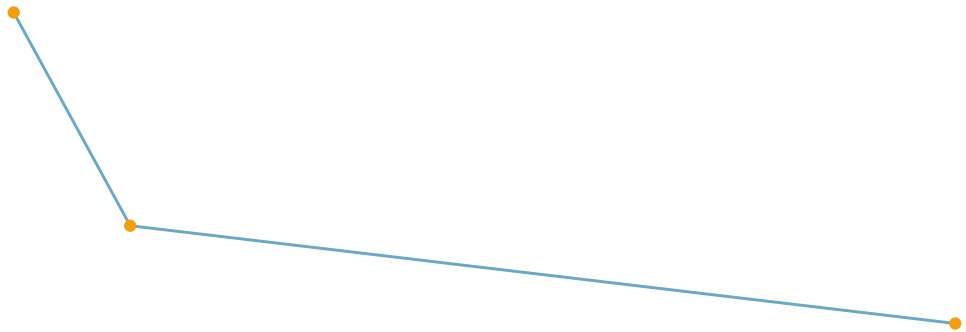
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

03/88 Apus (Aps) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

03/88 Apus (Aps) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 72370	221.965542	-79.044722	HIP 81065
HIP 81065	248.364417	-78.896972	HIP 72370, HIP 81852
HIP 81852	250.772583	-77.516583	HIP 81065

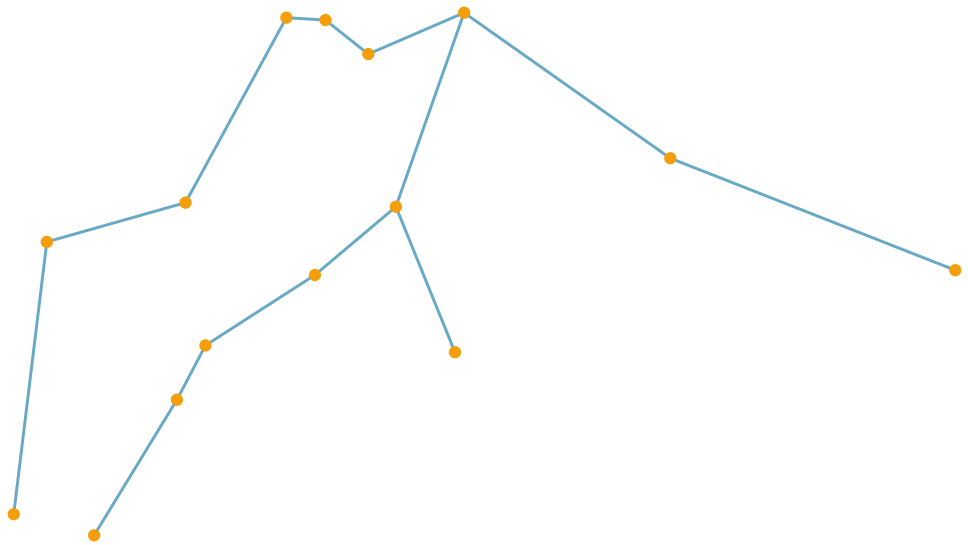
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

04/88 Aquarius (Aqr) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

04/88 Aquarius (Aqr) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 106278	322.889667	-5.571167	HIP 109074, HIP 102618
HIP 109074	331.445958	+0.319833	HIP 106278, HIP 110395, HIP 110003
HIP 110395	335.413750	-1.387361	HIP 109074, HIP 110960
HIP 110960	337.207500	+0.020056	HIP 110395, HIP 111497
HIP 111497	338.838875	+0.117361	HIP 110960, HIP 112961
HIP 112961	343.153583	-7.579667	HIP 111497, HIP 114855
HIP 114855	348.972000	-9.087694	HIP 112961, HIP 115438
HIP 115438	350.742917	-20.100333	HIP 114855
HIP 110003	334.208208	-7.783250	HIP 109074, HIP 109139, HIP 111123
HIP 109139	331.609208	-13.869528	HIP 110003
HIP 111123	337.661750	-10.677889	HIP 110003, HIP 112716
HIP 112716	342.397958	-13.592528	HIP 111123, HIP 113136
HIP 113136	343.662667	-15.820750	HIP 112716, HIP 114341
HIP 114341	347.361500	-21.172472	HIP 113136
HIP 102618	311.918875	-9.495694	HIP 106278

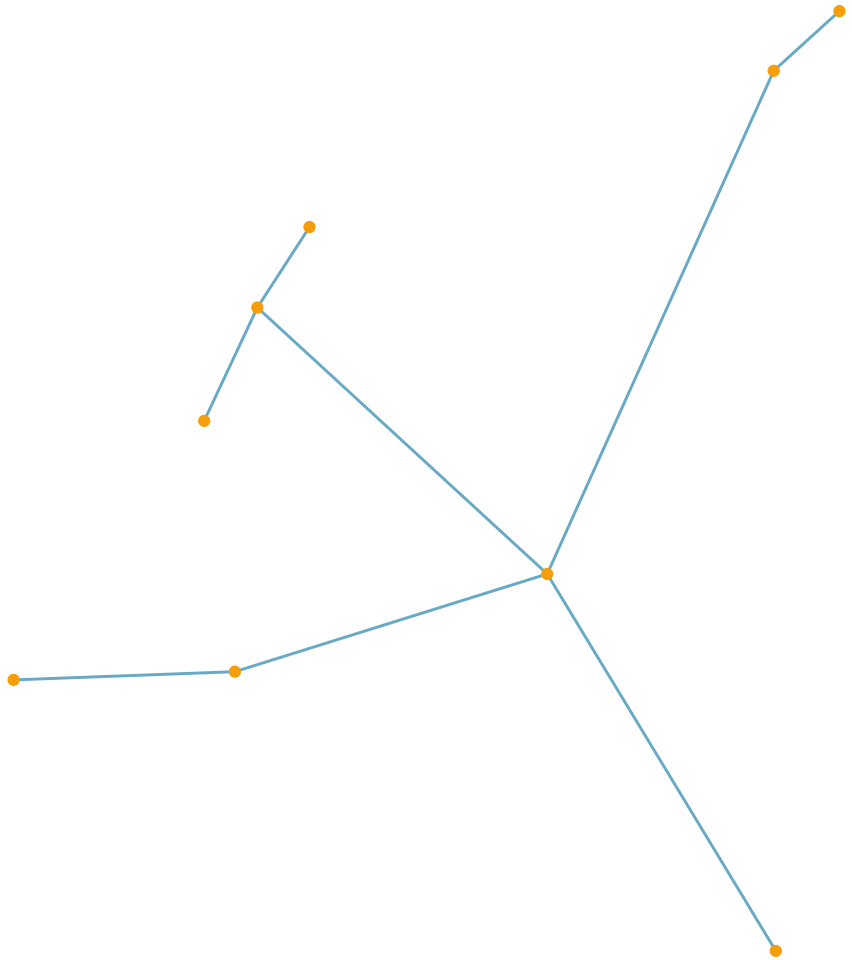
Effective release edges: 14 • unique plotted points: 15 • source: release_effective_edges.csv

05/88 Aquila (Aql) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

05/88 Aquila (Aql) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 98036	298.828208	+6.407944	HIP 97649
HIP 97649	297.694500	+8.867389	HIP 98036, HIP 97278, HIP 95501
HIP 97278	296.564875	+10.613278	HIP 97649
HIP 95501	291.373958	+3.114583	HIP 97649, HIP 97804, HIP 93747, HIP 93805
HIP 97804	298.118167	+1.005667	HIP 95501, HIP 99473
HIP 99473	302.826083	+0.821472	HIP 97804
HIP 93747	286.352542	+13.863722	HIP 95501, HIP 93244
HIP 93244	284.905792	+15.068472	HIP 93747
HIP 93805	286.562292	-4.882333	HIP 95501

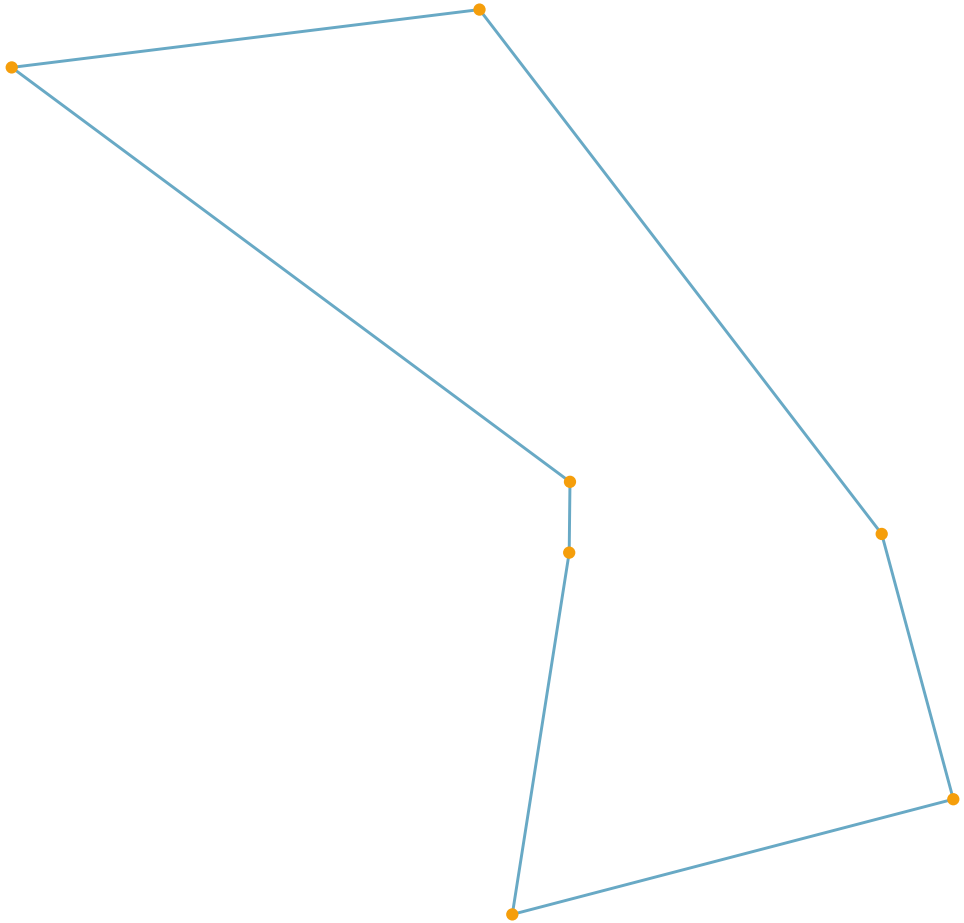
Effective release edges: 8 • unique plotted points: 9 • source: release_effective_edges.csv

06/88 Ara (Ara) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

06/88 Ara (Ara) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 88714	271.657833	-50.091444	HIP 85792, HIP 85258
HIP 85792	262.960500	-49.875972	HIP 88714, HIP 83081
HIP 83081	254.655125	-55.990056	HIP 85792, HIP 82363
HIP 82363	252.446292	-59.041306	HIP 83081, HIP 85727
HIP 85727	262.774917	-60.683611	HIP 82363, HIP 85267
HIP 85267	261.348583	-56.377694	HIP 85727, HIP 85258
HIP 85258	261.325000	-55.529833	HIP 85267, HIP 88714

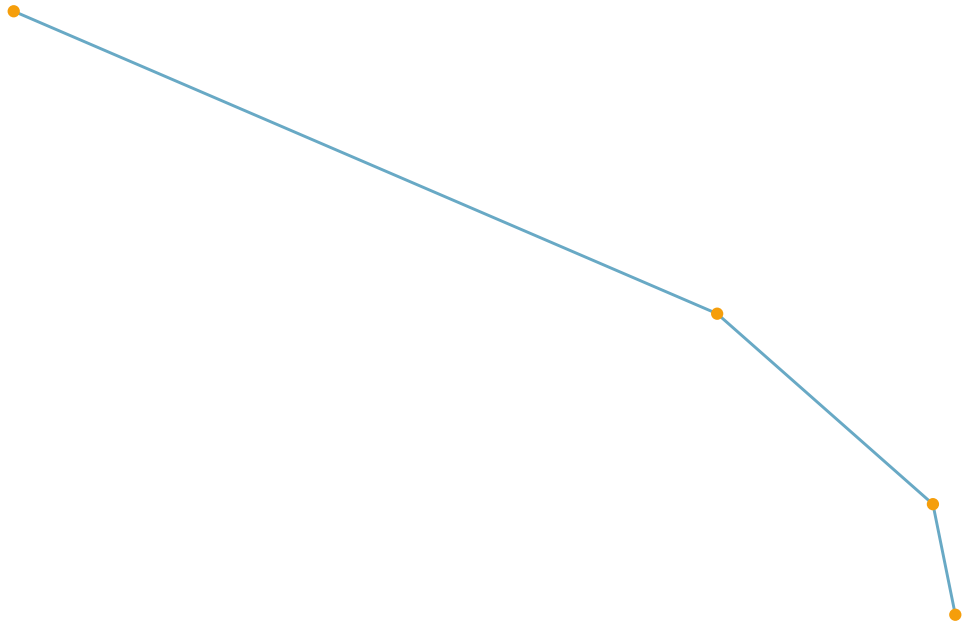
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

07/88 Aries (Ari) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

07/88 Aries (Ari) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 13209	42.495792	+27.260778	HIP 9884
HIP 9884	31.792875	+23.462778	HIP 13209, HIP 8903
HIP 8903	28.659792	+20.808306	HIP 9884, HIP 8832
HIP 8832	28.382375	+19.294083	HIP 8903

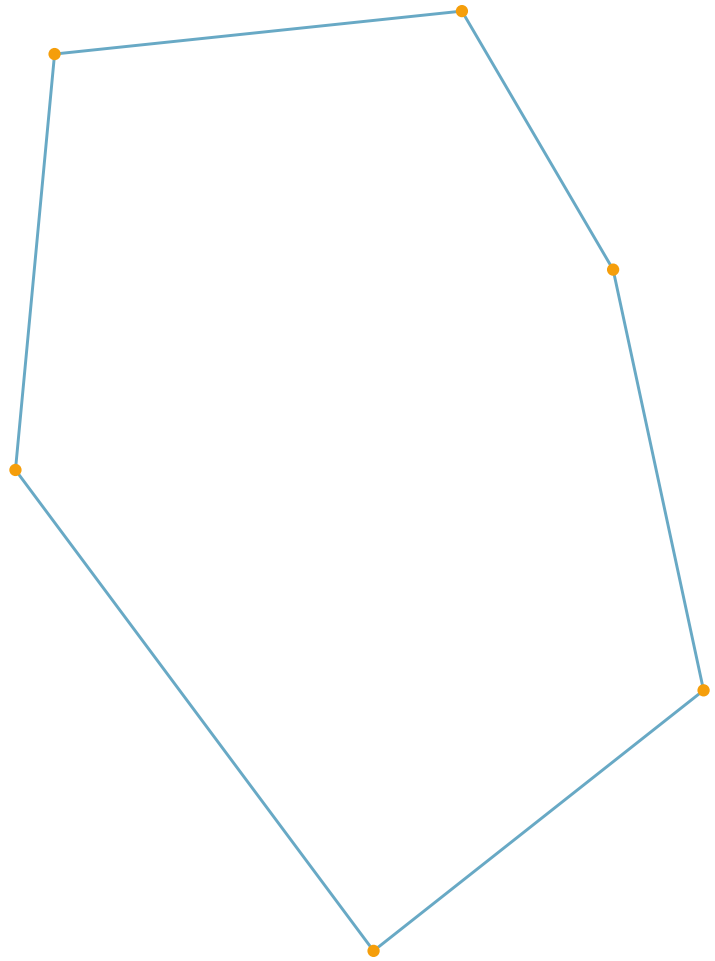
Effective release edges: 3 • unique plotted points: 4 • source: release_effective_edges.csv

08/88 Auriga (Aur) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

08/88 Auriga (Aur) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 28380	89.930167	+37.212778	HIP 28360, HIP 25428
HIP 28360	89.882375	+44.947444	HIP 28380, HIP 24608
HIP 24608	79.172083	+45.999028	HIP 28360, HIP 23453
HIP 23453	75.619500	+41.075889	HIP 24608, HIP 23015
HIP 23015	74.248417	+33.166139	HIP 23453, HIP 25428
HIP 25428	81.572917	+28.607861	HIP 23015, HIP 28380

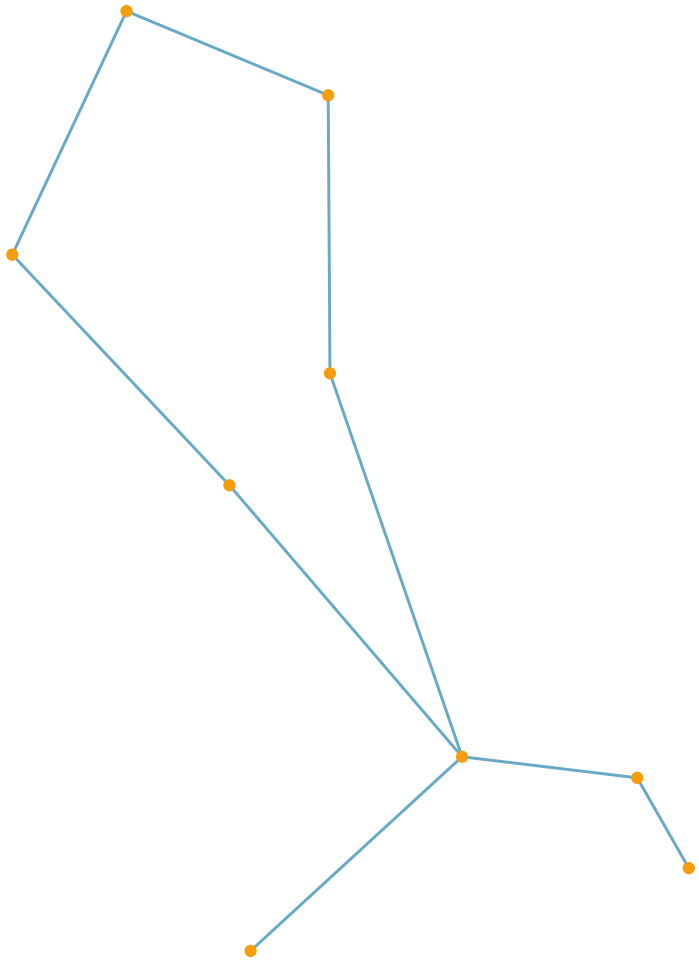
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

09/88 Boötes (Boo) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

09/88 Boötes (Boo) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 71795	220.287167	+13.728333	HIP 69673
HIP 69673	213.918125	+19.187278	HIP 71795, HIP 72105, HIP 71053, HIP 67927
HIP 72105	221.246875	+27.074167	HIP 69673, HIP 74666
HIP 74666	228.875417	+33.315111	HIP 72105, HIP 73555
HIP 73555	225.486625	+40.390639	HIP 74666, HIP 71075
HIP 71075	218.019833	+38.307889	HIP 73555, HIP 71053
HIP 71053	217.957750	+30.371139	HIP 71075, HIP 69673
HIP 67927	208.671333	+18.398583	HIP 69673, HIP 67459
HIP 67459	207.369583	+15.797806	HIP 67927

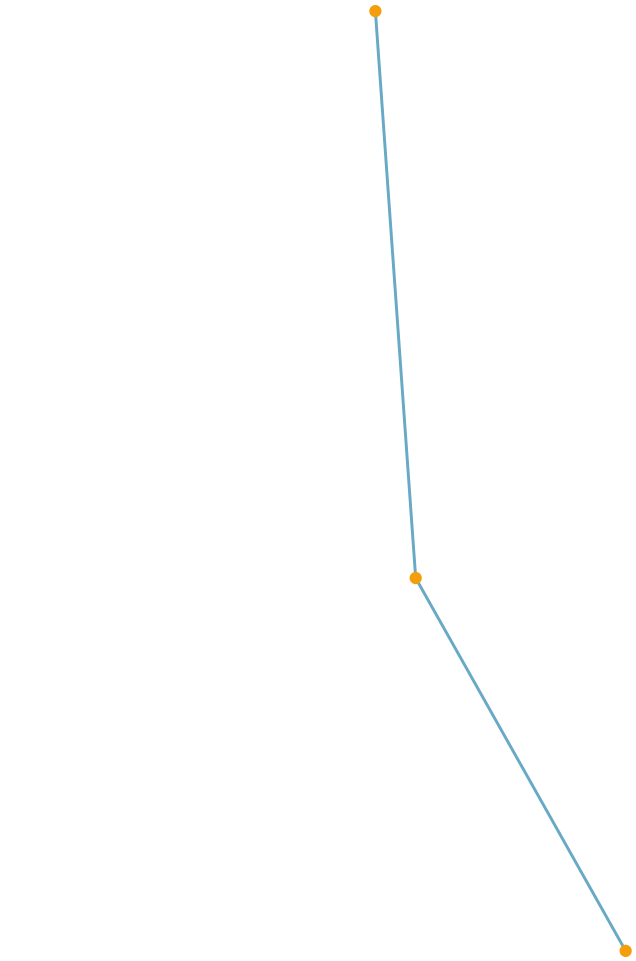
Effective release edges: 9 • unique plotted points: 9 • source: release_effective_edges.csv

10/88 Caelum (Cae) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

10/88 Caelum (Cae) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 21060	67.708750	-44.953750	HIP 21770
HIP 21770	70.140917	-41.863583	HIP 21060, HIP 21861
HIP 21861	70.514375	-37.144778	HIP 21770

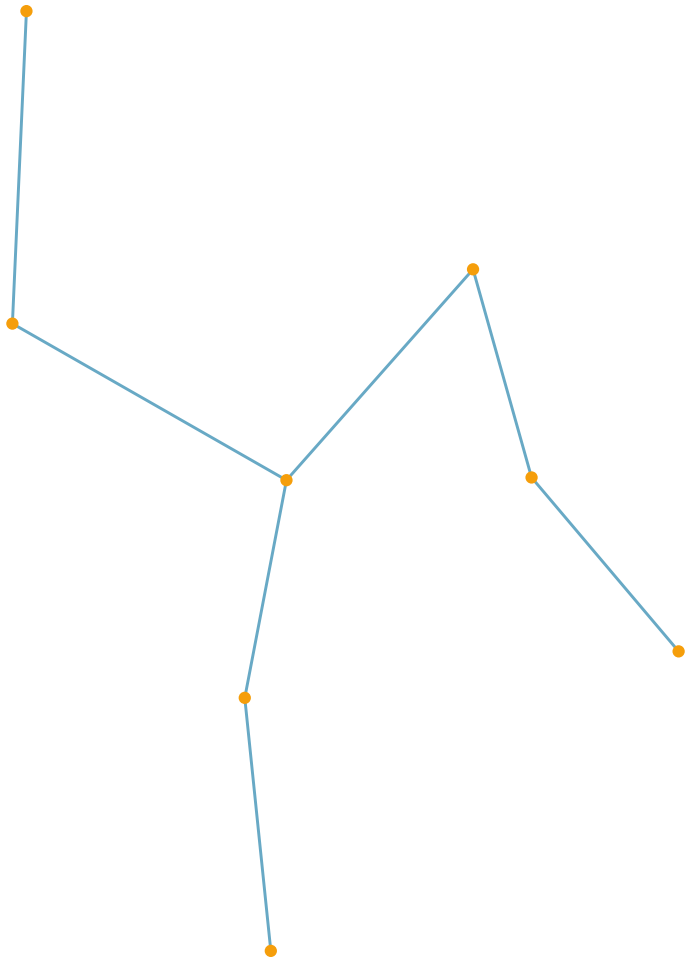
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

11/88 Camelopardalis (Cam) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

11/88 Camelopardalis (Cam) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 33694	105.016033	+76.977442	HIP 29997
HIP 29997	94.711583	+69.320034	HIP 33694, HIP 22783
HIP 22783	73.512544	+66.342660	HIP 29997, HIP 23522, HIP 17959
HIP 23522	75.854572	+60.442281	HIP 22783, HIP 23040
HIP 23040	74.321752	+53.752083	HIP 23522
HIP 17959	57.589483	+71.332368	HIP 22783, HIP 17884
HIP 17884	57.380356	+65.526006	HIP 17959, HIP 16228
HIP 16228	52.267227	+59.940335	HIP 17884

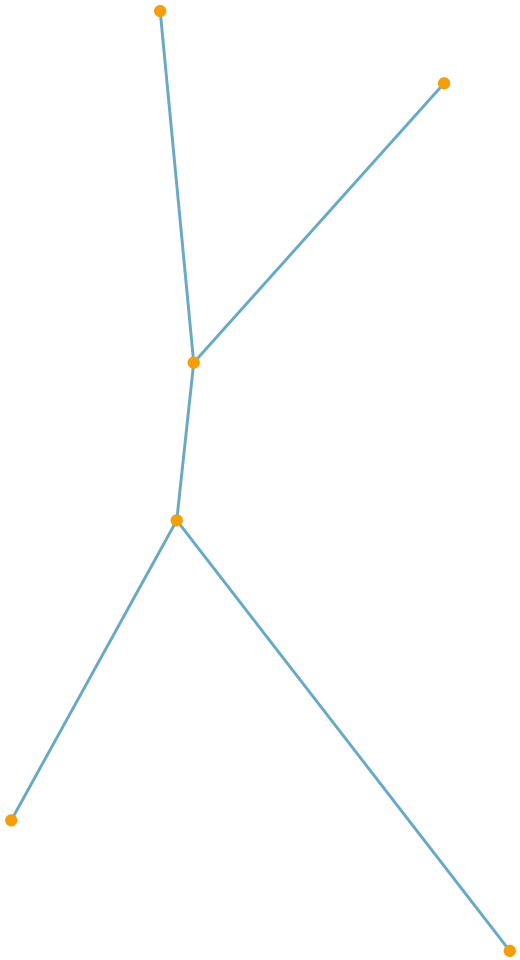
Effective release edges: 7 • unique plotted points: 8 • source: release_effective_edges.csv

12/88 Cancer (Cnc) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

12/88 Cancer (Cnc) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 43103	131.674292	+28.760000	HIP 42806
HIP 42806	130.821708	+21.468583	HIP 43103, HIP 40843, HIP 42911
HIP 40843	125.016125	+27.218611	HIP 42806
HIP 42911	131.171292	+18.154861	HIP 42806, HIP 40526, HIP 44066
HIP 40526	124.128958	+9.185667	HIP 42911
HIP 44066	134.621667	+11.857778	HIP 42911

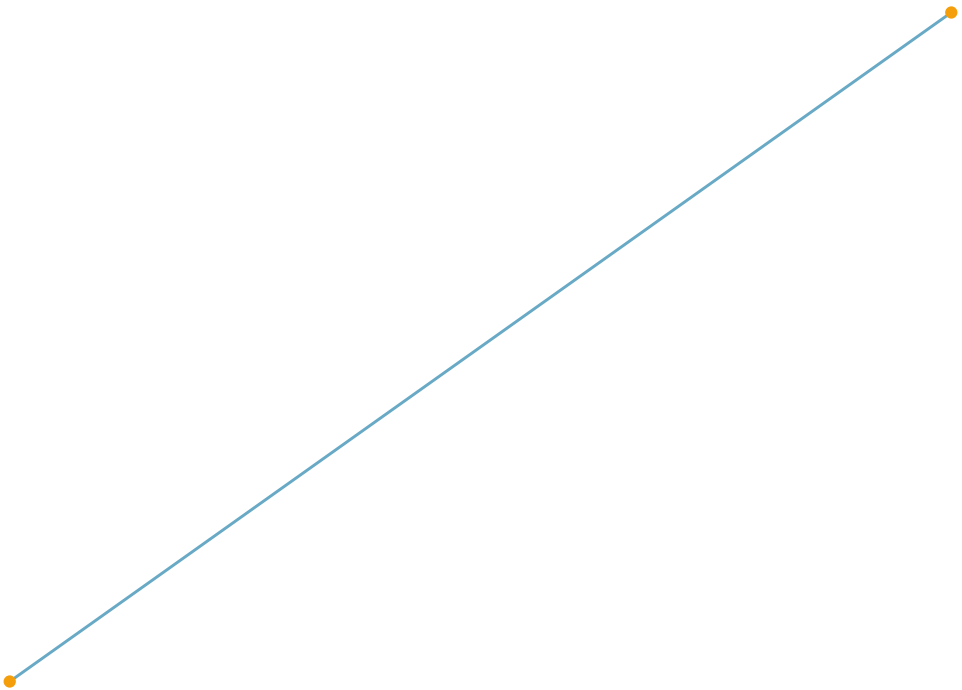
Effective release edges: 5 • unique plotted points: 6 • source: release_effective_edges.csv

13/88 Canes Venatici (CVn) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

13/88 Canes Venatici (CVn) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 61317	188.437875	+41.356778	HIP 63125
HIP 63125	194.007667	+38.318250	HIP 61317

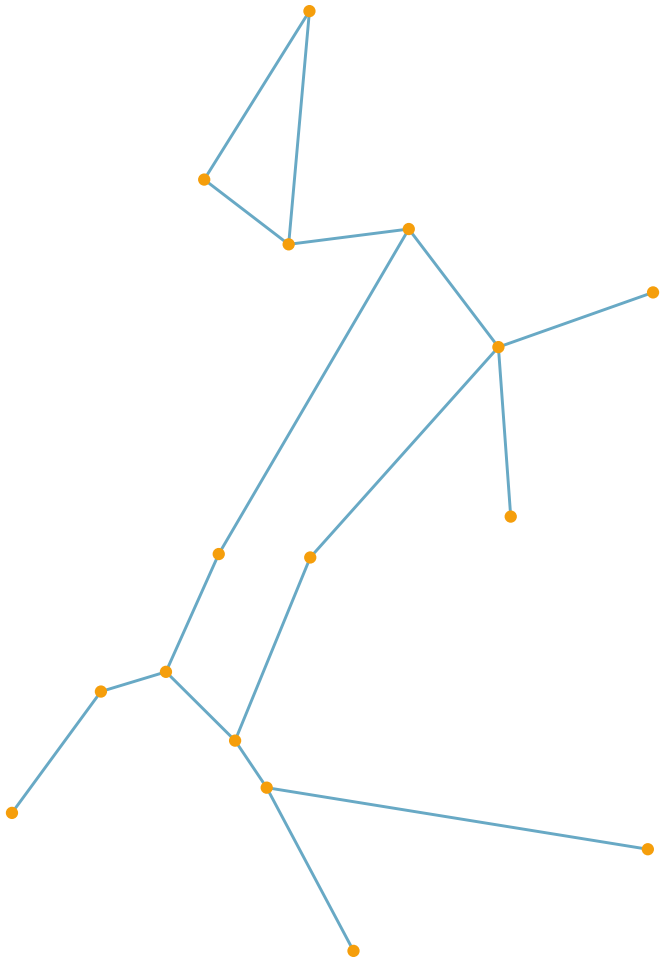
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

14/88 Canis Major (CMa) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

14/88 Canis Major (CMa) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 33160	103.547833	-12.038583	HIP 34045, HIP 33347
HIP 34045	105.939542	-15.633250	HIP 33160, HIP 33347
HIP 33347	104.034292	-17.054250	HIP 34045, HIP 32349, HIP 33160
HIP 32349	101.288542	-16.713139	HIP 33347, HIP 33977, HIP 31592
HIP 33977	105.756125	-23.833306	HIP 32349, HIP 34444
HIP 34444	107.097875	-26.393194	HIP 33977, HIP 35037, HIP 33856
HIP 35037	108.702750	-26.772694	HIP 34444, HIP 35904
HIP 35904	111.023792	-29.303111	HIP 35037
HIP 33579	104.656458	-28.972083	HIP 33856, HIP 32759, HIP 30122
HIP 33856	105.429792	-27.934833	HIP 33579, HIP 34444, HIP 33165
HIP 33165	103.554375	-23.928361	HIP 33856, HIP 31592
HIP 31592	99.170833	-19.255722	HIP 33165, HIP 31416, HIP 30324, HIP 32349
HIP 31416	98.764083	-22.964833	HIP 31592
HIP 30324	95.674958	-17.955917	HIP 31592
HIP 32759	102.460292	-32.508500	HIP 33579
HIP 30122	95.078292	-30.063389	HIP 33579

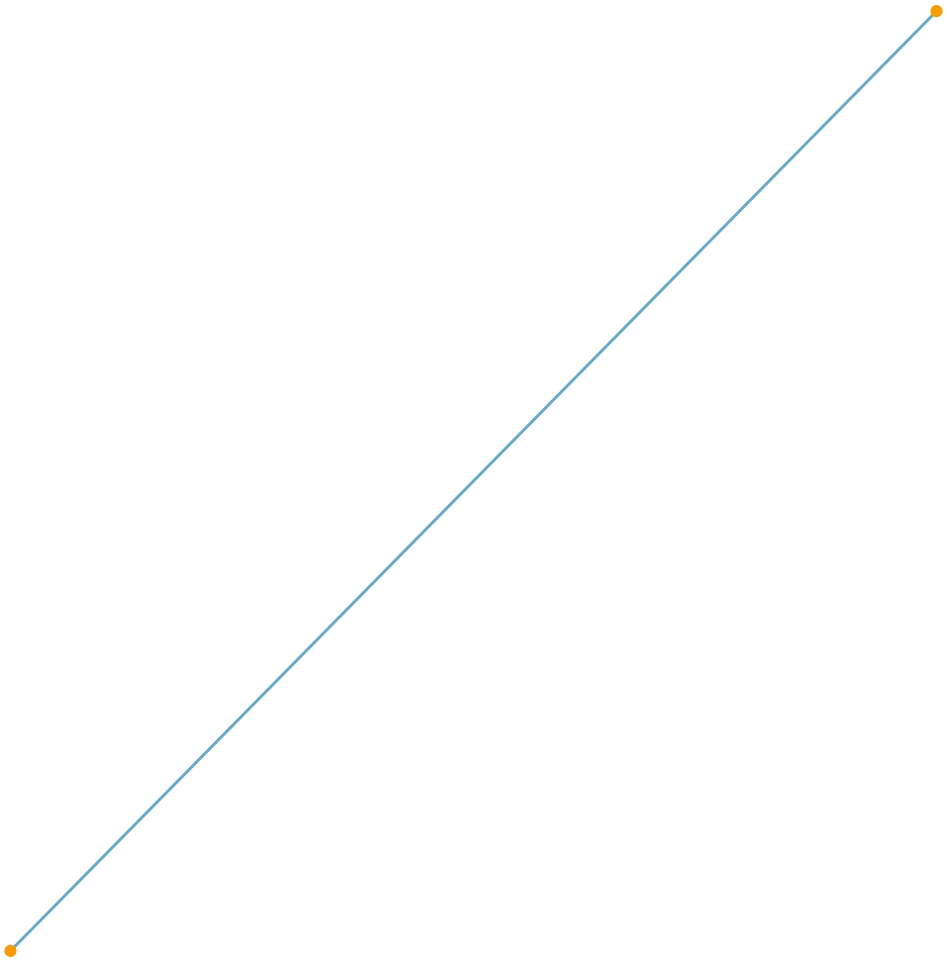
Effective release edges: 17 • unique plotted points: 16 • source: release_effective_edges.csv

15/88 Canis Minor (CMi) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

15/88 Canis Minor (CMi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 37279	114.827250	+5.227500	HIP 36188
HIP 36188	111.787792	+8.289417	HIP 37279

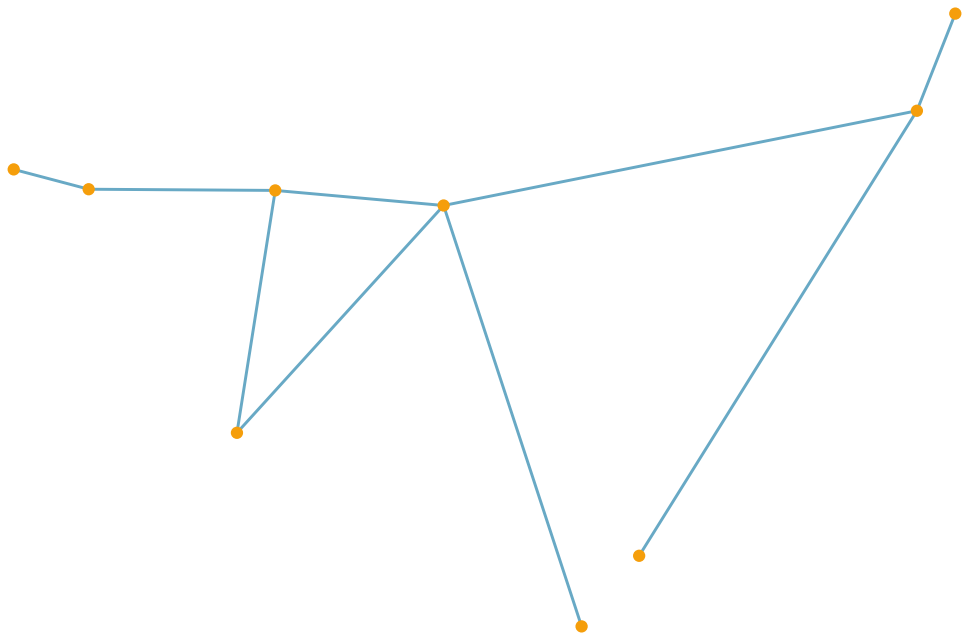
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

16/88 Capricornus (Cap) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

16/88 Capricornus (Cap) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 100064	304.513417	-12.544861	HIP 100345
HIP 100345	305.252708	-14.781389	HIP 100064, HIP 104139, HIP 102485
HIP 104139	316.486583	-17.232722	HIP 100345, HIP 105515, HIP 105881, HIP 102978
HIP 105515	320.561583	-16.834556	HIP 104139, HIP 106985, HIP 105881
HIP 106985	325.022250	-16.662250	HIP 105515, HIP 107556
HIP 107556	326.759542	-16.126556	HIP 106985
HIP 105881	321.666792	-22.411389	HIP 105515, HIP 104139
HIP 102485	311.524042	-25.270528	HIP 100345
HIP 102978	312.955417	-26.919139	HIP 104139

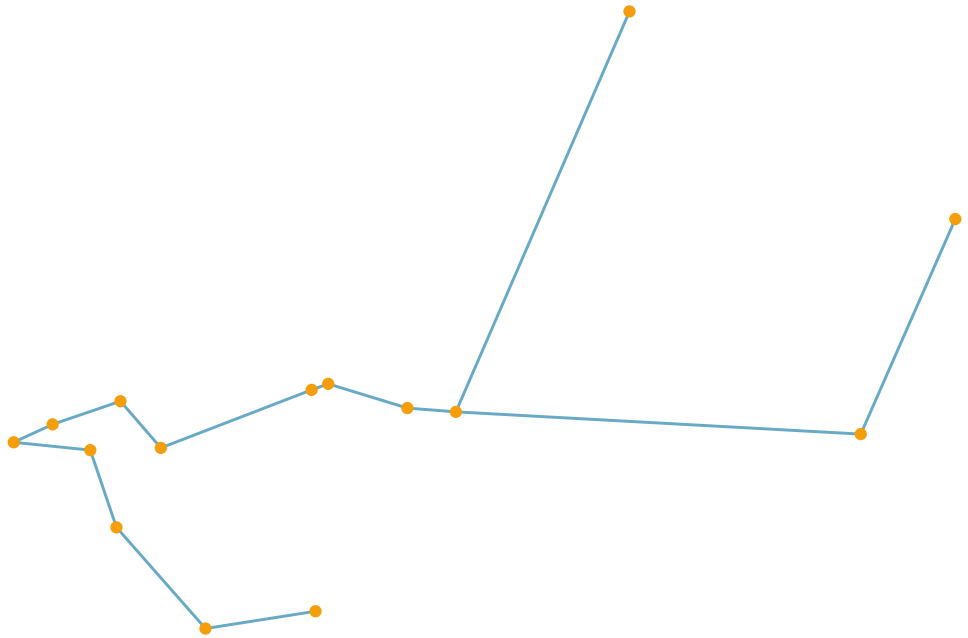
Effective release edges: 9 • unique plotted points: 9 • source: release_effective_edges.csv

17/88 Carina (Car) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

17/88 Carina (Car) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 45238	138.301000	-69.717472	HIP 50099
HIP 50099	153.434500	-70.037917	HIP 45238, HIP 52419
HIP 52419	160.739292	-64.394472	HIP 50099, HIP 52468
HIP 52468	160.884625	-60.566639	HIP 52419, HIP 54463
HIP 54463	167.147500	-58.975056	HIP 52468, HIP 53253
HIP 53253	163.373208	-58.853278	HIP 54463, HIP 51232
HIP 51232	156.969792	-58.739417	HIP 53253, HIP 50371
HIP 50371	154.270875	-61.332333	HIP 51232, HIP 45556
HIP 45556	139.272625	-59.275250	HIP 50371, HIP 45080
HIP 42568	130.154333	-59.761028	HIP 41037, HIP 45080
HIP 41037	125.628583	-59.509528	HIP 42568, HIP 30438, HIP 39429
HIP 30438	95.987875	-52.695722	HIP 41037, HIP 31685
HIP 45080	137.742125	-58.966917	HIP 45556, HIP 42568
HIP 31685	99.440292	-43.195917	HIP 30438
HIP 39429	120.896125	-40.003194	HIP 41037

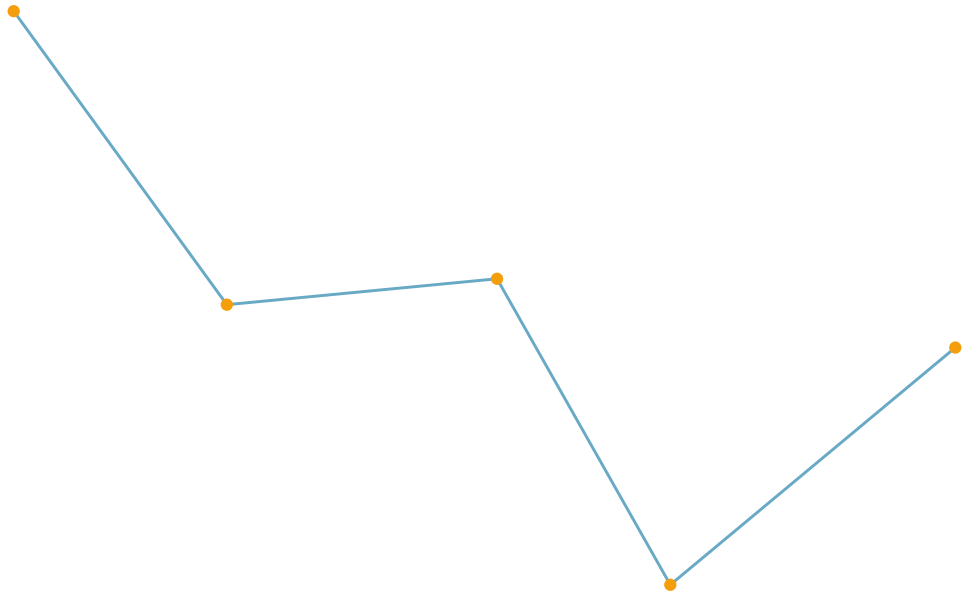
Effective release edges: 14 • unique plotted points: 15 • source: release_effective_edges.csv

18/88 Cassiopeia (Cas) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

18/88 Cassiopeia (Cas) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 8886	28.598667	+63.670139	HIP 6686
HIP 6686	21.452500	+60.235417	HIP 8886, HIP 4427
HIP 4427	14.177083	+60.716750	HIP 6686, HIP 3179
HIP 3179	10.126625	+56.537417	HIP 4427, HIP 746
HIP 746	2.292042	+59.150222	HIP 3179

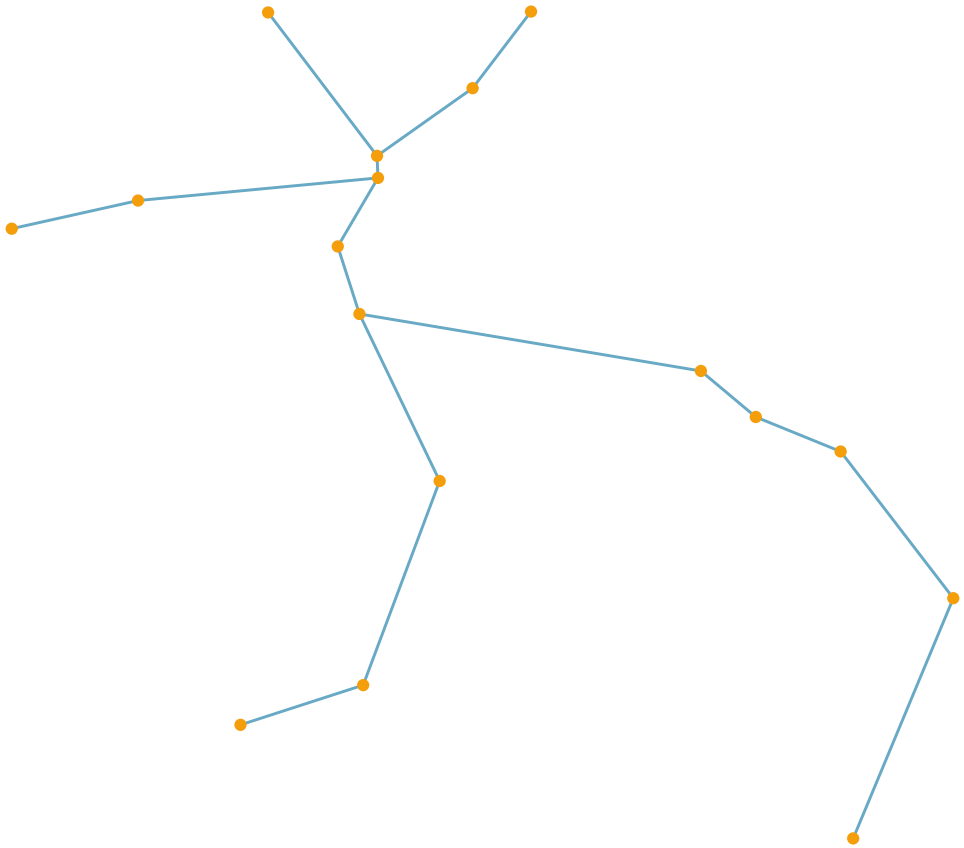
Effective release edges: 4 • unique plotted points: 5 • source: release_effective_edges.csv

19/88 Centaurus (Cen) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

19/88 Centaurus (Cen) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 71683	219.920417	-60.835139	HIP 68702
HIP 68702	210.956000	-60.372972	HIP 71683, HIP 66657
HIP 66657	204.971958	-53.466361	HIP 68702, HIP 68002
HIP 68002	208.885125	-47.288278	HIP 66657, HIP 68282, HIP 61932
HIP 68282	209.669875	-44.803528	HIP 68002, HIP 67472
HIP 67472	207.404208	-42.473694	HIP 68282, HIP 67464, HIP 71352
HIP 67464	207.376250	-41.687667	HIP 67472, HIP 65936, HIP 68933
HIP 65936	202.761125	-39.407278	HIP 67464, HIP 65109
HIP 65109	200.150292	-36.712083	HIP 65936
HIP 68933	211.672167	-36.368694	HIP 67464
HIP 71352	218.876875	-42.157750	HIP 67472, HIP 73334
HIP 73334	224.790417	-42.104139	HIP 71352
HIP 61932	190.380000	-48.959889	HIP 68002, HIP 60823
HIP 60823	187.010042	-50.230611	HIP 61932, HIP 59196
HIP 59196	182.089750	-50.722417	HIP 60823, HIP 56480
HIP 56480	173.690458	-54.264139	HIP 59196, HIP 56561
HIP 56561	173.945542	-63.019833	HIP 56480

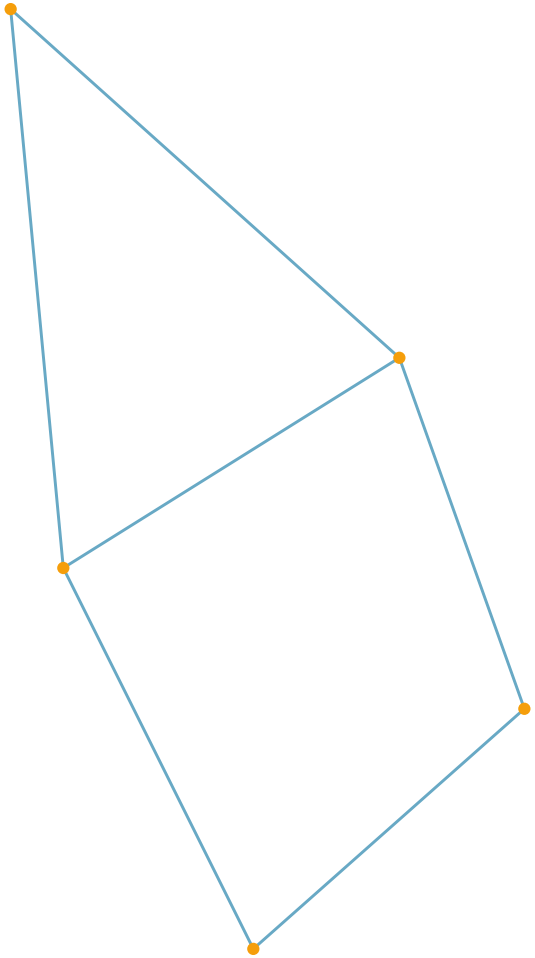
Effective release edges: 16 • unique plotted points: 17 • source: release_effective_edges.csv

20/88 Cepheus (Cep) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

20/88 Cepheus (Cep) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 109492	332.713583	+58.201250	HIP 112724, HIP 105199
HIP 112724	342.420458	+66.200722	HIP 109492, HIP 106032, HIP 116727
HIP 106032	322.164917	+70.560694	HIP 112724, HIP 105199, HIP 116727
HIP 105199	319.644083	+62.585444	HIP 106032, HIP 109492
HIP 116727	354.837417	+77.631972	HIP 112724, HIP 106032

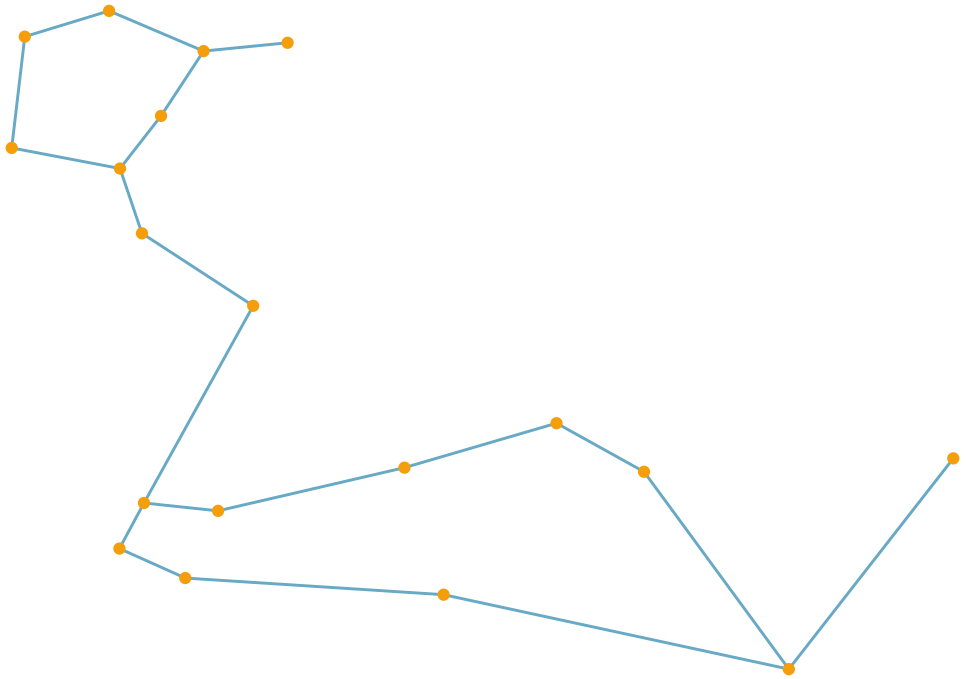
Effective release edges: 6 • unique plotted points: 5 • source: release_effective_edges.csv

21/88 Cetus (Cet) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

21/88 Cetus (Cet) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 10324	33.250042	+8.846750	HIP 11484
HIP 11484	37.039667	+8.460083	HIP 10324, HIP 12828, HIP 12093
HIP 8102	26.021375	-15.939556	HIP 3419, HIP 11783
HIP 3419	10.896792	-17.986694	HIP 8102, HIP 1562, HIP 5364
HIP 1562	4.857000	-8.823833	HIP 3419
HIP 5364	17.146917	-10.181917	HIP 3419, HIP 6537
HIP 6537	21.006042	-8.182750	HIP 5364, HIP 8645
HIP 8645	27.865042	-10.334944	HIP 6537, HIP 11345
HIP 11345	36.487542	-12.290444	HIP 8645, HIP 12390
HIP 12390	39.890542	-11.871583	HIP 11345, HIP 12770, HIP 10826
HIP 12770	41.030625	-13.858667	HIP 12390, HIP 11783
HIP 11783	38.022000	-15.244333	HIP 12770, HIP 8102
HIP 10826	34.836625	-2.977056	HIP 12390, HIP 12387
HIP 12387	39.870625	+0.328528	HIP 10826, HIP 12706
HIP 12706	40.825500	+3.236167	HIP 12387, HIP 14135, HIP 12093
HIP 14135	45.569917	+4.089917	HIP 12706, HIP 13954
HIP 13954	44.928750	+8.907389	HIP 14135, HIP 12828
HIP 12828	41.234875	+10.114222	HIP 13954, HIP 11484
HIP 12093	38.968708	+5.593306	HIP 11484, HIP 12706

Effective release edges: 20 • unique plotted points: 19 • source: release_effective_edges.csv

22/88 Chamaeleon (Cha) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

22/88 Chamaeleon (Cha) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 40702	124.630292	-76.919972	HIP 51839
HIP 51839	158.867583	-78.607806	HIP 40702, HIP 60000
HIP 60000	184.587250	-79.312278	HIP 51839

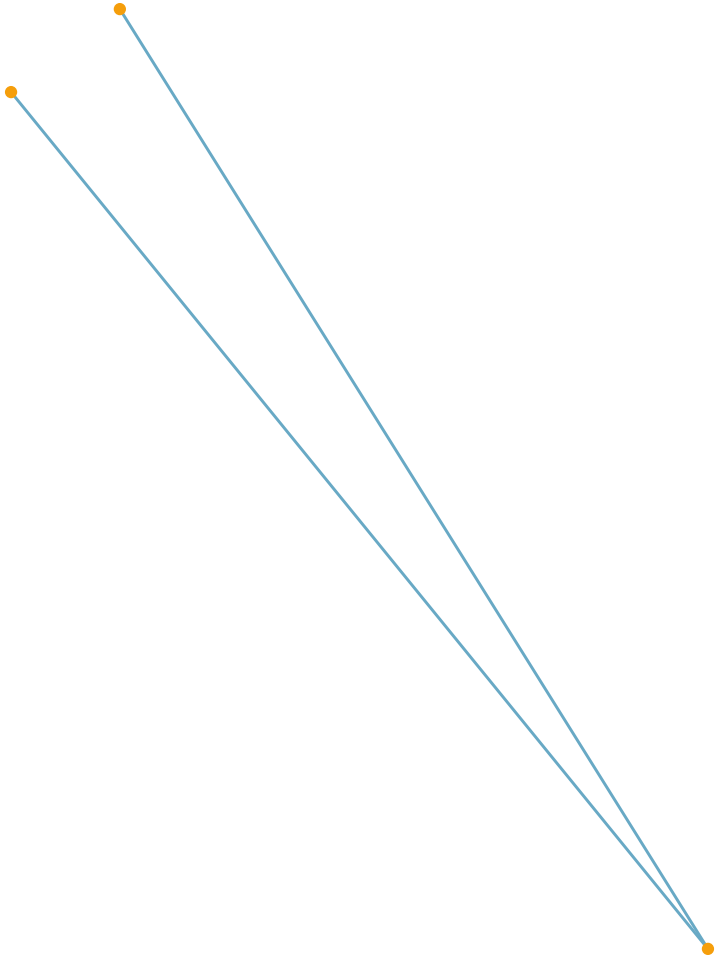
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

23/88 Circinus (Cir) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

23/88 Circinus (Cir) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 71908	220.627875	-64.974583	HIP 75323, HIP 74824
HIP 75323	230.844417	-59.320694	HIP 71908
HIP 74824	229.379000	-58.800889	HIP 71908

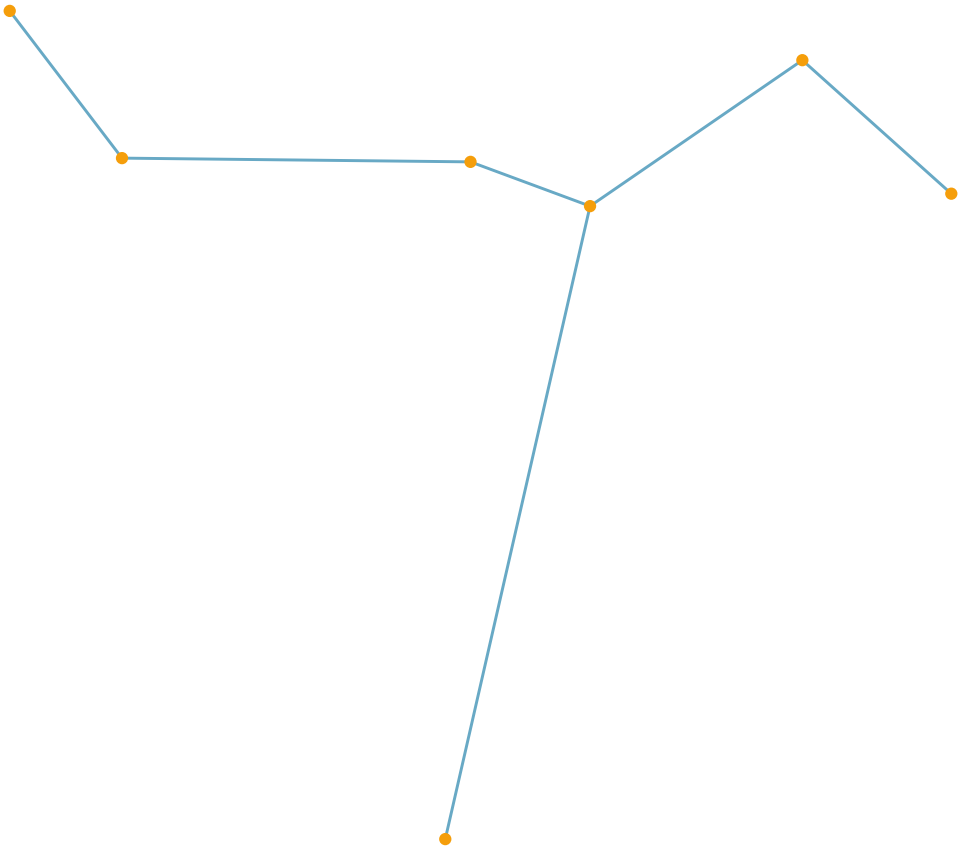
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

24/88 Columba (Col) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

24/88 Columba (Col) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 30277	95.528542	-33.436278	HIP 29807
HIP 29807	94.138083	-35.140722	HIP 30277, HIP 28199
HIP 28199	89.384208	-35.283306	HIP 29807, HIP 27628
HIP 27628	87.739792	-35.769306	HIP 28199, HIP 28328, HIP 26634
HIP 28328	89.786625	-42.815111	HIP 27628
HIP 26634	84.912250	-34.074056	HIP 27628, HIP 25859
HIP 25859	82.803083	-35.470444	HIP 26634

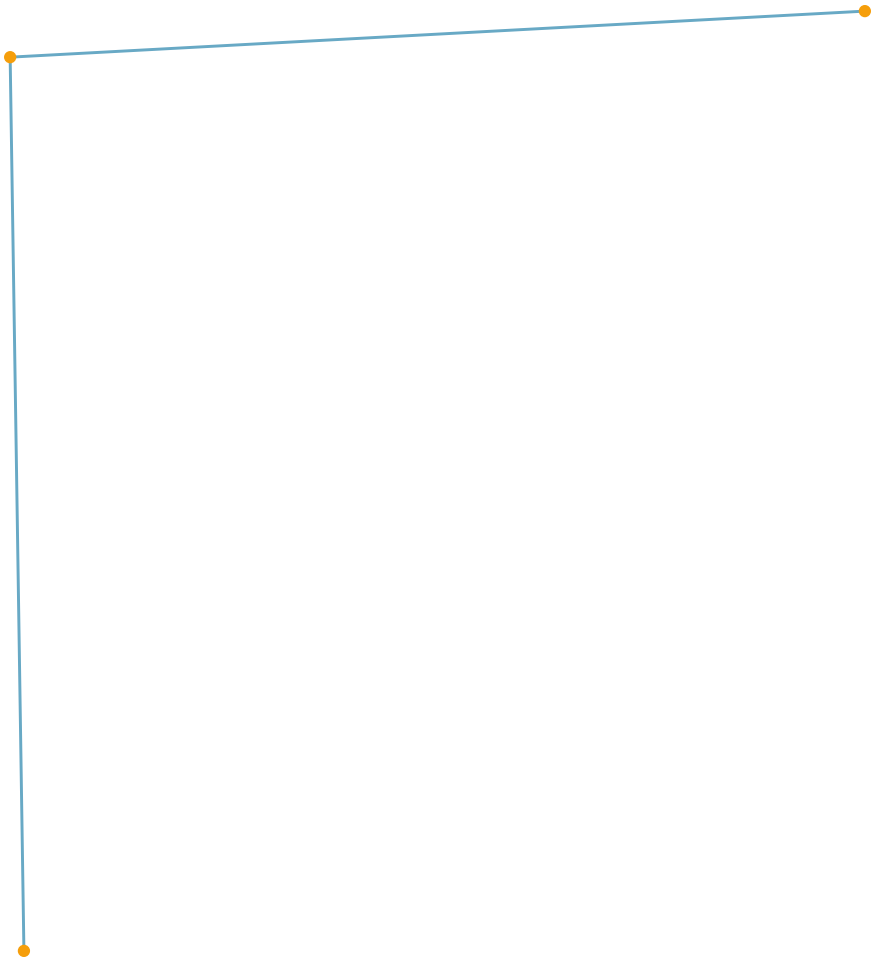
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

25/88 Coma Berenices (Com) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

25/88 Coma Berenices (Com) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 64241	197.498125	+17.529111	HIP 64394
HIP 64394	197.970500	+27.876028	HIP 64241, HIP 60742
HIP 60742	186.734708	+28.268611	HIP 64394

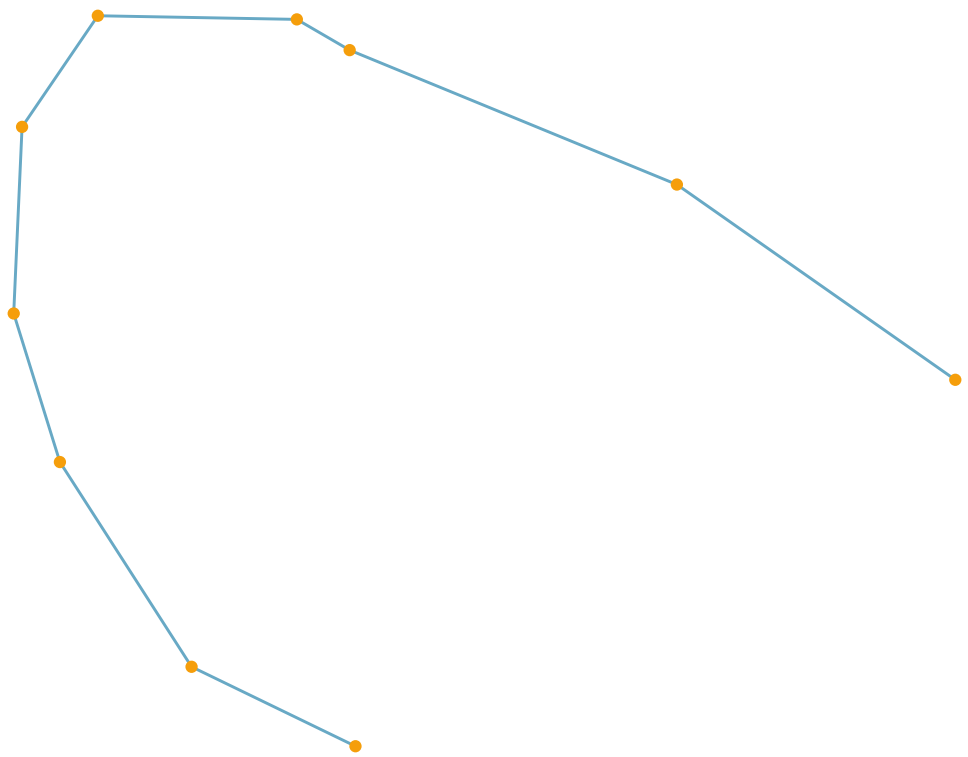
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

26/88 Corona Australis (CrA) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

26/88 Corona Australis (CrA) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 91875	280.945583	-38.323306	HIP 92989, HIP 90887
HIP 92989	284.168708	-37.343194	HIP 91875, HIP 93174
HIP 93174	284.681125	-37.107083	HIP 92989, HIP 93825
HIP 93825	286.604333	-37.062750	HIP 93174, HIP 94114
HIP 94114	287.367833	-37.904250	HIP 93825, HIP 94160
HIP 94160	287.507292	-39.340694	HIP 94114, HIP 94005
HIP 94005	287.087208	-40.496639	HIP 94160, HIP 93542
HIP 93542	285.778458	-42.095000	HIP 94005, HIP 92953
HIP 92953	284.070708	-42.710611	HIP 93542
HIP 90887	278.088792	-39.703917	HIP 91875

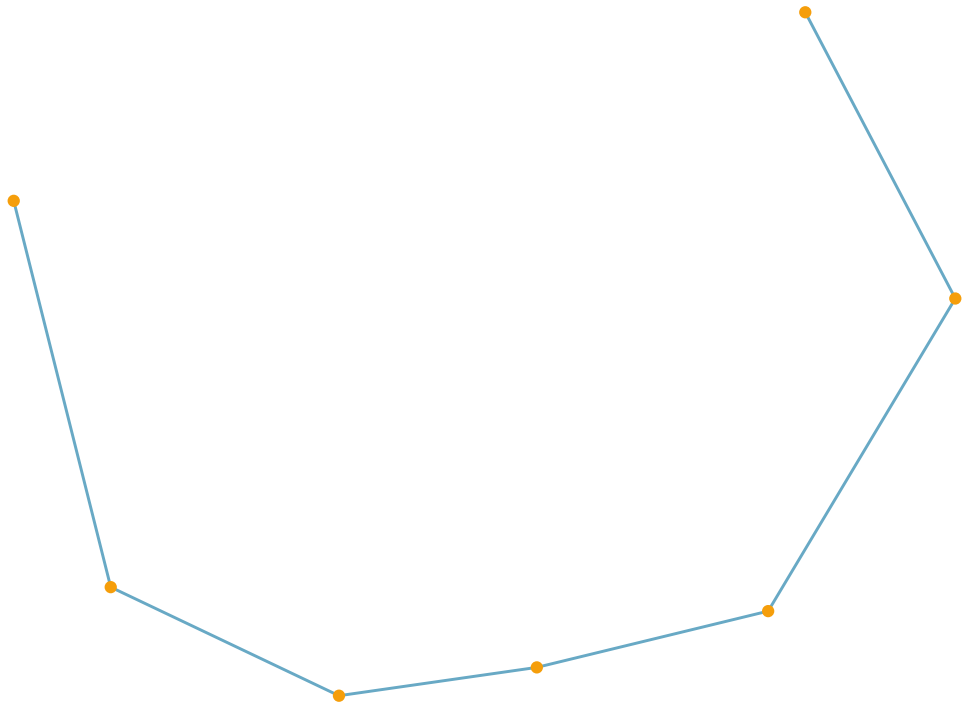
Effective release edges: 9 • unique plotted points: 10 • source: release_effective_edges.csv

27/88 Corona Borealis (CrB) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

27/88 Corona Borealis (CrB) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 76127	233.232500	+31.359167	HIP 75695
HIP 75695	231.957708	+29.105500	HIP 76127, HIP 76267
HIP 76267	233.671625	+26.714917	HIP 75695, HIP 76952
HIP 76952	235.686000	+26.295528	HIP 76267, HIP 77512
HIP 77512	237.398750	+26.068556	HIP 76952, HIP 78159
HIP 78159	239.397083	+26.878028	HIP 77512, HIP 78493
HIP 78493	240.360792	+29.851083	HIP 78159

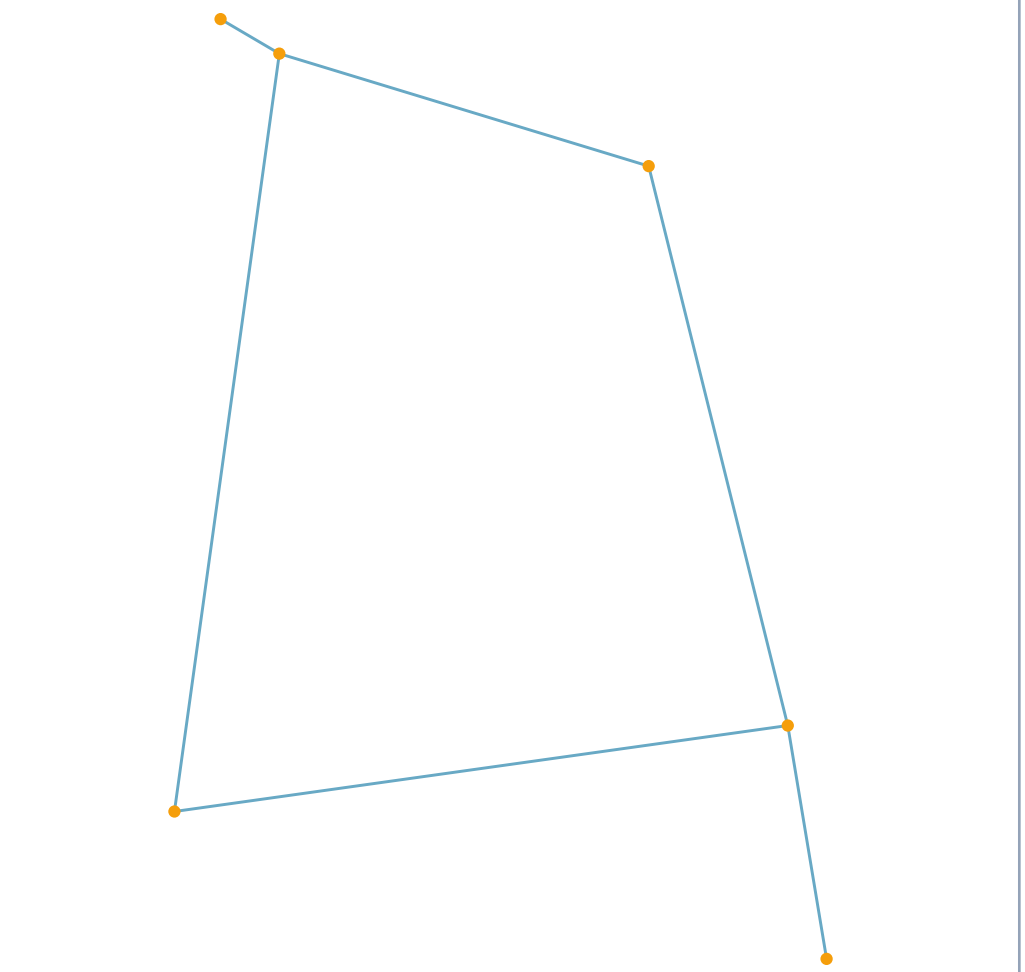
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

28/88 Corvus (Crv) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

28/88 Corvus (Crv) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 61174	188.018667	-16.195861	HIP 60965
HIP 60965	187.466583	-16.515083	HIP 61174, HIP 59803, HIP 61359
HIP 59803	183.951958	-17.541972	HIP 60965, HIP 59316
HIP 59316	182.531375	-22.619806	HIP 59803, HIP 59199, HIP 61359
HIP 59199	182.103125	-24.728778	HIP 59316
HIP 61359	188.596792	-23.396611	HIP 59316, HIP 60965

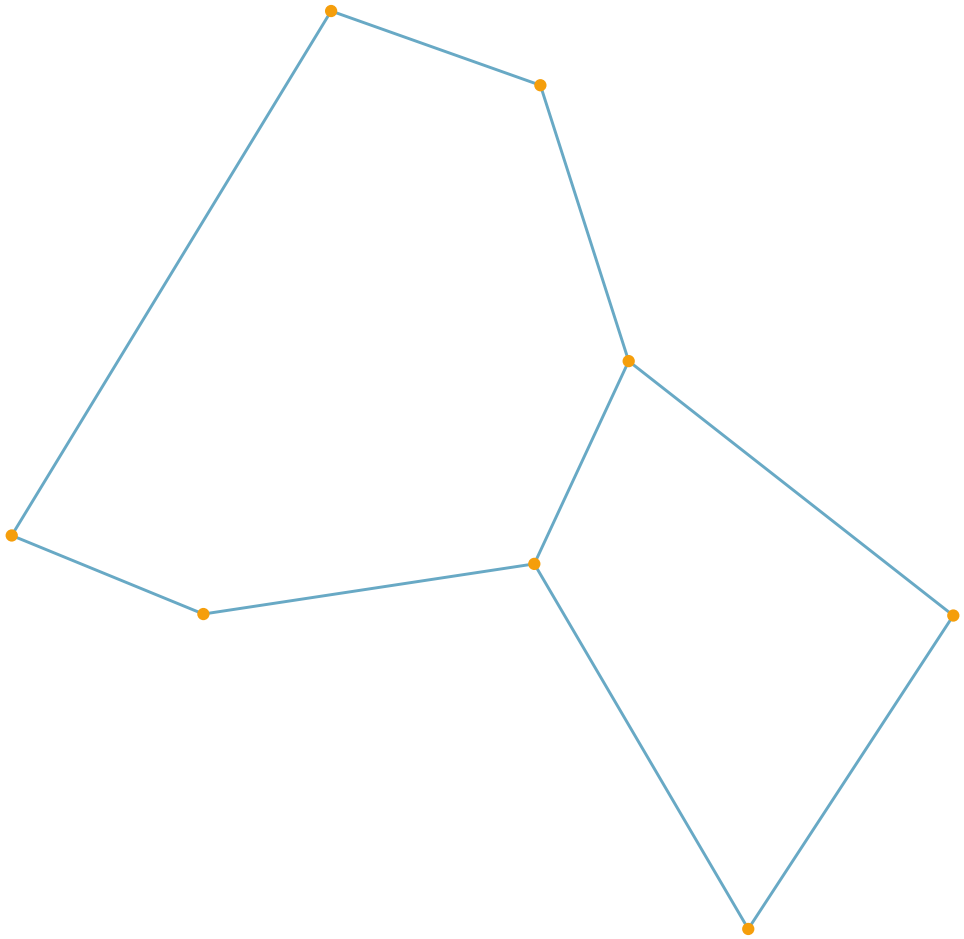
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

29/88 Crater (Crt) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

29/88 Crater (Crt) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 53740	164.944792	-18.299111	HIP 54682, HIP 55282
HIP 54682	167.914542	-22.825611	HIP 53740, HIP 55705
HIP 55705	171.220750	-17.684028	HIP 54682, HIP 55282, HIP 57283
HIP 55282	169.835500	-14.779056	HIP 55705, HIP 53740, HIP 55687
HIP 55687	171.152542	-10.859389	HIP 55282, HIP 56633
HIP 56633	174.170625	-9.802250	HIP 55687, HIP 58188
HIP 58188	179.004083	-17.150806	HIP 56633, HIP 57283
HIP 57283	176.190667	-18.350611	HIP 58188, HIP 55705

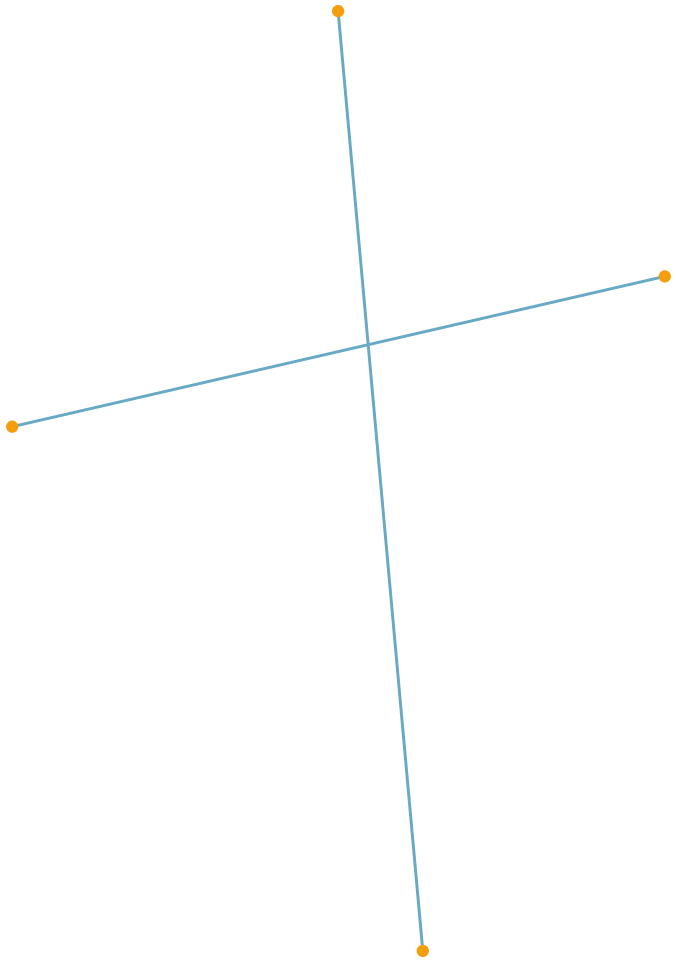
Effective release edges: 9 • unique plotted points: 8 • source: release_effective_edges.csv

30/88 Crux (Cru) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

30/88 Crux (Cru) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 61084	187.791375	-57.112556	HIP 60718
HIP 60718	186.649750	-63.099056	HIP 61084
HIP 62434	191.930500	-59.688722	HIP 59747
HIP 59747	183.786500	-58.748889	HIP 62434

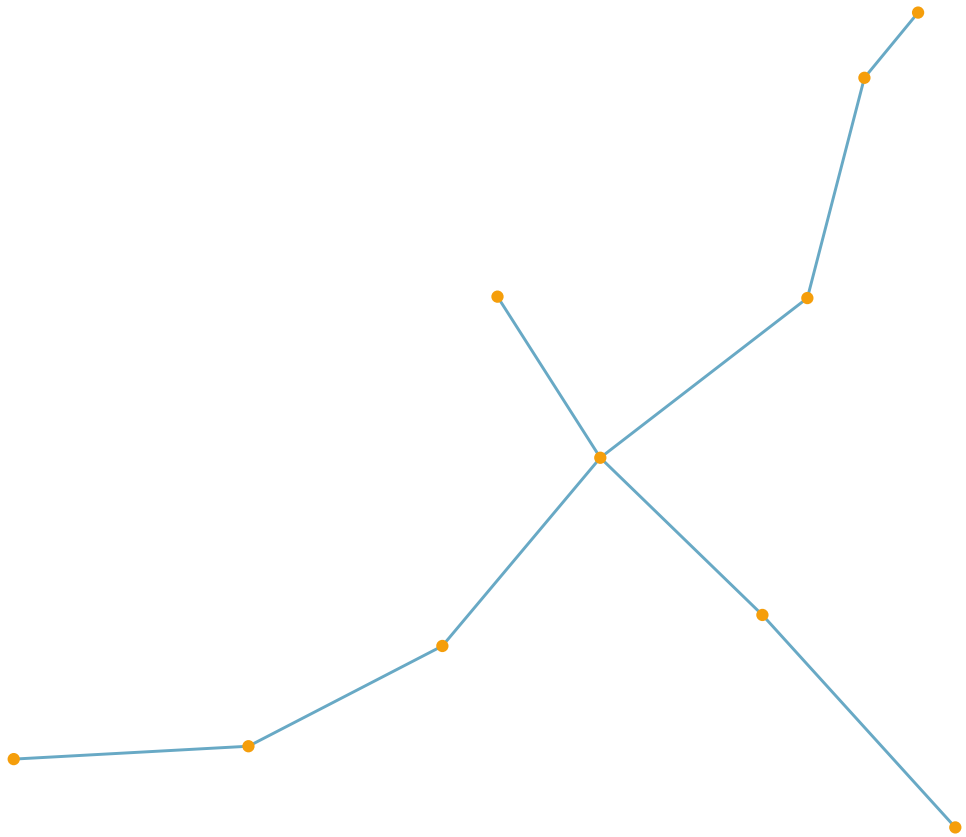
Effective release edges: 2 • unique plotted points: 4 • source: release_effective_edges.csv

31/88 Cygnus (Cyg) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

31/88 Cygnus (Cyg) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 94779	289.275458	+53.368167	HIP 95853
HIP 95853	292.426417	+51.729472	HIP 94779, HIP 97165
HIP 97165	296.243500	+45.130694	HIP 95853, HIP 100453
HIP 100453	305.557083	+40.256694	HIP 97165, HIP 102098, HIP 102488, HIP 98110
HIP 102098	310.357958	+45.280333	HIP 100453
HIP 102488	311.551792	+33.969444	HIP 100453, HIP 104732
HIP 104732	318.234083	+30.227083	HIP 102488, HIP 107310
HIP 107310	326.035000	+28.743222	HIP 104732
HIP 98110	299.076667	+35.083500	HIP 100453, HIP 95947
HIP 95947	292.680375	+27.959694	HIP 98110

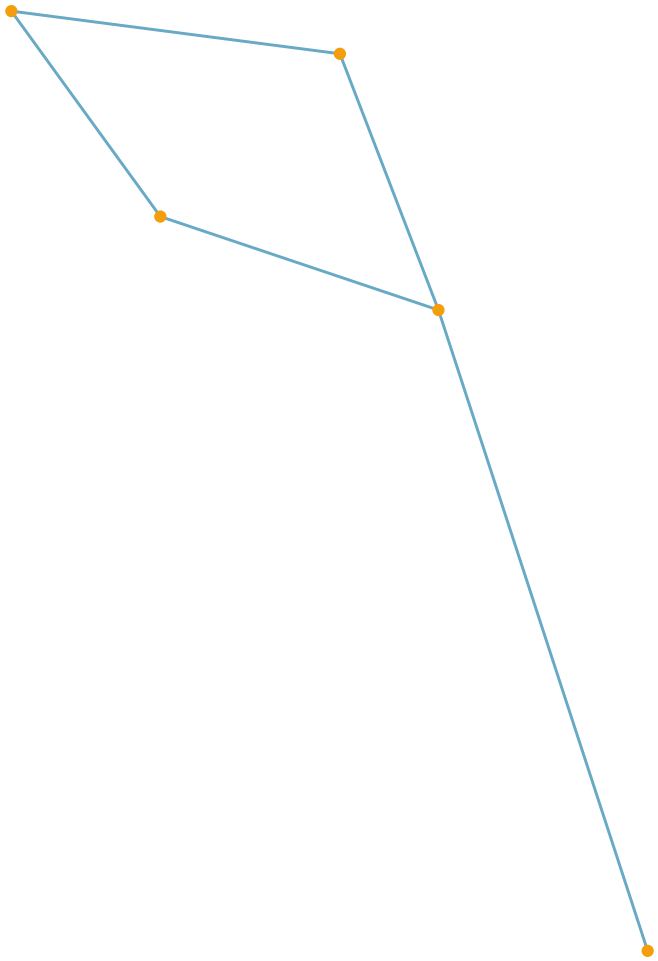
Effective release edges: 9 • unique plotted points: 10 • source: release_effective_edges.csv

32/88 Delphinus (Del) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

32/88 Delphinus (Del) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 101421	308.303167	+11.303333	HIP 101769
HIP 101769	309.386958	+14.595194	HIP 101421, HIP 101958, HIP 102281
HIP 101958	309.909375	+15.912056	HIP 101769, HIP 102532
HIP 102532	311.664667	+16.124778	HIP 101958, HIP 102281
HIP 102281	310.864792	+15.074694	HIP 102532, HIP 101769

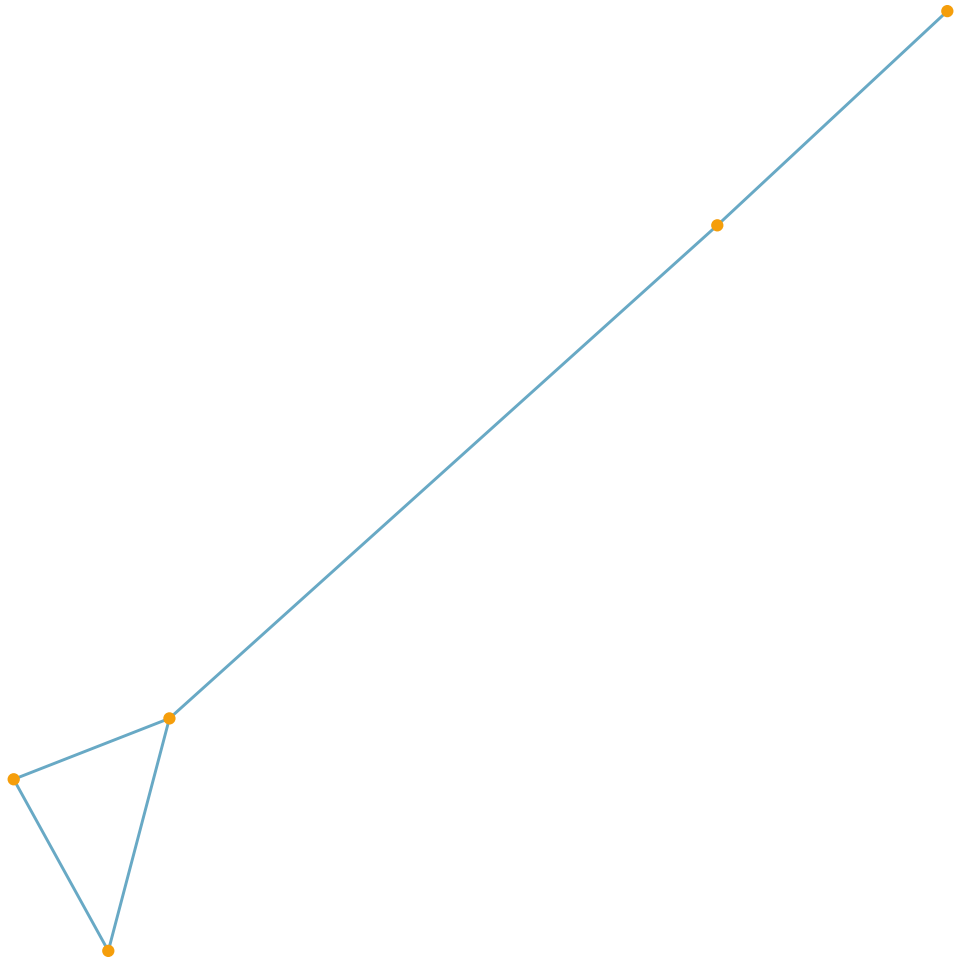
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

33/88 Dorado (Dor) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

33/88 Dorado (Dor) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 27100	86.193417	-65.735528	HIP 27890, HIP 26069
HIP 27890	88.524583	-63.091028	HIP 27100, HIP 26069
HIP 26069	83.406333	-62.489861	HIP 27890, HIP 27100, HIP 21281
HIP 21281	68.498833	-55.045000	HIP 26069, HIP 19893
HIP 19893	64.006208	-51.487083	HIP 21281

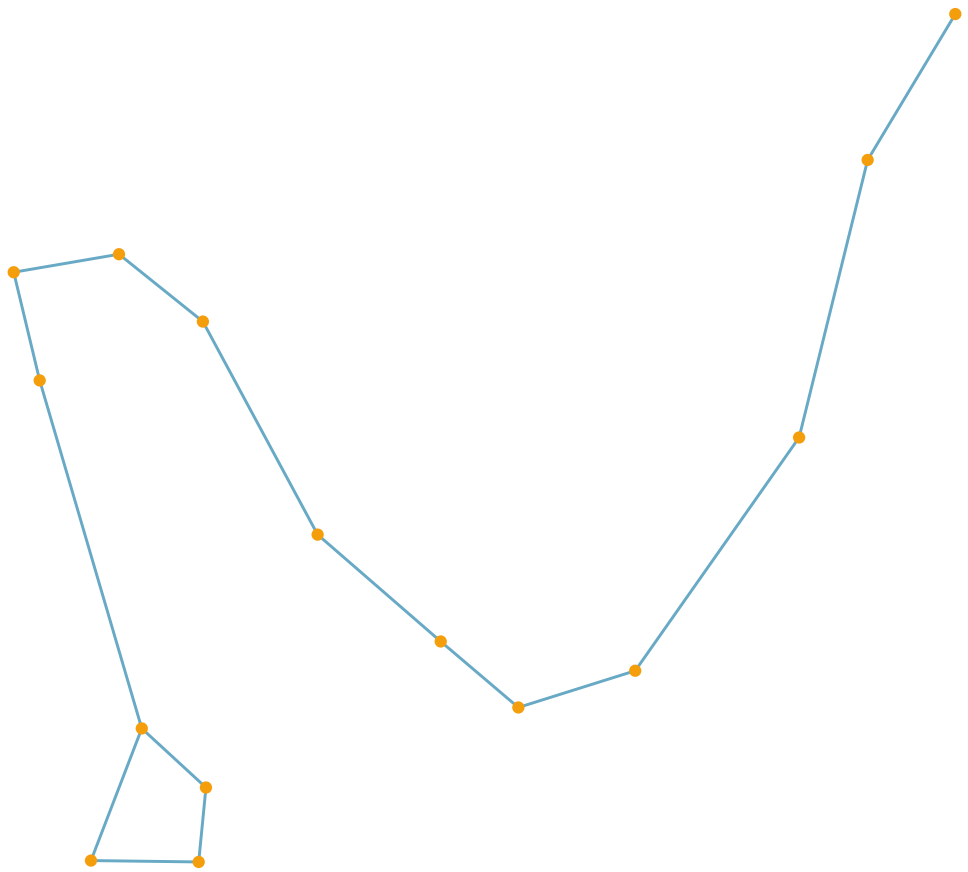
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

34/88 Draco (Dra) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

34/88 Draco (Dra) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 87585	268.381792	+56.872444	HIP 87833, HIP 85829, HIP 94376
HIP 87833	269.151583	+51.488944	HIP 87585, HIP 85670
HIP 85670	262.608250	+52.301361	HIP 87833, HIP 85829
HIP 85829	263.066167	+55.172806	HIP 85670, HIP 87585
HIP 94376	288.138125	+67.661306	HIP 87585, HIP 97433
HIP 97433	297.042542	+70.267833	HIP 94376, HIP 94648
HIP 94648	288.888708	+73.355222	HIP 97433, HIP 89937
HIP 89937	275.259750	+72.733694	HIP 94648, HIP 83895
HIP 83895	257.196792	+65.714639	HIP 89937, HIP 80331
HIP 80331	245.997958	+61.514083	HIP 83895, HIP 78527
HIP 78527	240.473750	+58.564444	HIP 80331, HIP 75458
HIP 75458	231.232417	+58.966028	HIP 78527, HIP 68756
HIP 68756	211.097625	+64.375806	HIP 75458, HIP 61281
HIP 61281	188.371000	+69.788222	HIP 68756, HIP 56211
HIP 56211	172.851208	+69.331111	HIP 61281

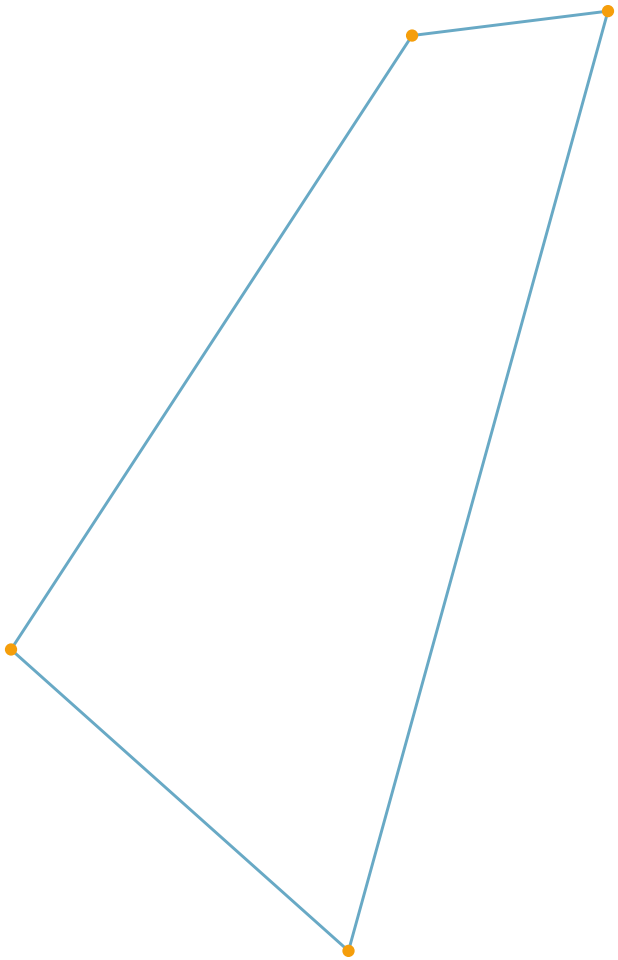
Effective release edges: 15 • unique plotted points: 15 • source: release_effective_edges.csv

35/88 Equuleus (Equ) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

35/88 Equuleus (Equ) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 104521	317.585292	+10.131944	HIP 104858, HIP 104987
HIP 104858	318.619958	+10.007722	HIP 104521, HIP 105570
HIP 105570	320.723250	+6.811111	HIP 104858, HIP 104987
HIP 104987	318.955833	+5.248083	HIP 105570, HIP 104521

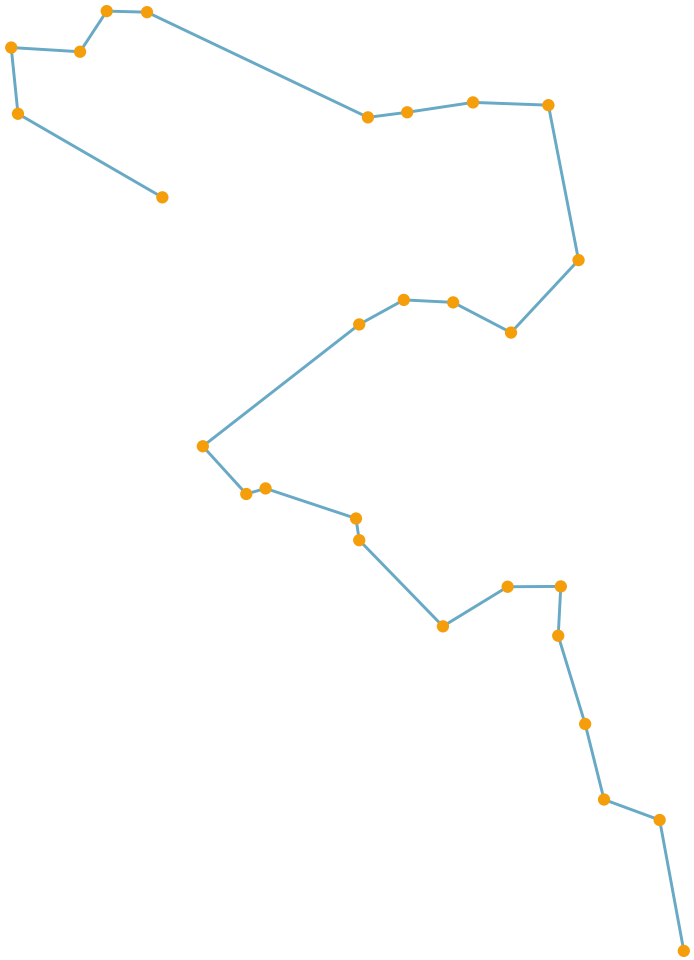
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

36/88 Eridanus (Eri) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

36/88 Eridanus (Eri) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 7588	24.428125	-57.236667	HIP 9007
HIP 9007	28.986792	-51.609583	HIP 7588, HIP 10602
HIP 10602	34.127083	-51.512111	HIP 9007, HIP 11407
HIP 11407	36.746250	-47.703833	HIP 10602, HIP 12413
HIP 12413	39.949667	-42.891639	HIP 11407, HIP 12486
HIP 12486	40.166375	-39.855306	HIP 12413, HIP 13847
HIP 13847	44.565500	-40.304722	HIP 12486, HIP 15510
HIP 15510	49.971750	-43.071556	HIP 13847, HIP 17797
HIP 17797	57.149250	-37.620139	HIP 15510, HIP 17874
HIP 17874	57.363667	-36.200111	HIP 17797, HIP 20042
HIP 20042	64.473417	-33.798333	HIP 17874, HIP 20535
HIP 20535	66.009042	-34.017000	HIP 20042, HIP 21393
HIP 21393	68.887792	-30.562306	HIP 20535, HIP 17651
HIP 17651	56.712458	-23.248444	HIP 21393, HIP 16611
HIP 16611	53.446875	-21.632806	HIP 17651, HIP 15474
HIP 15474	49.879042	-21.757944	HIP 16611, HIP 14146
HIP 14146	45.598292	-23.624333	HIP 15474, HIP 12843
HIP 12843	41.274917	-18.572639	HIP 14146, HIP 13701
HIP 13701	44.106667	-8.897611	HIP 12843, HIP 15197
HIP 15197	48.958458	-8.819833	HIP 13701, HIP 16537
HIP 16537	53.235083	-9.458306	HIP 15197, HIP 17378
HIP 17378	55.812333	-9.765194	HIP 16537, HIP 21444
HIP 21444	69.079750	-3.352444	HIP 17378, HIP 22109
HIP 22109	71.375583	-3.254611	HIP 21444, HIP 22701
HIP 22701	73.223667	-5.452750	HIP 22109, HIP 23875
HIP 23875	76.962625	-5.086250	HIP 22701, HIP 23972
HIP 23972	77.286583	-8.754083	HIP 23875, HIP 21594
HIP 21594	69.545292	-14.303583	HIP 23972

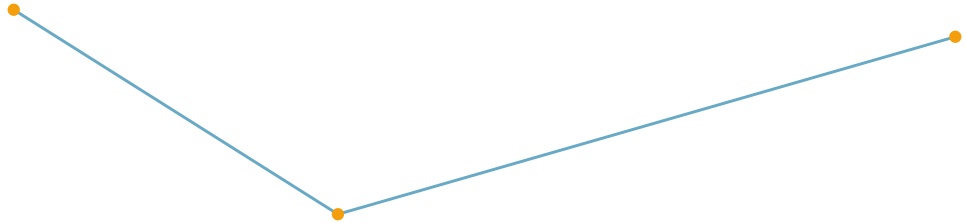
Effective release edges: 27 • unique plotted points: 28 • source: release_effective_edges.csv

37/88 Fornax (For) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

37/88 Fornax (For) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
Dalim (Alpha Fornacis)	48.018875	-28.987611	Beta Fornacis
Beta Fornacis	42.272583	-32.405889	Dalim (Alpha Fornacis), Nu Fornacis
Nu Fornacis	31.122667	-29.296806	Beta Fornacis

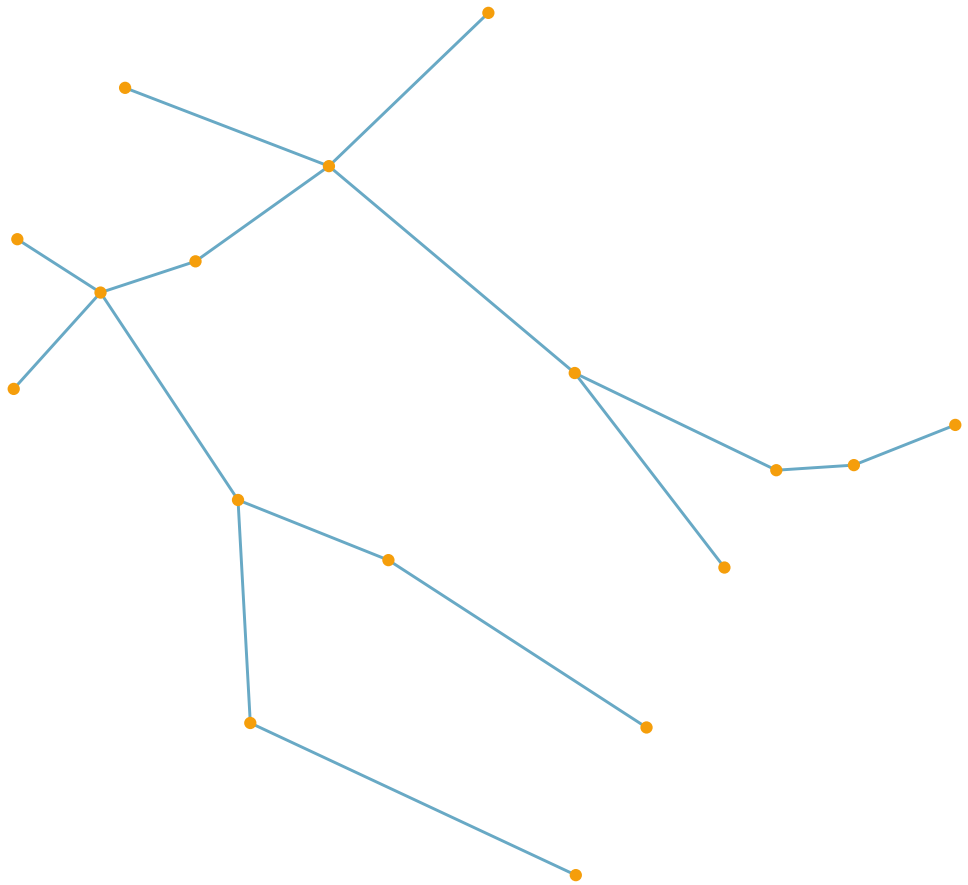
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

38/88 Gemini (Gem) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

38/88 Gemini (Gem) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 31681	99.427917	+16.399417	HIP 34088
HIP 34088	106.027250	+20.570306	HIP 31681, HIP 35550
HIP 35550	110.030792	+21.982333	HIP 34088, HIP 35350, HIP 36962
HIP 35350	109.523375	+16.540472	HIP 35550, HIP 32362
HIP 32362	101.322625	+12.896056	HIP 35350
HIP 36962	113.980708	+26.896000	HIP 35550, HIP 37740, HIP 37826, HIP 36046
HIP 37740	116.111958	+24.398139	HIP 36962
HIP 37826	116.330667	+28.026306	HIP 36962
HIP 36046	111.432000	+27.798278	HIP 36962, HIP 34693
HIP 34693	107.784958	+30.245278	HIP 36046, HIP 36850, HIP 33018, HIP 32246
HIP 36850	113.650000	+31.888639	HIP 34693
HIP 33018	103.197250	+33.961361	HIP 34693
HIP 32246	100.983042	+25.131167	HIP 34693, HIP 30883, HIP 30343
HIP 30883	97.240792	+20.212167	HIP 32246
HIP 30343	95.739958	+22.513861	HIP 32246, HIP 29655
HIP 29655	93.719583	+22.506833	HIP 30343, HIP 28734
HIP 28734	91.030083	+23.263639	HIP 29655

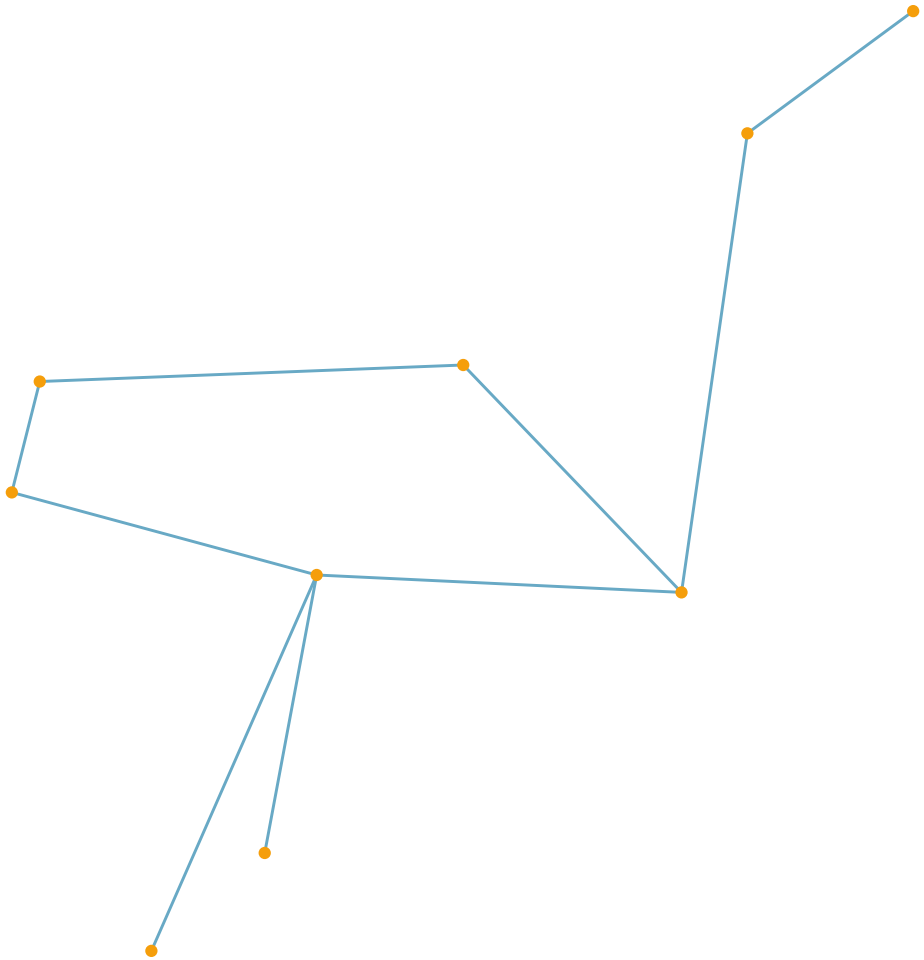
Effective release edges: 16 • unique plotted points: 17 • source: release_effective_edges.csv

39/88 Grus (Gru) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

39/88 Grus (Gru) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 114131	346.719875	-43.520333	HIP 110997, HIP 114421
HIP 110997	337.317292	-43.495556	HIP 114131, HIP 109268
HIP 109268	332.057833	-46.960611	HIP 110997, HIP 112122, HIP 109111
HIP 112122	340.666375	-46.884556	HIP 109268, HIP 114421, HIP 113638, HIP 112623
HIP 114421	347.589292	-45.246639	HIP 112122, HIP 114131
HIP 113638	345.220292	-52.754111	HIP 112122
HIP 112623	342.138333	-51.316694	HIP 112122
HIP 109111	331.528750	-39.543056	HIP 109268, HIP 108085
HIP 108085	328.481875	-37.364833	HIP 109111

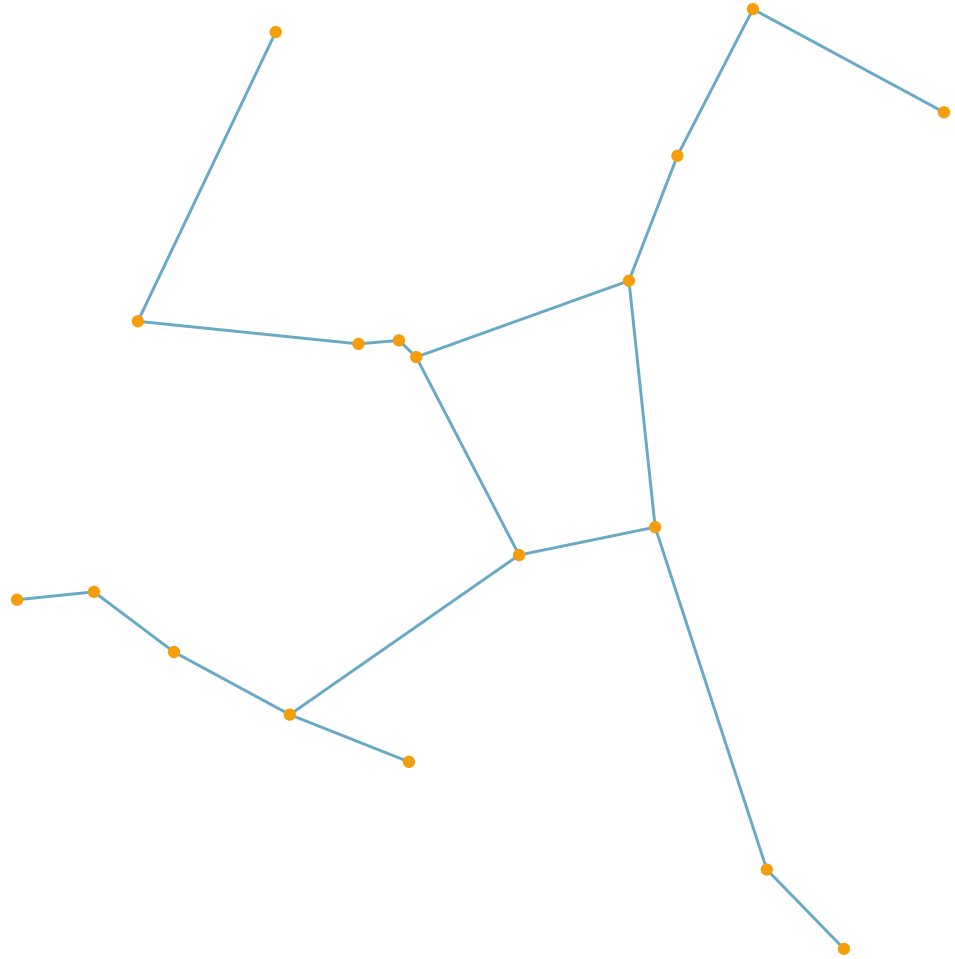
Effective release edges: 9 • unique plotted points: 9 • source: release_effective_edges.csv

40/88 Hercules (Her) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

40/88 Hercules (Her) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 86414	264.866208	+46.006333	HIP 87808
HIP 87808	269.063250	+37.250528	HIP 86414, HIP 85112
HIP 85112	260.920708	+37.145917	HIP 87808, HIP 84606
HIP 84606	259.417875	+37.291333	HIP 85112, HIP 84380
HIP 84380	258.761875	+36.809167	HIP 84606, HIP 81833, HIP 83207
HIP 81833	250.723917	+38.922472	HIP 84380, HIP 81126, HIP 81693
HIP 81126	248.525792	+42.436889	HIP 81833, HIP 79992
HIP 79992	244.935208	+46.313278	HIP 81126, HIP 77760
HIP 77760	238.167458	+42.450000	HIP 79992
HIP 81693	250.322833	+31.601889	HIP 81833, HIP 80816, HIP 83207
HIP 80816	247.555250	+21.489639	HIP 81693, HIP 80170
HIP 80170	245.480167	+19.153028	HIP 80816
HIP 83207	255.072542	+30.926333	HIP 81693, HIP 85693, HIP 84380
HIP 85693	262.684583	+26.110611	HIP 83207, HIP 84379, HIP 86974
HIP 84379	258.758000	+24.839583	HIP 85693
HIP 86974	266.615500	+27.722500	HIP 87933, HIP 85693
HIP 87933	269.440958	+29.247917	HIP 86974, HIP 88794
HIP 88794	271.885625	+28.762472	HIP 87933

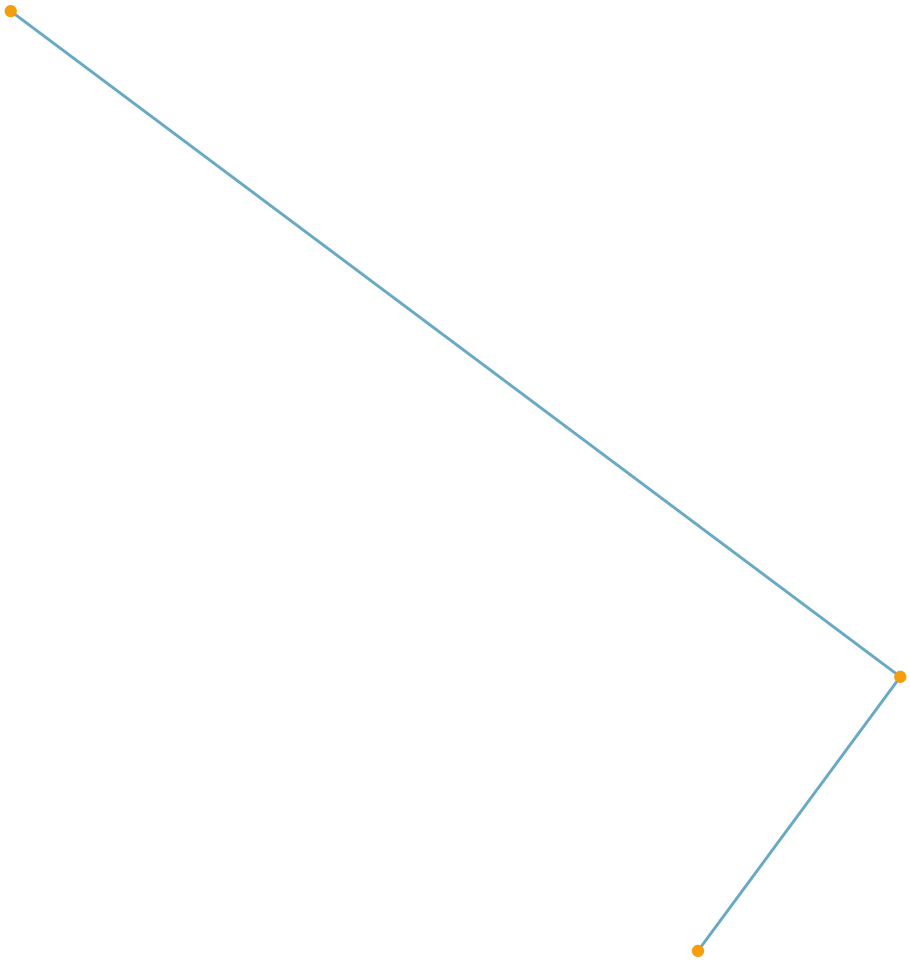
Effective release edges: 18 • unique plotted points: 18 • source: release_effective_edges.csv

41/88 Horologium (Hor) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

41/88 Horologium (Hor) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 19747	63.500333	-42.293861	HIP 12484
HIP 12484	40.164917	-54.549917	HIP 19747, HIP 14240
HIP 14240	45.903750	-59.737611	HIP 12484

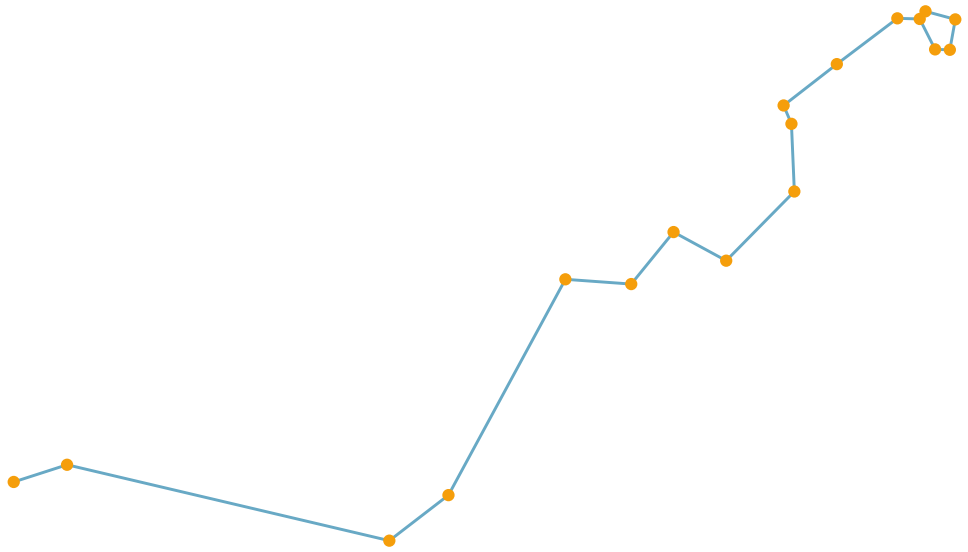
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

42/88 Hydra (Hya) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

42/88 Hydra (Hya) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 42799	130.806208	+3.398667	HIP 42402, HIP 43234
HIP 42402	129.689375	+3.341472	HIP 42799, HIP 42313
HIP 42313	129.414208	+5.703806	HIP 42402, HIP 43109
HIP 43109	131.694375	+6.418917	HIP 42313, HIP 43234
HIP 43234	132.108250	+5.837889	HIP 43109, HIP 42799, HIP 43813
HIP 43813	133.848667	+5.945528	HIP 43234, HIP 45336
HIP 45336	138.590792	+2.315028	HIP 43813, HIP 46776
HIP 46776	142.995542	-1.184667	HIP 45336, HIP 46509
HIP 46509	142.286833	-2.768944	HIP 46776, HIP 46390
HIP 46390	141.896875	-8.658694	HIP 46509, HIP 48356
HIP 48356	147.869500	-14.846556	HIP 46390, HIP 49841
HIP 49841	152.647500	-12.353833	HIP 48356, HIP 51069
HIP 51069	156.522958	-16.836083	HIP 49841, HIP 52943
HIP 52943	162.405958	-16.194139	HIP 51069, HIP 56343
HIP 56343	173.251083	-31.857528	HIP 52943, HIP 57936
HIP 57936	178.227333	-33.908139	HIP 56343, HIP 64166
HIP 64166	197.263667	-23.117972	HIP 57936, HIP 64962
HIP 64962	199.730208	-23.171417	HIP 64166

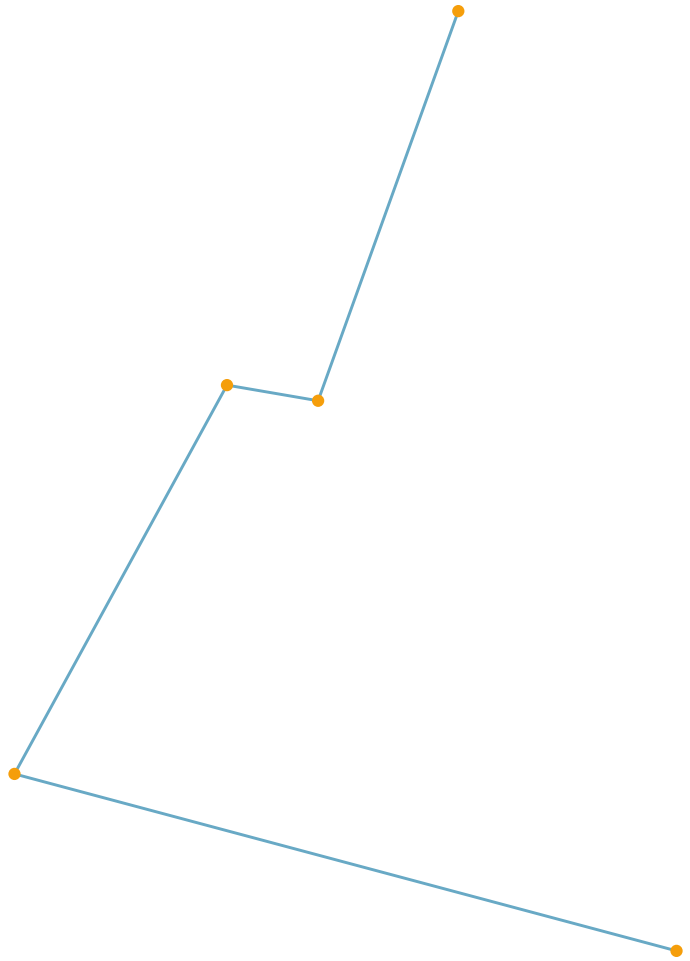
Effective release edges: 18 • unique plotted points: 18 • source: release_effective_edges.csv

43/88 Hydrus (Hyi) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

43/88 Hydrus (Hyi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 2021	6.413333	-77.255028	HIP 17678
HIP 17678	56.809292	-74.239250	HIP 2021, HIP 12394
HIP 12394	39.896750	-68.266944	HIP 17678, HIP 11001
HIP 11001	35.437583	-68.659417	HIP 12394, HIP 9236
HIP 9236	29.691125	-61.569917	HIP 11001

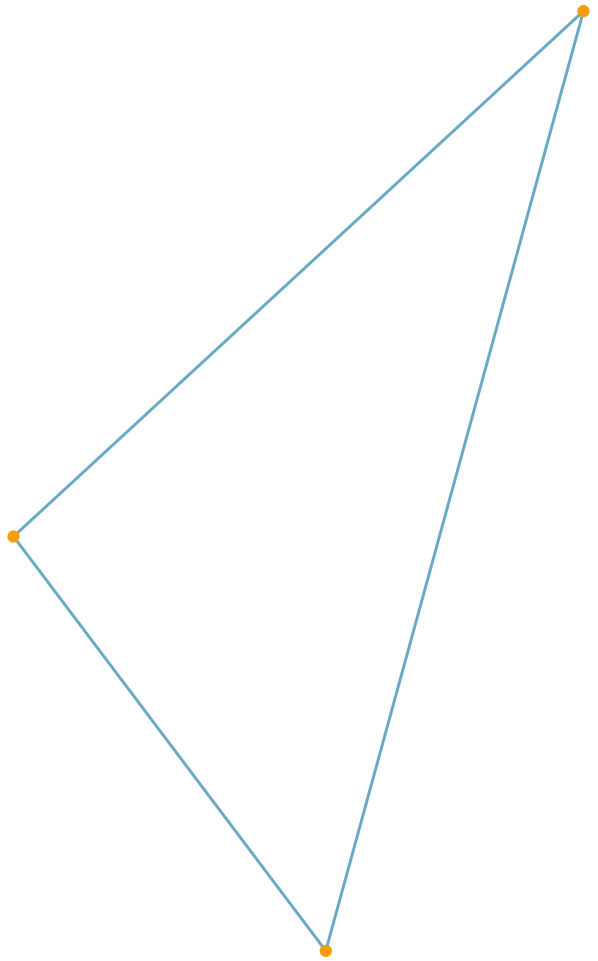
Effective release edges: 4 • unique plotted points: 5 • source: release_effective_edges.csv

44/88 Indus (Ind) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

44/88 Indus (Ind) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 105319	319.966167	-53.449278	HIP 101772, HIP 103227
HIP 101772	309.391625	-47.291667	HIP 105319, HIP 103227
HIP 103227	313.702417	-58.454083	HIP 101772, HIP 105319

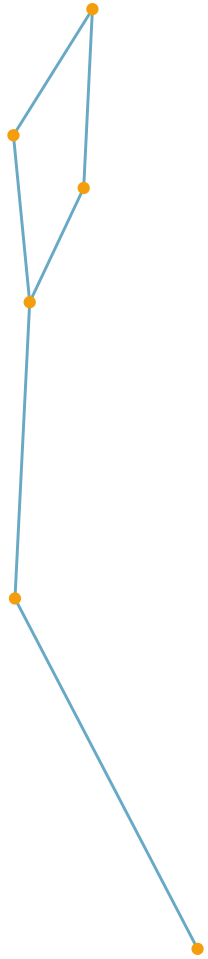
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

45/88 Lacerta (Lac) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

45/88 Lacerta (Lac) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 109937	333.992375	+37.748722	HIP 111104
HIP 111104	337.621917	+43.123389	HIP 109937, HIP 111022
HIP 111022	337.382583	+47.706889	HIP 111104, HIP 110609, HIP 111169
HIP 110609	336.129167	+49.476389	HIP 111022, HIP 110538
HIP 110538	335.890167	+52.229500	HIP 110609, HIP 111169
HIP 111169	337.822417	+50.282444	HIP 110538, HIP 111022

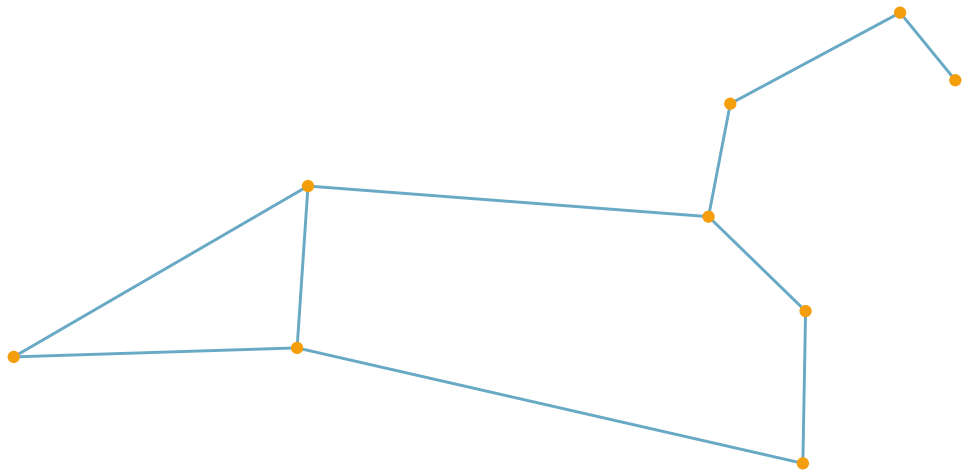
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

46/88 Leo (Leo) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

46/88 Leo (Leo) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 57632	177.266167	+14.572333	HIP 54879, HIP 54872
HIP 54879	168.560167	+15.429750	HIP 57632, HIP 49669, HIP 54872
HIP 49669	152.093583	+11.967194	HIP 54879, HIP 49583
HIP 49583	151.833125	+16.762667	HIP 49669, HIP 50583
HIP 50583	154.992333	+19.841861	HIP 49583, HIP 54872, HIP 50335
HIP 54872	168.526708	+20.524028	HIP 50583, HIP 57632, HIP 54879
HIP 50335	154.172500	+23.417333	HIP 50583, HIP 48455
HIP 48455	148.191500	+26.007083	HIP 50335, HIP 47908
HIP 47908	146.462917	+23.774278	HIP 48455

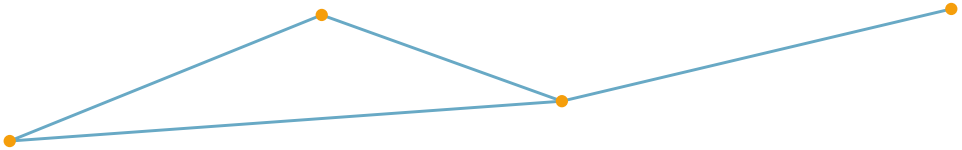
Effective release edges: 10 • unique plotted points: 9 • source: release_effective_edges.csv

47/88 Leo Minor (LMi) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

47/88 Leo Minor (LMi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 53229	163.327667	+34.215556	HIP 51233, HIP 49593
HIP 51233	156.971208	+36.707472	HIP 53229, HIP 49593
HIP 49593	151.857208	+35.244694	HIP 51233, HIP 46952, HIP 53229
HIP 46952	143.555750	+36.397611	HIP 49593

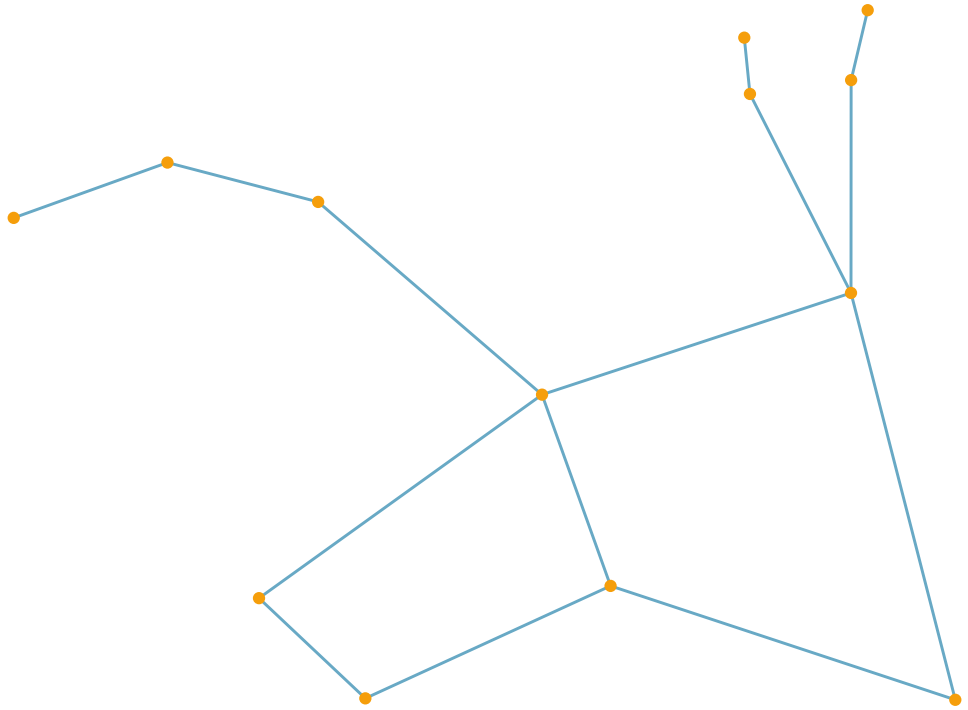
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

48/88 Lepus (Lep) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

48/88 Lepus (Lep) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 28910	91.538875	-14.935278	HIP 28103
HIP 28103	89.101333	-14.168028	HIP 28910, HIP 27288
HIP 27288	86.738958	-14.821944	HIP 28103, HIP 25985
HIP 25985	83.182542	-17.822306	HIP 27288, HIP 24305, HIP 27654, HIP 25606
HIP 24305	78.232792	-16.205417	HIP 25985, HIP 24845, HIP 24327, HIP 23685
HIP 27654	87.829792	-20.877500	HIP 25985, HIP 27072
HIP 27072	86.116542	-22.447500	HIP 27654, HIP 25606
HIP 25606	82.061375	-20.759222	HIP 27072, HIP 23685, HIP 25985
HIP 23685	76.365208	-22.370861	HIP 25606, HIP 24305
HIP 24845	79.893875	-13.176778	HIP 24305, HIP 24873
HIP 24327	78.307875	-12.941278	HIP 24305, HIP 24244
HIP 24244	78.074542	-11.869139	HIP 24327
HIP 24873	79.995958	-12.315611	HIP 24845

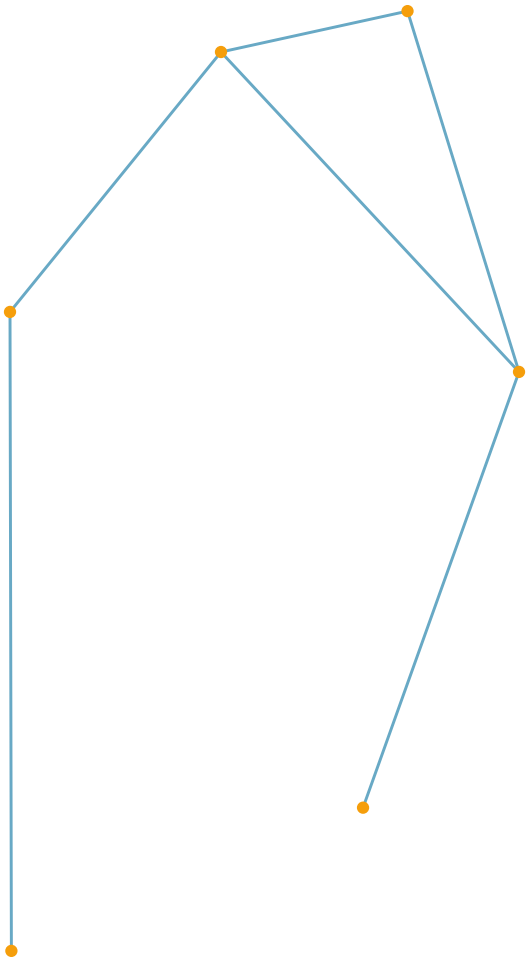
Effective release edges: 14 • unique plotted points: 13 • source: release_effective_edges.csv

49/88 Libra (Lib) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

49/88 Libra (Lib) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
Delta Librae	225.296700	-8.519700	Beta Librae, Alpha Librae
Alpha Librae	222.719600	-16.041800	Delta Librae, Beta Librae, Sigma Librae
Sigma Librae	226.017600	-25.281900	Alpha Librae
Gamma Librae	233.881600	-14.789500	Beta Librae, Upsilon Librae
Upsilon Librae	234.256100	-28.135100	Gamma Librae
Beta Librae	229.251700	-9.382900	Delta Librae, Alpha Librae, Gamma Librae

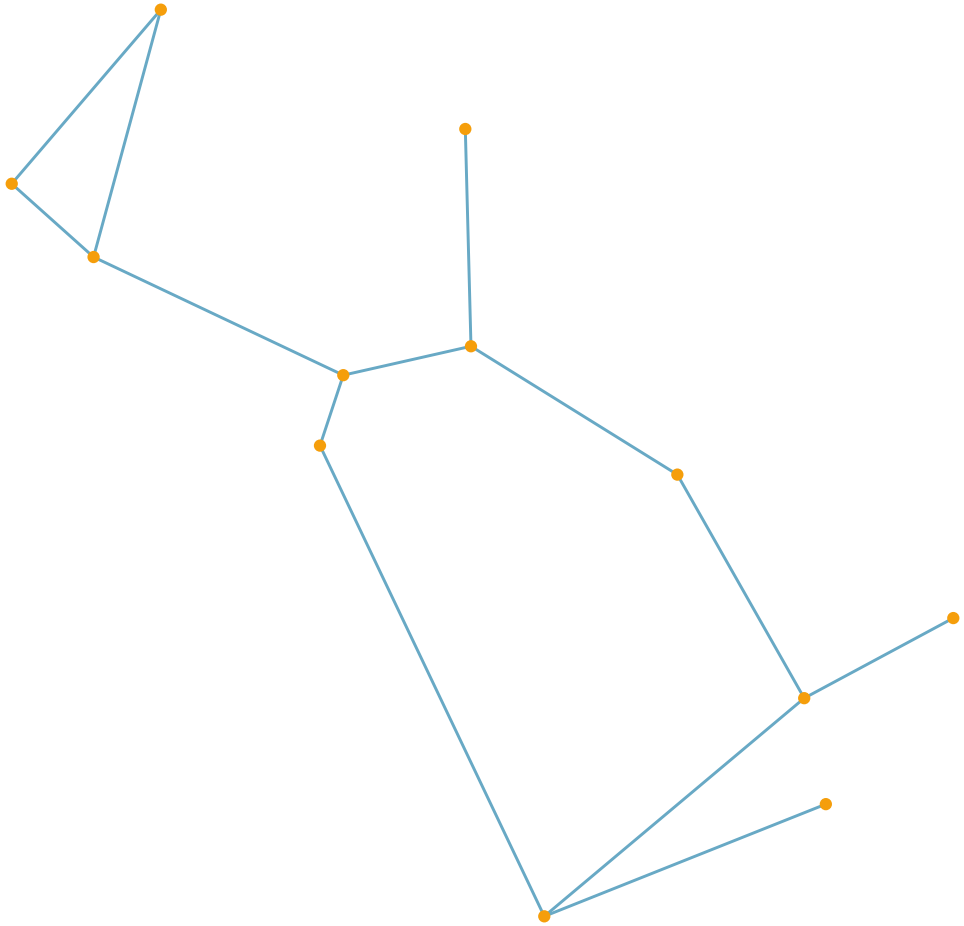
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

50/88 Lupus (Lup) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

50/88 Lupus (Lup) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 77634	237.739750	-33.627111	HIP 78970, HIP 78384
HIP 78970	241.817375	-36.755528	HIP 77634, HIP 78384
HIP 78384	240.030583	-38.396639	HIP 78970, HIP 77634, HIP 76297
HIP 76297	233.785250	-41.166694	HIP 78384, HIP 75141, HIP 76552
HIP 75141	230.343083	-40.647472	HIP 76297, HIP 75177, HIP 73273
HIP 75177	230.451833	-36.261167	HIP 75141
HIP 73273	224.633125	-43.133861	HIP 75141, HIP 71860
HIP 76552	234.513833	-42.567472	HIP 76297, HIP 74395
HIP 74395	228.071667	-52.099083	HIP 76552, HIP 71860, HIP 71536
HIP 71860	220.482375	-47.388139	HIP 74395, HIP 70576, HIP 73273
HIP 71536	219.471875	-49.425750	HIP 74395
HIP 70576	216.545000	-45.379250	HIP 71860

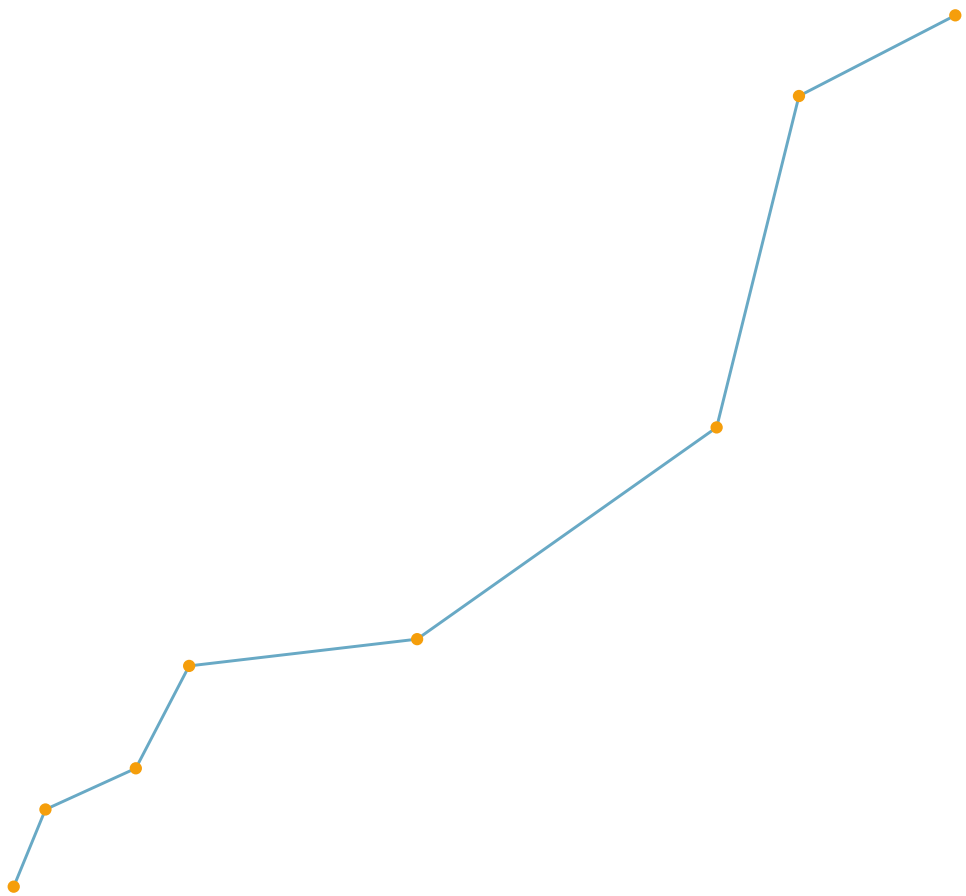
Effective release edges: 13 • unique plotted points: 12 • source: release_effective_edges.csv

51/88 Lynx (Lyn) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

51/88 Lynx (Lyn) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 45860	140.264417	+34.392528	HIP 45688
HIP 45688	139.711125	+36.802889	HIP 45860, HIP 44700
HIP 44700	136.632458	+38.452250	HIP 45688, HIP 44248
HIP 44248	135.161458	+41.783444	HIP 44700, HIP 41075
HIP 41075	125.708875	+43.188361	HIP 44248, HIP 36145
HIP 36145	111.678583	+49.211639	HIP 41075, HIP 33449
HIP 33449	104.319167	+58.423056	HIP 36145, HIP 30060
HIP 30060	94.905792	+59.010917	HIP 33449

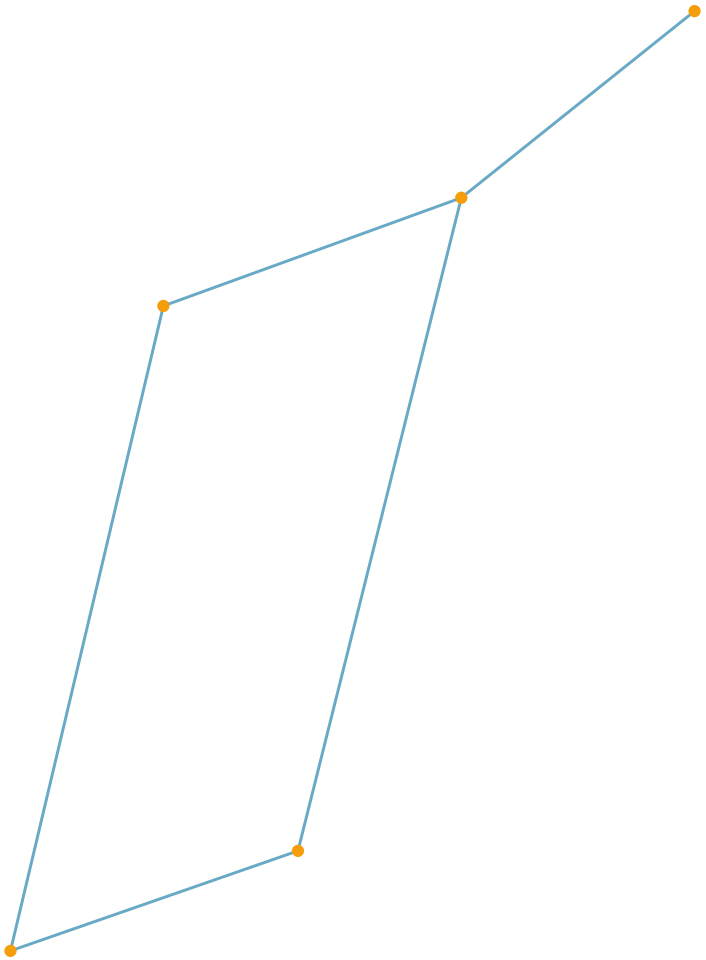
Effective release edges: 7 • unique plotted points: 8 • source: release_effective_edges.csv

52/88 Lyra (Lyr) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

52/88 Lyra (Lyr) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 91262	279.234125	+38.783000	HIP 91971
HIP 91971	281.193083	+37.605056	HIP 91262, HIP 92420, HIP 92791
HIP 92420	282.519958	+33.362667	HIP 91971, HIP 93194
HIP 93194	284.735917	+32.689556	HIP 92420, HIP 92791
HIP 92791	283.626208	+36.898611	HIP 93194, HIP 91971

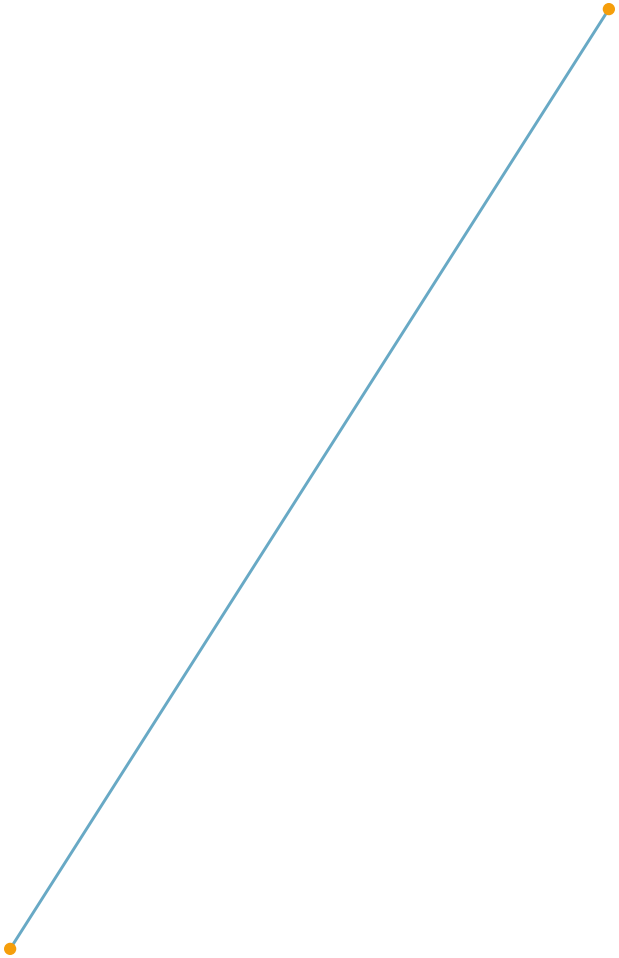
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

53/88 Mensa (Men) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

53/88 Mensa (Men) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 25918	82.969417	-76.341667	HIP 21949
HIP 21949	70.766458	-70.931111	HIP 25918

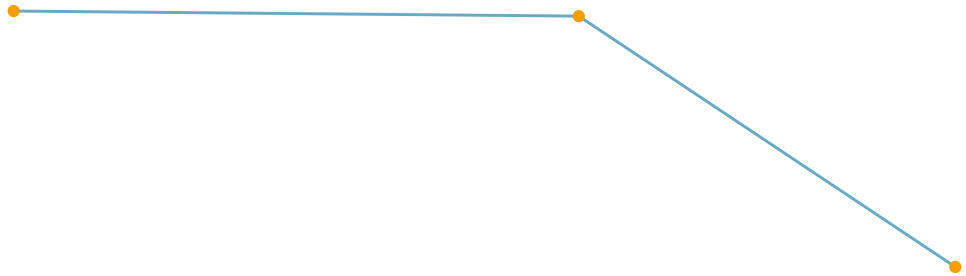
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

54/88 Microscopium (Mic) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

54/88 Microscopium (Mic) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 105140	319.484375	-32.172472	HIP 103738
HIP 103738	315.322750	-32.257778	HIP 105140, HIP 102831
HIP 102831	312.492000	-33.779667	HIP 103738

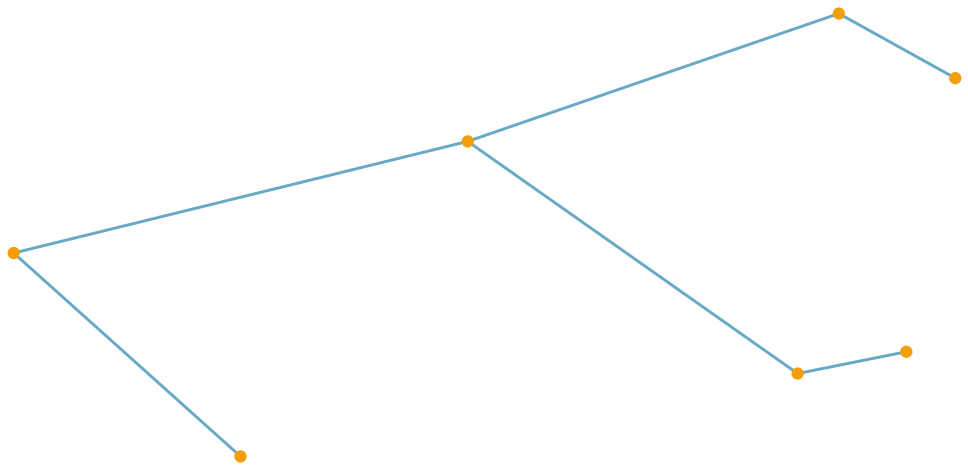
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

55/88 Monoceros (Mon) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

55/88 Monoceros (Mon) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 29651	93.713917	-6.274722	HIP 30867
HIP 30867	97.204458	-7.033056	HIP 29651, HIP 34769
HIP 34769	107.966083	+0.492778	HIP 30867, HIP 30419, HIP 39863
HIP 30419	95.942083	+4.592833	HIP 34769, HIP 29151
HIP 29151	92.241250	+2.499722	HIP 30419
HIP 39863	122.148583	-2.983778	HIP 34769, HIP 37447
HIP 37447	115.312000	-9.551083	HIP 39863

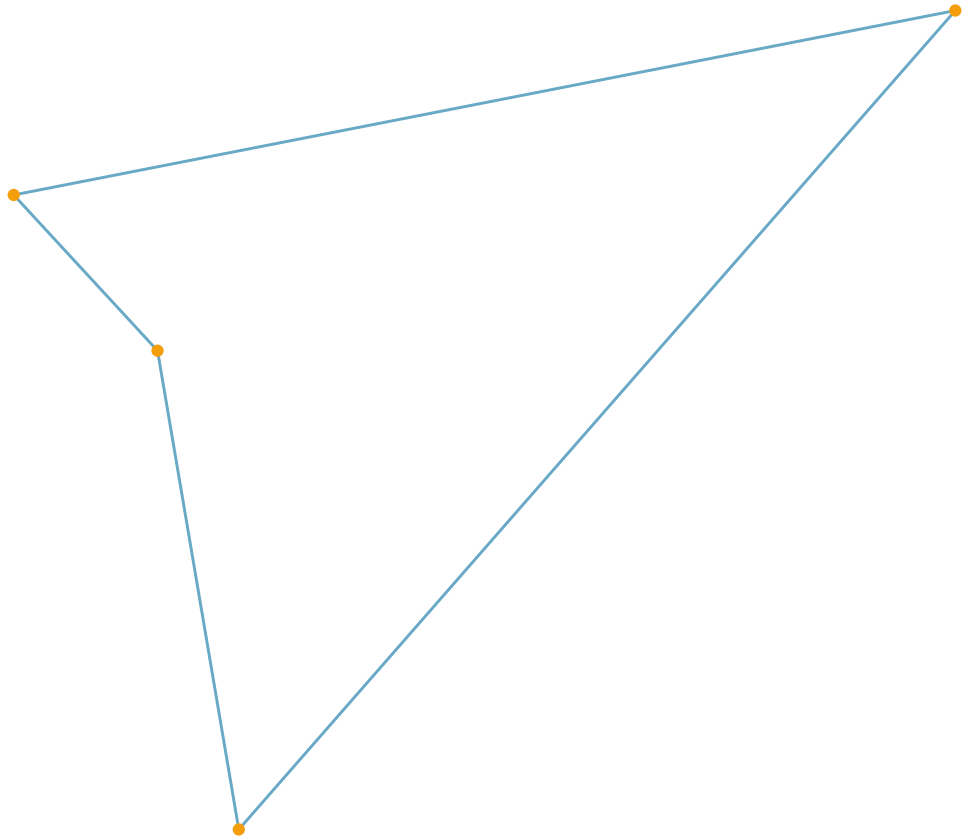
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

56/88 Musca (Mus) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

56/88 Musca (Mus) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 62322	191.570292	-68.108083	HIP 57363, HIP 61585
HIP 57363	176.402375	-66.728833	HIP 62322, HIP 61199
HIP 61199	188.117125	-72.132972	HIP 57363, HIP 61585
HIP 61585	189.296167	-69.135528	HIP 61199, HIP 62322

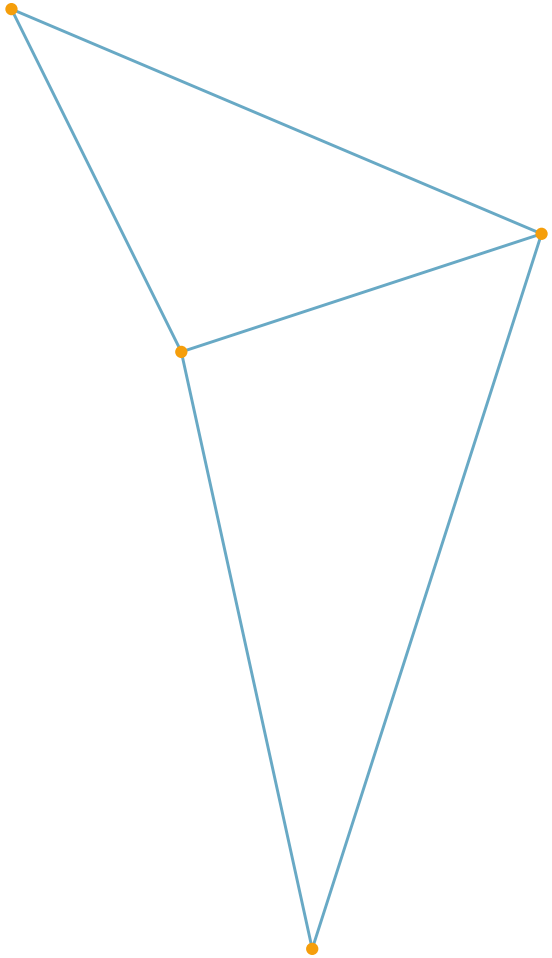
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

57/88 Norma (Nor) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

57/88 Norma (Nor) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 79509	243.369708	-54.630417	HIP 80000, HIP 78639
HIP 80000	244.960708	-50.155389	HIP 79509, HIP 80582, HIP 78639
HIP 80582	246.796042	-47.554722	HIP 80000, HIP 78639
HIP 78639	240.803583	-49.229722	HIP 80582, HIP 80000, HIP 79509

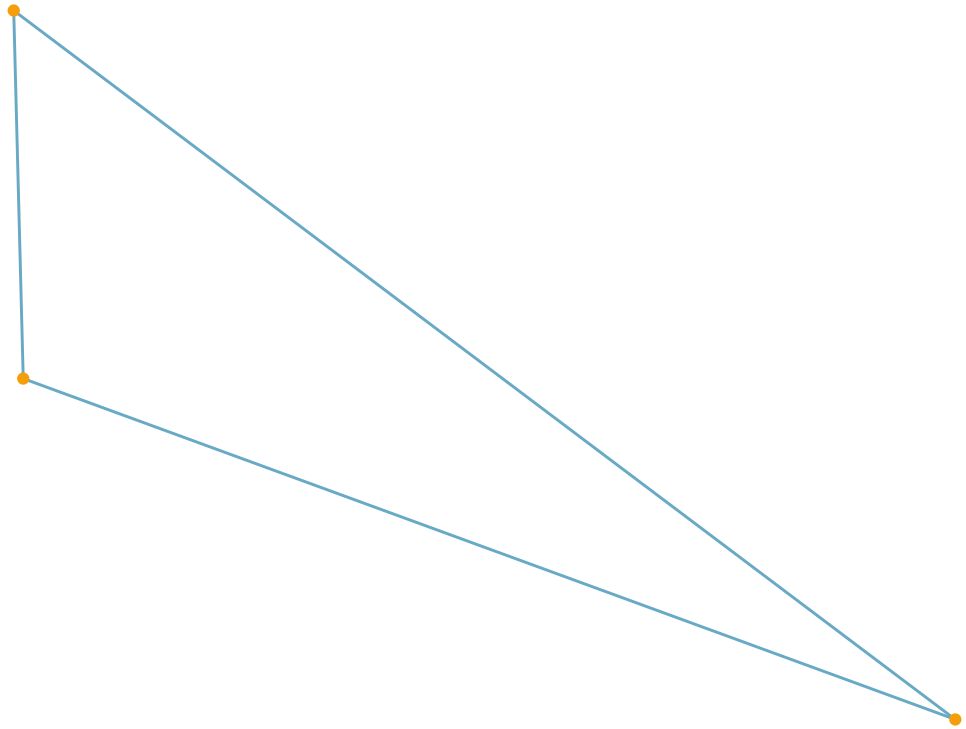
Effective release edges: 5 • unique plotted points: 4 • source: release_effective_edges.csv

58/88 Octans (Oct) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

58/88 Octans (Oct) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 107089	325.368625	-77.389472	HIP 112405, HIP 70638
HIP 112405	341.515500	-81.381611	HIP 107089, HIP 70638
HIP 70638	216.732250	-83.667861	HIP 112405, HIP 107089

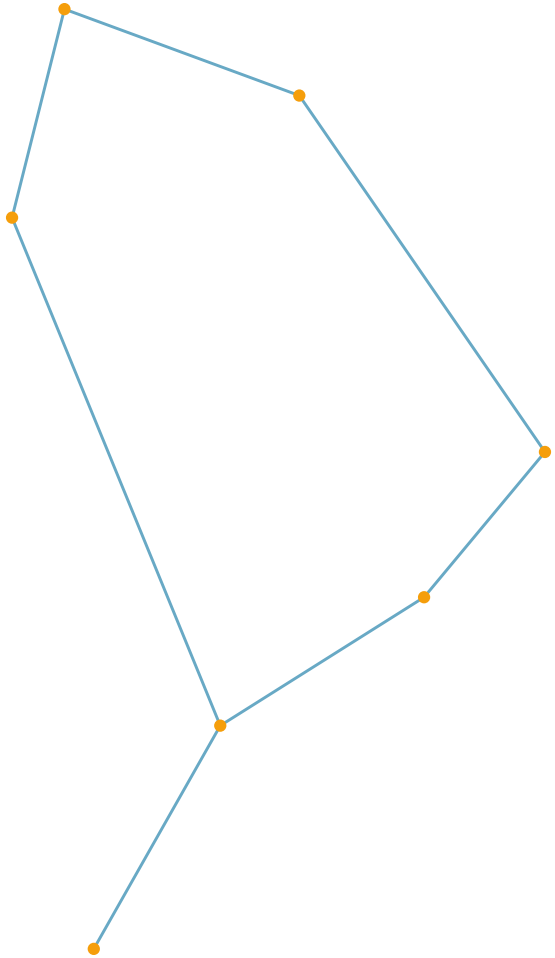
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

59/88 Ophiuchus (Oph) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

59/88 Ophiuchus (Oph) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 86032	263.733333	+12.560583	HIP 86742, HIP 83000
HIP 86742	265.868250	+4.566917	HIP 86032, HIP 84012
HIP 84012	257.594417	-15.725139	HIP 86742, HIP 81377, HIP 85755
HIP 83000	254.417792	+9.375056	HIP 86032, HIP 79882
HIP 79882	244.580167	-4.692611	HIP 83000, HIP 81377
HIP 81377	249.289708	-10.567139	HIP 79882, HIP 84012
HIP 85755	262.853958	-23.962583	HIP 84012

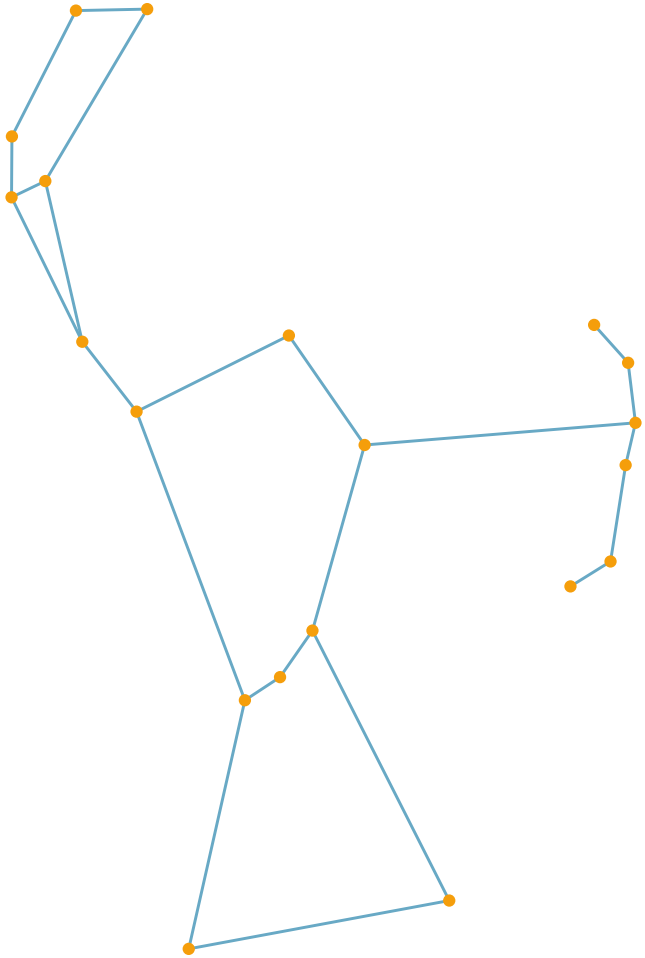
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

60/88 Orion (Ori) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

60/88 Orion (Ori) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 26727	85.189667	-1.942583	HIP 26311, HIP 27989, HIP 27366
HIP 26311	84.053375	-1.201917	HIP 26727, HIP 25930
HIP 25930	83.001667	+0.299083	HIP 26311, HIP 24436, HIP 25336
HIP 29434	93.013667	+16.130444	HIP 29426, HIP 28716
HIP 29426	92.985000	+14.208806	HIP 29434, HIP 29038, HIP 28614
HIP 28716	90.979917	+20.138472	HIP 29434, HIP 27913
HIP 27913	88.596167	+20.276417	HIP 28716, HIP 29038
HIP 29038	91.893000	+14.768528	HIP 29426, HIP 27913, HIP 28614
HIP 28614	90.595792	+9.647361	HIP 29426, HIP 27989, HIP 29038
HIP 27989	88.792875	+7.407028	HIP 28614, HIP 26727, HIP 26207
HIP 27366	86.939125	-9.669611	HIP 26727, HIP 24436
HIP 24436	78.634458	-8.201639	HIP 27366, HIP 25930
HIP 25336	81.282792	+6.349722	HIP 25930, HIP 26207, HIP 22449
HIP 26207	83.784500	+9.934167	HIP 25336, HIP 27989
HIP 22449	72.458917	+6.961250	HIP 25336, HIP 22549, HIP 22509
HIP 22549	72.801542	+5.605111	HIP 22449, HIP 22730
HIP 22730	73.344833	+2.508278	HIP 22549, HIP 23123
HIP 23123	74.637083	+1.714028	HIP 22730
HIP 22509	72.653000	+8.900250	HIP 22449, HIP 22845
HIP 22845	73.723750	+10.151139	HIP 22509

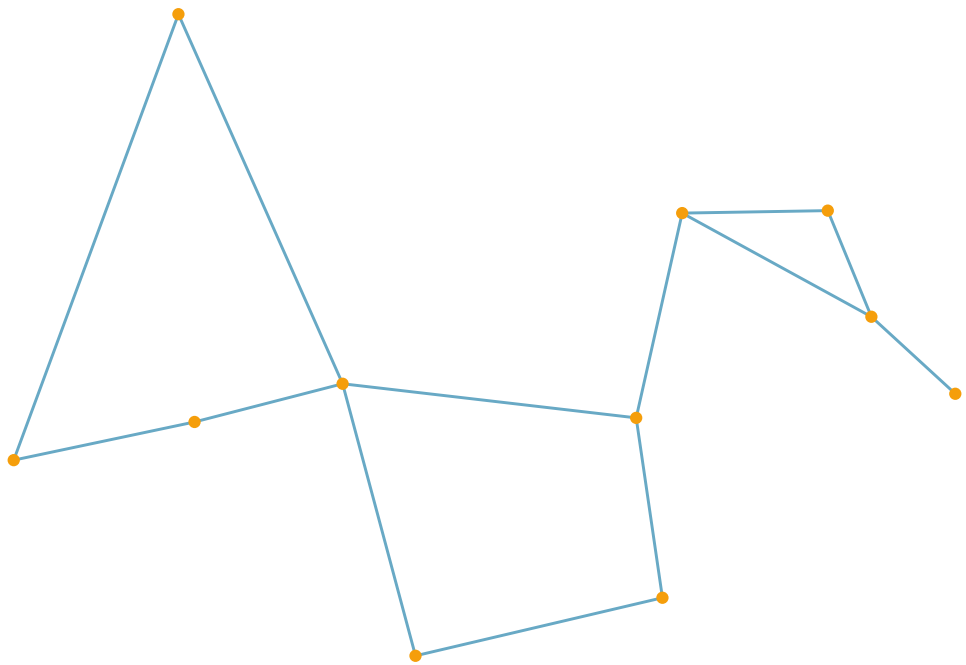
Effective release edges: 23 • unique plotted points: 20 • source: release_effective_edges.csv

61/88 Pavo (Pav) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

61/88 Pavo (Pav) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 100751	306.411875	-56.734889	HIP 105858, HIP 99240
HIP 105858	321.610375	-65.368139	HIP 100751, HIP 102395
HIP 102395	311.239833	-66.203250	HIP 105858, HIP 99240
HIP 99240	302.174417	-66.179333	HIP 102395, HIP 100751, HIP 98495, HIP 93015
HIP 98495	300.147458	-72.910194	HIP 99240, HIP 91792
HIP 91792	280.758875	-71.427722	HIP 98495, HIP 93015
HIP 93015	284.237667	-67.233528	HIP 91792, HIP 99240, HIP 92609
HIP 92609	283.054333	-62.187556	HIP 93015, HIP 90098, HIP 88866
HIP 90098	275.806750	-61.493917	HIP 92609, HIP 88866
HIP 88866	272.144958	-63.668056	HIP 90098, HIP 92609, HIP 86929
HIP 86929	266.433333	-64.723722	HIP 88866

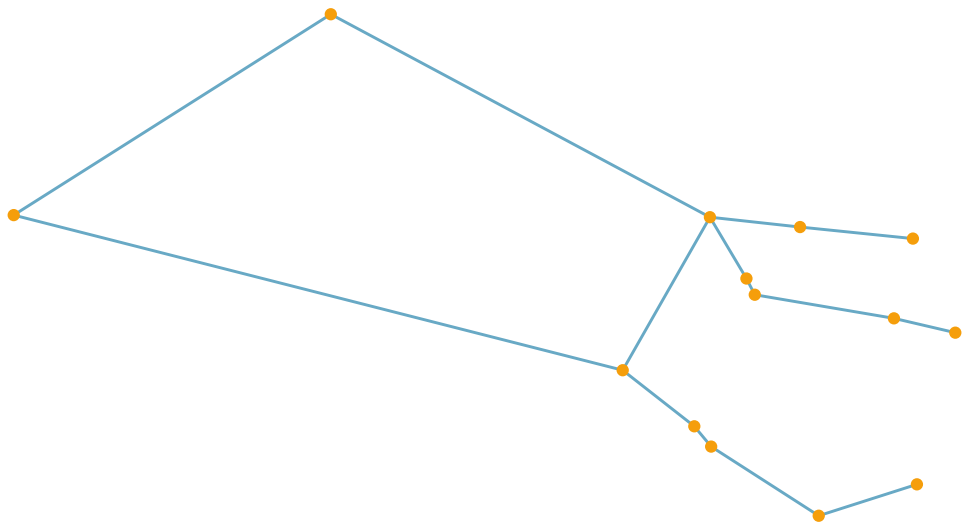
Effective release edges: 13 • unique plotted points: 11 • source: release_effective_edges.csv

62/88 Pegasus (Peg) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

62/88 Pegasus (Peg) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 1067	3.308958	+15.183611	HIP 113963, HIP 677
HIP 113963	346.190083	+15.205361	HIP 1067, HIP 112447, HIP 113881
HIP 113881	345.943042	+28.082444	HIP 112158, HIP 112748, HIP 677, HIP 113963
HIP 112158	340.750542	+30.221306	HIP 113881, HIP 109352
HIP 109352	332.307000	+33.172500	HIP 112158
HIP 112748	342.500417	+24.601694	HIP 113881, HIP 112440
HIP 112440	341.632667	+23.565667	HIP 112748, HIP 109176
HIP 109176	331.751958	+25.345056	HIP 112440, HIP 107354
HIP 107354	326.161250	+25.645000	HIP 109176
HIP 112447	341.672667	+12.174083	HIP 113963, HIP 112029
HIP 112029	340.365292	+10.831389	HIP 112447, HIP 109427
HIP 109427	332.549250	+6.197778	HIP 112029, HIP 107315
HIP 107315	326.046417	+9.875000	HIP 109427
HIP 677	2.096542	+29.090833	HIP 113881, HIP 1067

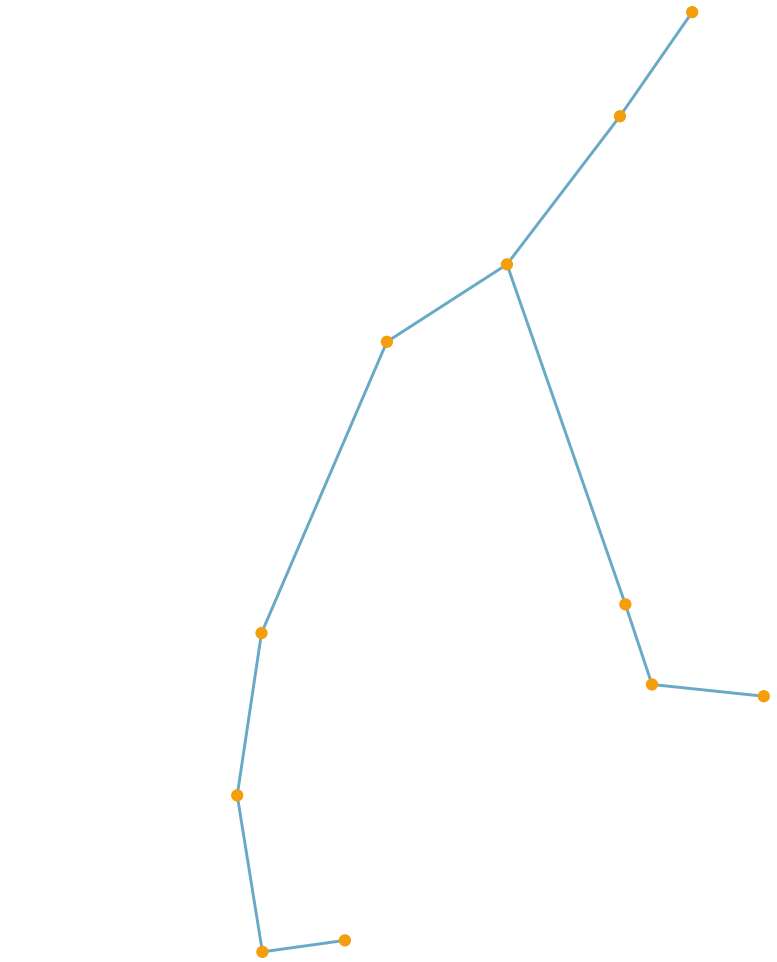
Effective release edges: 14 • unique plotted points: 14 • source: release_effective_edges.csv

63/88 Perseus (Per) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

63/88 Perseus (Per) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 17448	56.079708	+32.288278	HIP 18246
HIP 18246	58.533000	+31.883667	HIP 17448, HIP 18614
HIP 18614	59.741250	+35.791028	HIP 18246, HIP 18532
HIP 18532	59.463417	+40.010278	HIP 18614, HIP 17358
HIP 17358	55.731167	+47.787667	HIP 18532, HIP 15863
HIP 15863	51.080625	+49.861250	HIP 17358, HIP 14328, HIP 14576
HIP 14328	46.199125	+53.506444	HIP 15863, HIP 13268
HIP 13268	42.674125	+55.895528	HIP 14328
HIP 14576	47.042208	+40.955639	HIP 15863, HIP 14354
HIP 14354	46.293750	+38.840528	HIP 14576, HIP 13254
HIP 13254	42.645458	+38.318917	HIP 14354

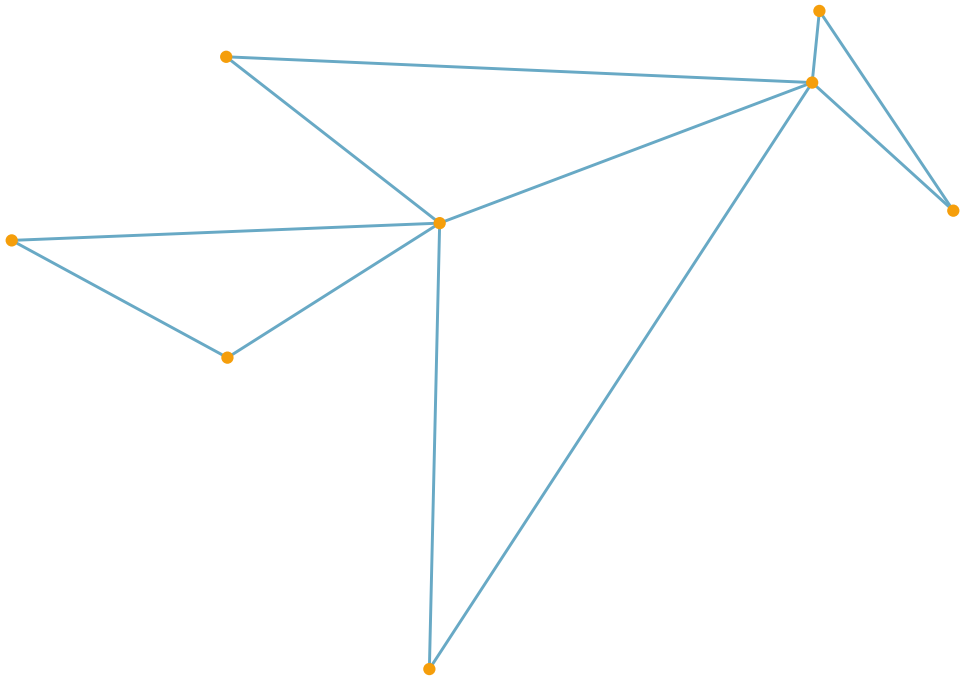
Effective release edges: 10 • unique plotted points: 11 • source: release_effective_edges.csv

64/88 Phoenix (Phe) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

64/88 Phoenix (Phe) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 5348	17.096083	-55.245833	HIP 5165, HIP 2072
HIP 5165	16.521292	-46.718500	HIP 5348, HIP 2072, HIP 7083, HIP 8837, HIP 6867
HIP 2072	6.550500	-43.679917	HIP 5165, HIP 5348, HIP 6867, HIP 2081, HIP 765
HIP 7083	22.812417	-49.073083	HIP 5165, HIP 8837
HIP 8837	28.411750	-46.302444	HIP 7083, HIP 5165
HIP 6867	22.091417	-43.317722	HIP 5165, HIP 2072
HIP 2081	6.570292	-42.305111	HIP 2072, HIP 765
HIP 765	2.352250	-45.747000	HIP 2081, HIP 2072

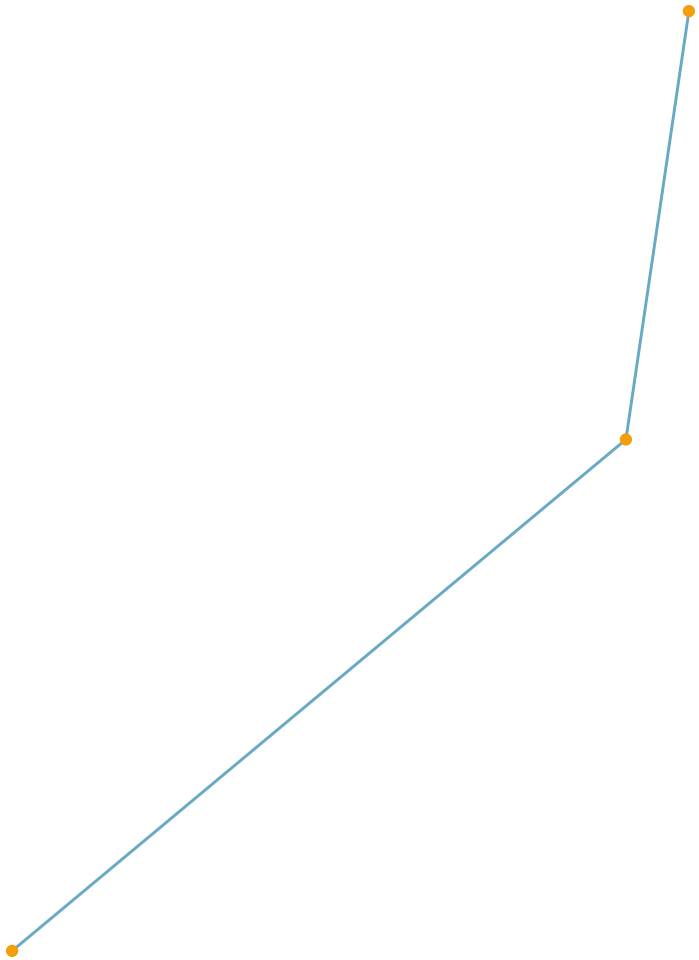
Effective release edges: 11 • unique plotted points: 8 • source: release_effective_edges.csv

65/88 Pictor (Pic) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

65/88 Pictor (Pic) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 32607	102.048083	-61.941972	HIP 27530
HIP 27530	87.456583	-56.166500	HIP 32607, HIP 27321
HIP 27321	86.821167	-51.066722	HIP 27530

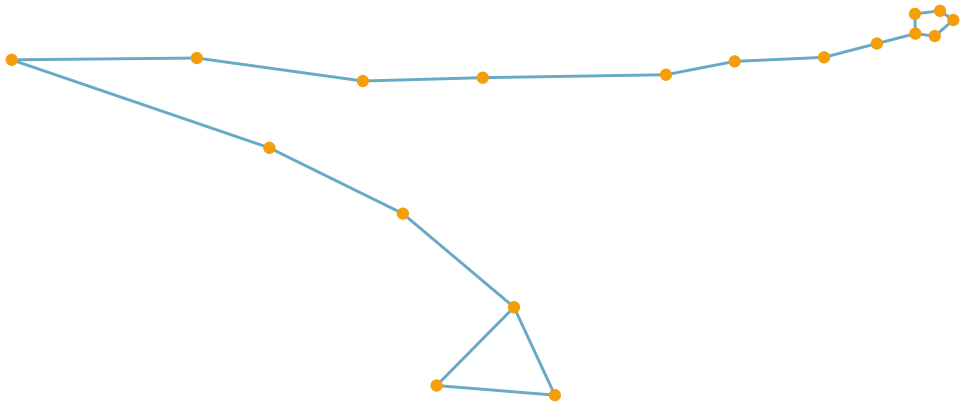
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

66/88 Pisces (Psc) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

66/88 Pisces (Psc) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 4889	15.704542	+31.804333	HIP 5742, HIP 6193
HIP 5742	18.437250	+24.583778	HIP 4889, HIP 6193, HIP 7097
HIP 6193	19.866583	+27.264083	HIP 4889, HIP 5742
HIP 7097	22.870792	+15.345833	HIP 5742, HIP 8198
HIP 8198	26.348292	+9.157639	HIP 7097, HIP 9487
HIP 9487	30.511667	+2.763750	HIP 8198, HIP 8833
HIP 8833	28.388917	+3.187472	HIP 9487, HIP 7884
HIP 7884	25.357958	+5.487611	HIP 8833, HIP 7007
HIP 7007	22.545583	+6.143944	HIP 7884, HIP 4906
HIP 4906	15.736083	+7.890083	HIP 7007, HIP 3760
HIP 3760	12.072250	+7.299917	HIP 4906, HIP 1645
HIP 1645	5.149417	+8.190250	HIP 3760, HIP 118268
HIP 118268	359.827500	+6.863583	HIP 1645, HIP 116771
HIP 116771	354.986750	+5.627361	HIP 118268, HIP 116928, HIP 115830
HIP 116928	355.512000	+1.780417	HIP 116771, HIP 115738
HIP 115738	351.732958	+1.255833	HIP 116928, HIP 114971
HIP 114971	349.289542	+3.282250	HIP 115738, HIP 115830
HIP 115830	351.992375	+6.379111	HIP 114971, HIP 116771

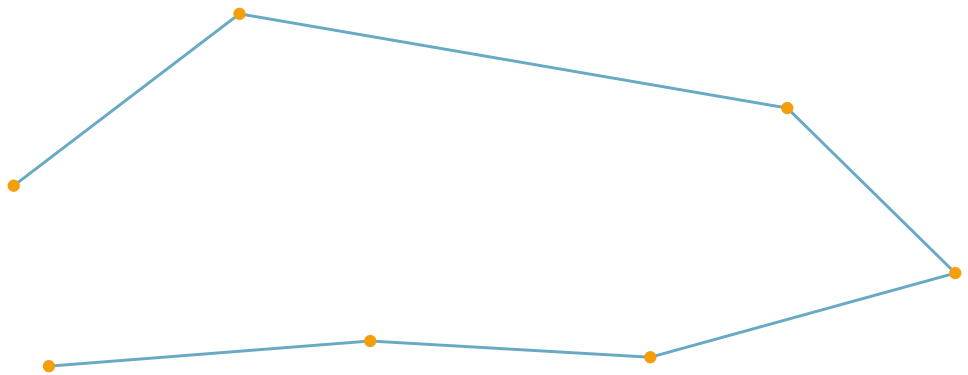
Effective release edges: 19 • unique plotted points: 18 • source: release_effective_edges.csv

67/88 Piscis Austrinus (PsA) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

67/88 Piscis Austrinus (PsA) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 113368	344.411792	-29.621833	HIP 111954
HIP 111954	340.163875	-27.043611	HIP 113368, HIP 108661
HIP 108661	330.209250	-28.453750	HIP 111954, HIP 107608
HIP 107608	326.934042	-30.898306	HIP 108661, HIP 109422
HIP 109422	332.535333	-32.548444	HIP 107608, HIP 111188
HIP 111188	337.876208	-32.346028	HIP 109422, HIP 113246
HIP 113246	343.987042	-32.539694	HIP 111188

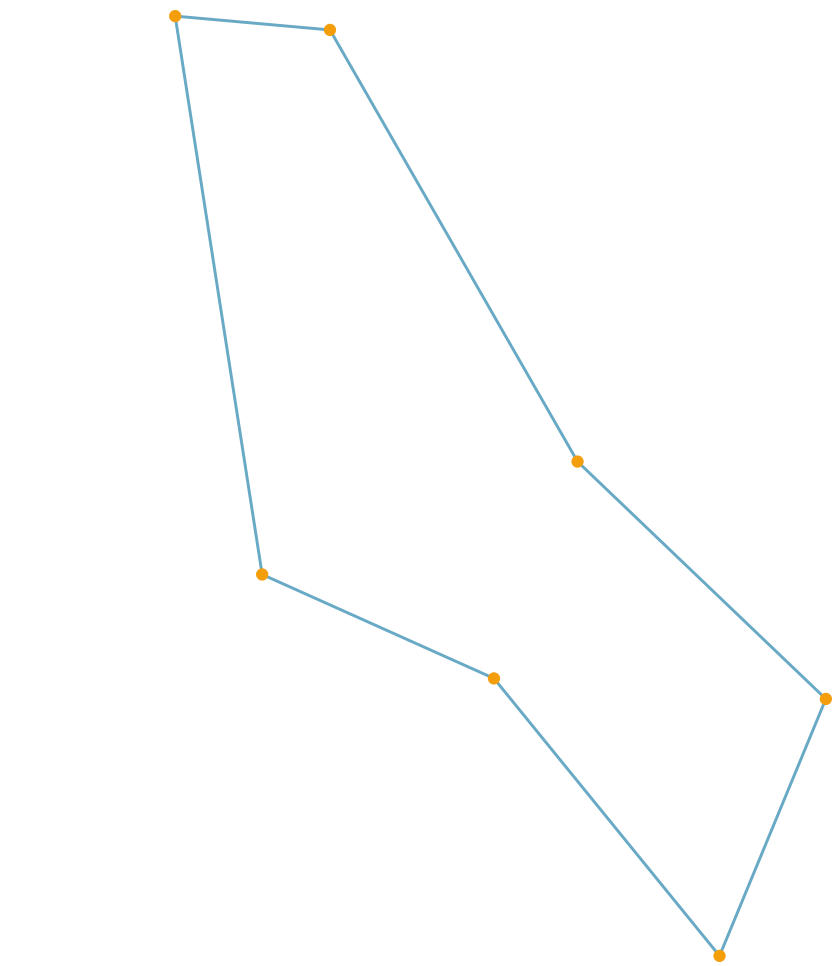
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

68/88 Puppis (Pup) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

68/88 Puppis (Pup) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 39757	121.886250	-24.304444	HIP 38146, HIP 39429
HIP 38146	117.257083	-24.912250	HIP 39757, HIP 35264
HIP 35264	109.285667	-37.097500	HIP 38146, HIP 31685
HIP 31685	99.440292	-43.195917	HIP 35264, HIP 32768
HIP 32768	102.483917	-50.614389	HIP 31685, HIP 36377
HIP 36377	112.307833	-43.301889	HIP 32768, HIP 39429
HIP 39429	120.896125	-40.003194	HIP 36377, HIP 39757

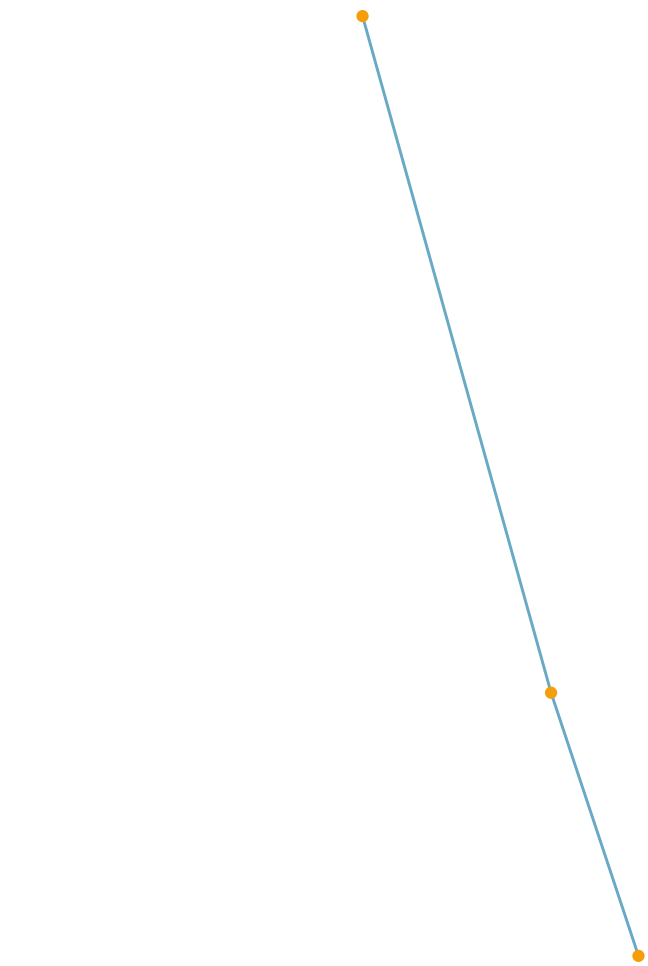
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

69/88 Pyxis (Pyx) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

69/88 Pyxis (Pyx) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 42515	130.025583	-35.308306	HIP 42828
HIP 42828	130.898125	-33.186417	HIP 42515, HIP 43409
HIP 43409	132.633375	-27.710056	HIP 42828

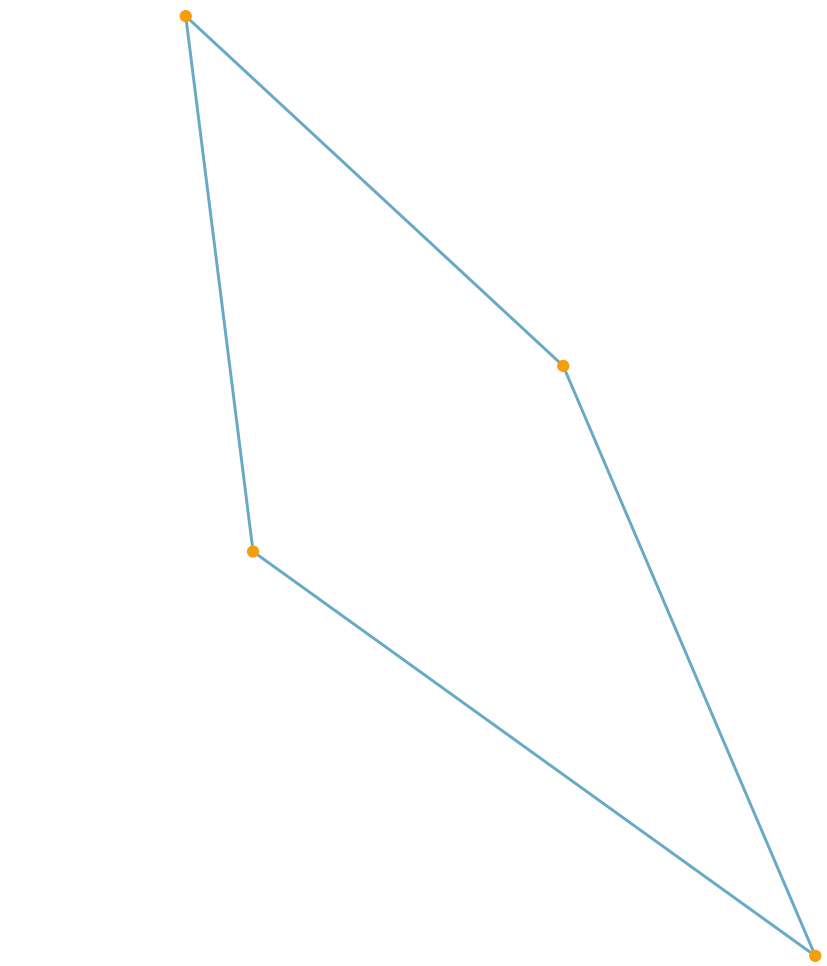
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

70/88 Reticulum (Ret) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

70/88 Reticulum (Ret) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 19780	63.605958	-62.473972	HIP 19921, HIP 17440
HIP 19921	64.121167	-59.301750	HIP 19780, HIP 18597
HIP 18597	59.686417	-61.400139	HIP 19921, HIP 17440
HIP 17440	56.048125	-64.807083	HIP 18597, HIP 19780

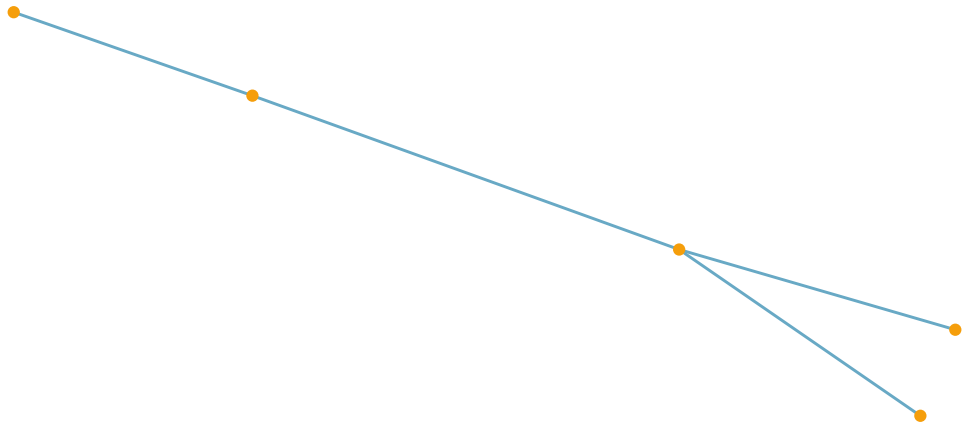
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

71/88 Sagitta (Sge) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

71/88 Sagitta (Sge) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 96837	295.262208	+17.476111	HIP 97365
HIP 97365	296.846958	+18.534250	HIP 96837, HIP 96757, HIP 98337
HIP 96757	295.024083	+18.013944	HIP 97365
HIP 98337	299.689125	+19.492083	HIP 97365, HIP 98920
HIP 98920	301.289458	+19.990889	HIP 98337

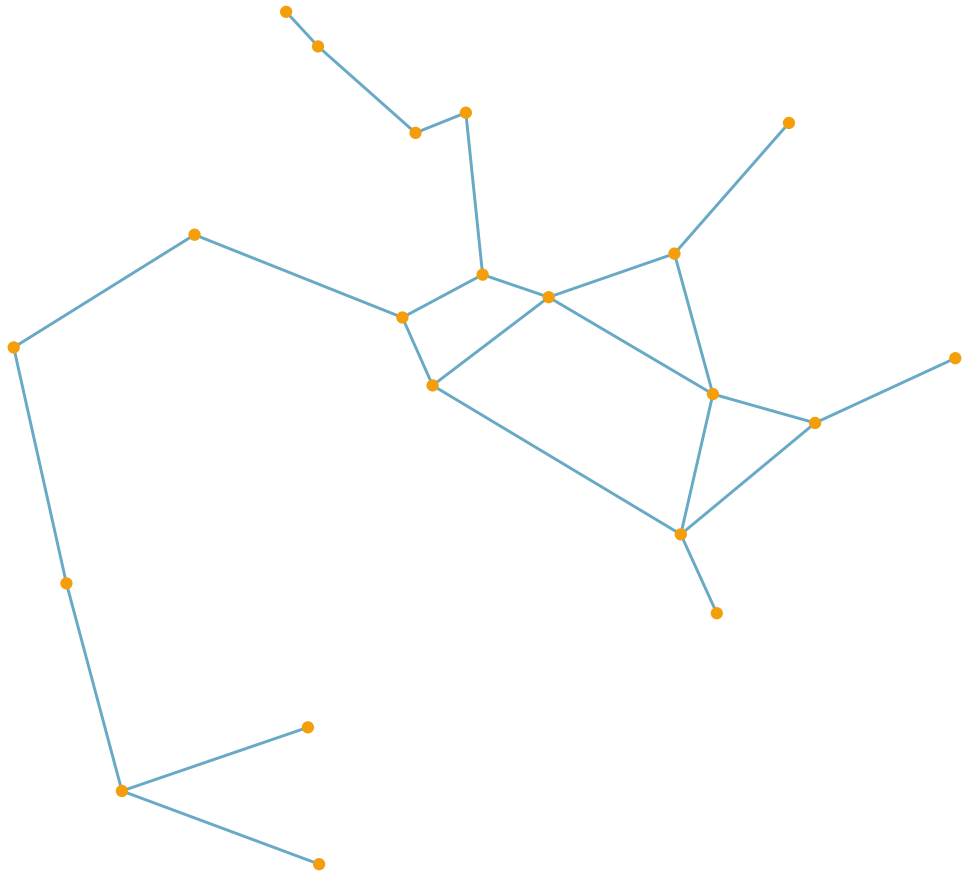
Effective release edges: 4 • unique plotted points: 5 • source: release_effective_edges.csv

72/88 Sagittarius (Sgr) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

72/88 Sagittarius (Sgr) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 89931	275.248417	-29.828028	HIP 90496, HIP 88635, HIP 90185, HIP 92041
HIP 90496	276.992792	-25.421250	HIP 89931, HIP 92041, HIP 89341
HIP 89642	274.407208	-36.761278	HIP 90185
HIP 90185	276.043125	-34.384306	HIP 89642, HIP 88635, HIP 89931, HIP 93506
HIP 88635	271.452167	-30.423639	HIP 90185, HIP 87072, HIP 89931
HIP 87072	266.890125	-27.830750	HIP 88635
HIP 93506	285.653000	-29.880111	HIP 90185, HIP 92041, HIP 93864
HIP 92041	281.413958	-26.990778	HIP 93506, HIP 89931, HIP 90496, HIP 92855
HIP 89341	273.440875	-21.058833	HIP 90496
HIP 93864	286.735167	-27.669806	HIP 93506, HIP 92855, HIP 96406
HIP 92855	283.816333	-26.296583	HIP 93864, HIP 92041, HIP 93085
HIP 93085	284.432417	-21.106611	HIP 92855, HIP 93683
HIP 93683	286.170542	-21.741361	HIP 93085, HIP 94820
HIP 94820	289.408708	-18.952889	HIP 93683, HIP 95168
HIP 95168	290.418250	-17.847250	HIP 94820
HIP 96406	294.006875	-24.719028	HIP 93864, HIP 98688
HIP 98688	300.664417	-27.709889	HIP 96406, HIP 98412
HIP 98412	299.934042	-35.276250	HIP 98688, HIP 98032
HIP 98032	298.815333	-41.868417	HIP 98412, HIP 95347, HIP 95294
HIP 95347	290.971458	-40.615639	HIP 98032
HIP 95294	290.804417	-44.799639	HIP 98032

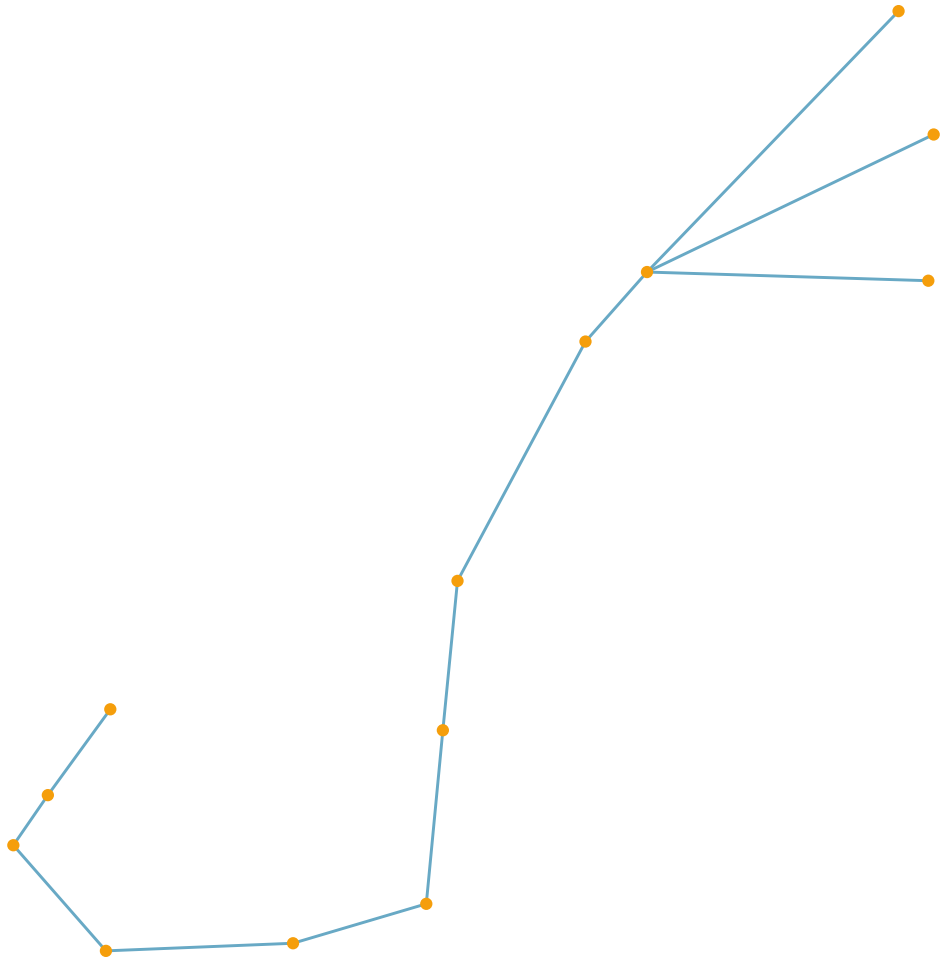
Effective release edges: 24 • unique plotted points: 21 • source: release_effective_edges.csv

73/88 Scorpius (Sco) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

73/88 Scorpius (Sco) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 85927	263.402208	-37.103750	HIP 86670
HIP 86670	265.622000	-39.029917	HIP 85927, HIP 87073
HIP 87073	266.896167	-40.126972	HIP 86670, HIP 86228
HIP 86228	264.329708	-42.997833	HIP 87073, HIP 84143
HIP 84143	258.038250	-43.238500	HIP 86228, HIP 82671
HIP 82671	253.498875	-42.362028	HIP 84143, HIP 82514
HIP 82514	252.967667	-38.047333	HIP 82671, HIP 82396
HIP 82396	252.542667	-34.292611	HIP 82514, HIP 81266
HIP 81266	248.970667	-28.215972	HIP 82396, HIP 80763
HIP 80763	247.351958	-26.431944	HIP 81266, HIP 78401, HIP 78265, HIP 78820
HIP 78401	240.083375	-22.621611	HIP 80763
HIP 78265	239.713000	-26.114056	HIP 80763
HIP 78820	241.359292	-19.805389	HIP 80763

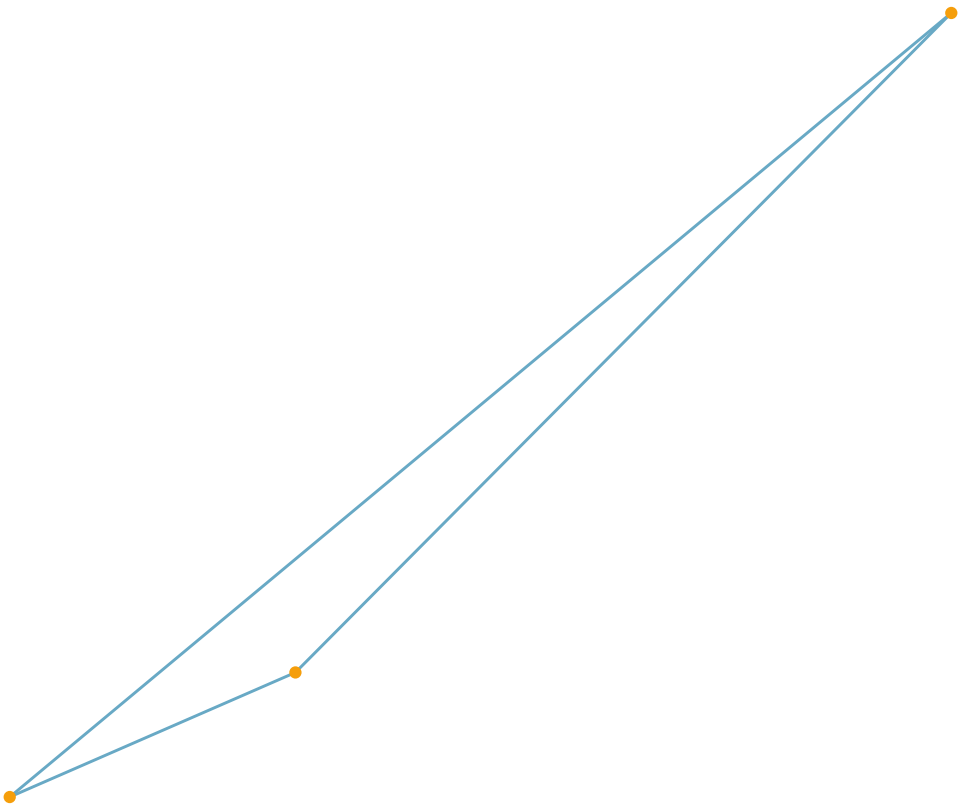
Effective release edges: 12 • unique plotted points: 13 • source: release_effective_edges.csv

74/88 Sculptor (Scl) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

74/88 Sculptor (Scl) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 116231	353.242458	-37.818361	HIP 4577, HIP 115102
HIP 4577	14.651458	-29.357472	HIP 116231, HIP 115102
HIP 115102	349.705958	-32.531833	HIP 4577, HIP 116231

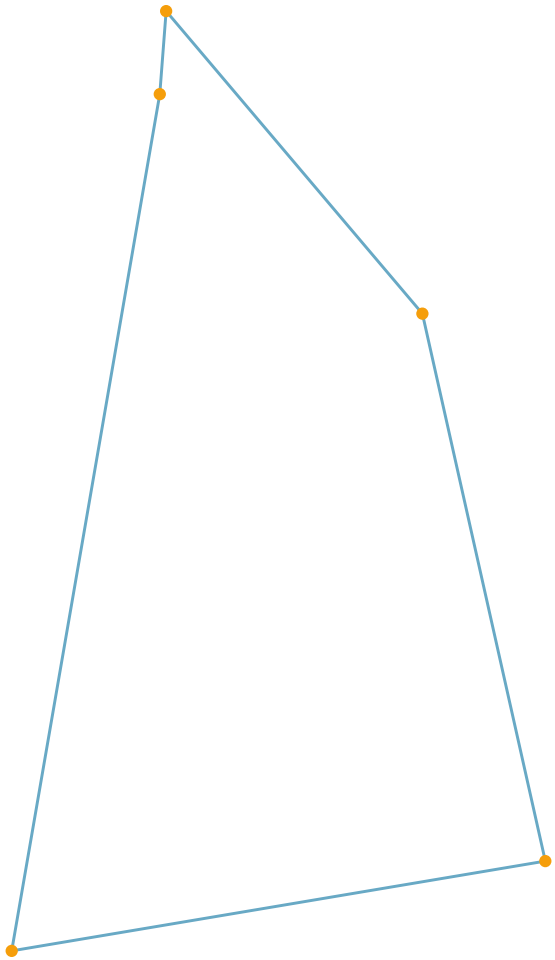
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

75/88 Scutum (Sct) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

75/88 Scutum (Sct) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 92175	281.793667	-4.747833	HIP 92202, HIP 91117
HIP 92202	281.870750	-5.705083	HIP 92175, HIP 92814
HIP 92814	283.679667	-15.603028	HIP 92202, HIP 90595
HIP 90595	277.299375	-14.565806	HIP 92814, HIP 91117
HIP 91117	278.801833	-8.243306	HIP 90595, HIP 92175

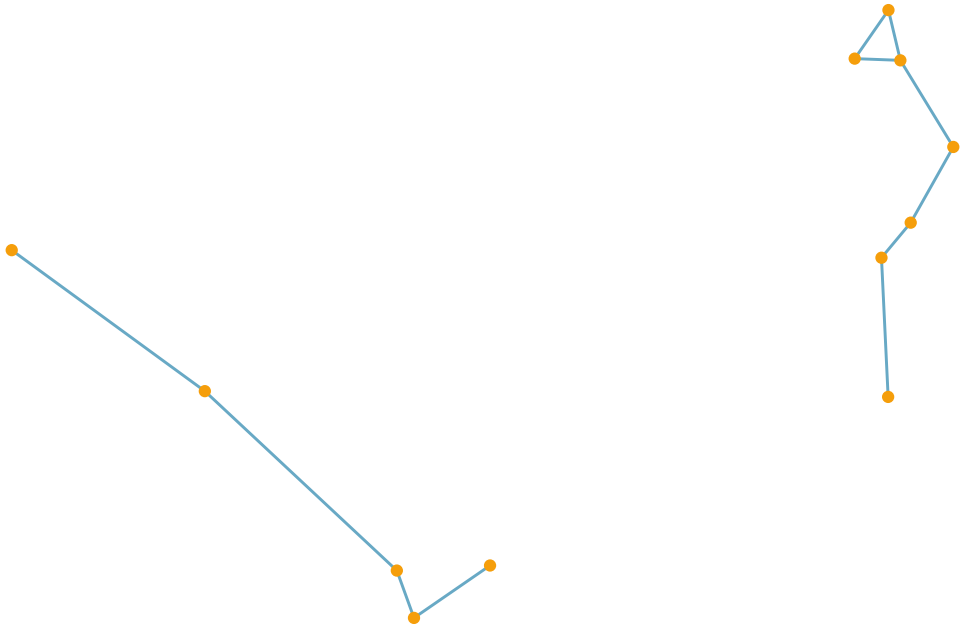
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

76/88 Serpens (Ser) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

76/88 Serpens (Ser) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 77516	237.405292	-3.430139	HIP 77622
HIP 77622	237.703708	+4.477583	HIP 77516, HIP 77070
HIP 77070	236.066667	+6.425528	HIP 77622, HIP 76276
HIP 76276	233.700792	+10.538861	HIP 77070, HIP 77233
HIP 77233	236.546708	+15.421917	HIP 76276, HIP 78072, HIP 77450
HIP 78072	239.112458	+15.664722	HIP 77233, HIP 77450
HIP 77450	237.185042	+18.141778	HIP 78072, HIP 77233
HIP 92946	284.054833	+4.203528	HIP 89962
HIP 89962	275.328833	-2.897111	HIP 92946, HIP 86565
HIP 86565	265.353833	-12.875167	HIP 89962, HIP 86263
HIP 86263	264.396792	-15.398417	HIP 86565, HIP 84880
HIP 84880	260.206833	-12.846889	HIP 86263

Effective release edges: 11 • unique plotted points: 12 • source: release_effective_edges.csv

77/88 Sextans (Sex) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

77/88 Sextans (Sex) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 51437	157.572917	+0.636972	HIP 49641
HIP 49641	151.984583	+0.371639	HIP 51437

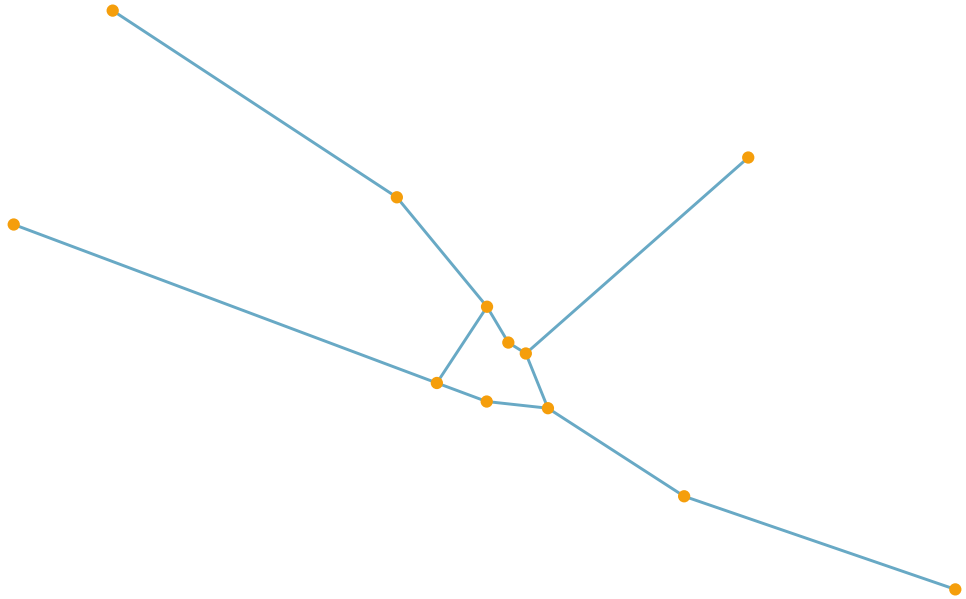
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

78/88 Taurus (Tau) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

78/88 Taurus (Tau) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 25428	81.572917	+28.607861	HIP 21881
HIP 21881	70.561250	+22.956972	HIP 25428, HIP 20889
HIP 20889	67.153875	+19.180528	HIP 21881, HIP 21421, HIP 20648
HIP 21421	68.980000	+16.509750	HIP 26451, HIP 20889, HIP 20894
HIP 26451	84.411167	+21.142583	HIP 21421
HIP 20205	64.948042	+15.627694	HIP 20455, HIP 18724, HIP 20894
HIP 20455	65.733458	+17.542583	HIP 20205, HIP 20648, HIP 17847
HIP 18724	60.170083	+12.490389	HIP 20205, HIP 15900
HIP 15900	51.203500	+9.029056	HIP 18724
HIP 20894	67.165292	+15.870944	HIP 21421, HIP 20205
HIP 20648	66.372167	+17.928000	HIP 20889, HIP 20455
HIP 17847	57.290542	+24.053528	HIP 20455

Effective release edges: 12 • unique plotted points: 12 • source: release_effective_edges.csv

79/88 Telescopium (Tel) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

79/88 Telescopium (Tel) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 90568	277.207250	-49.070028	HIP 90422
HIP 90422	276.743458	-45.968333	HIP 90568

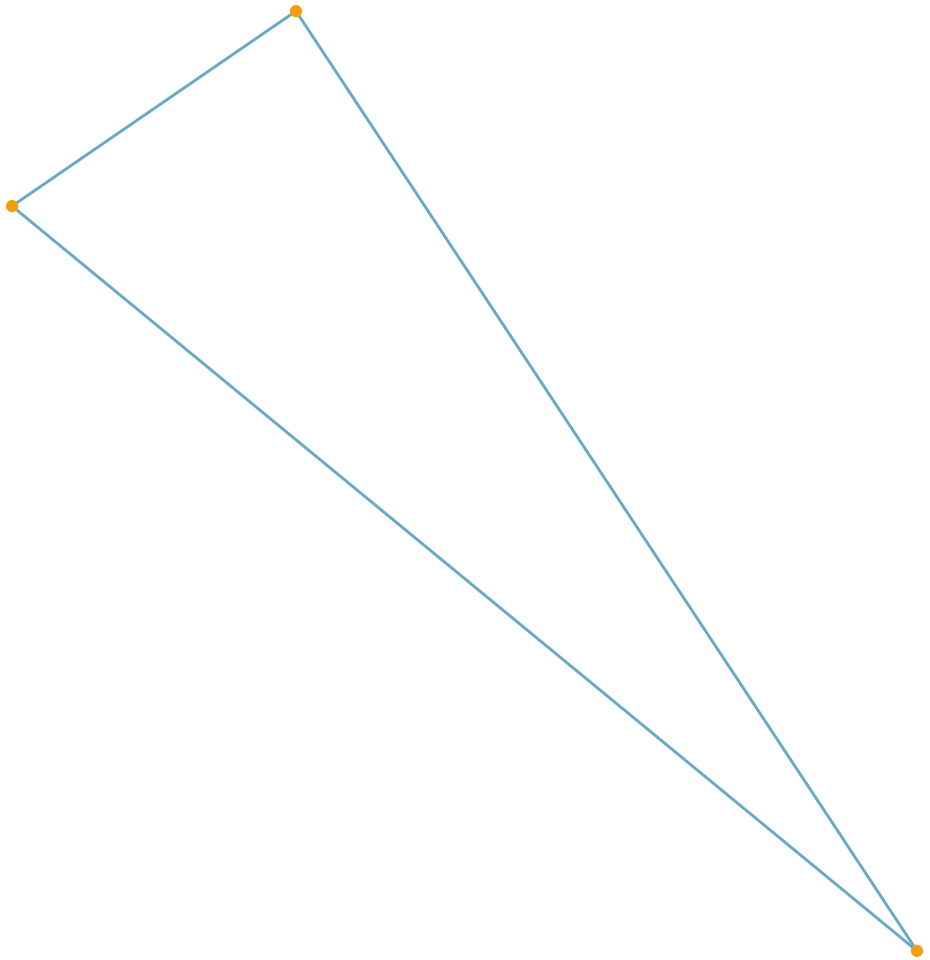
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

80/88 Triangulum (Tri) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

80/88 Triangulum (Tri) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 10670	34.328500	+33.847333	HIP 10064, HIP 8796
HIP 10064	32.385500	+34.987389	HIP 10670, HIP 8796
HIP 8796	28.270417	+29.579389	HIP 10064, HIP 10670

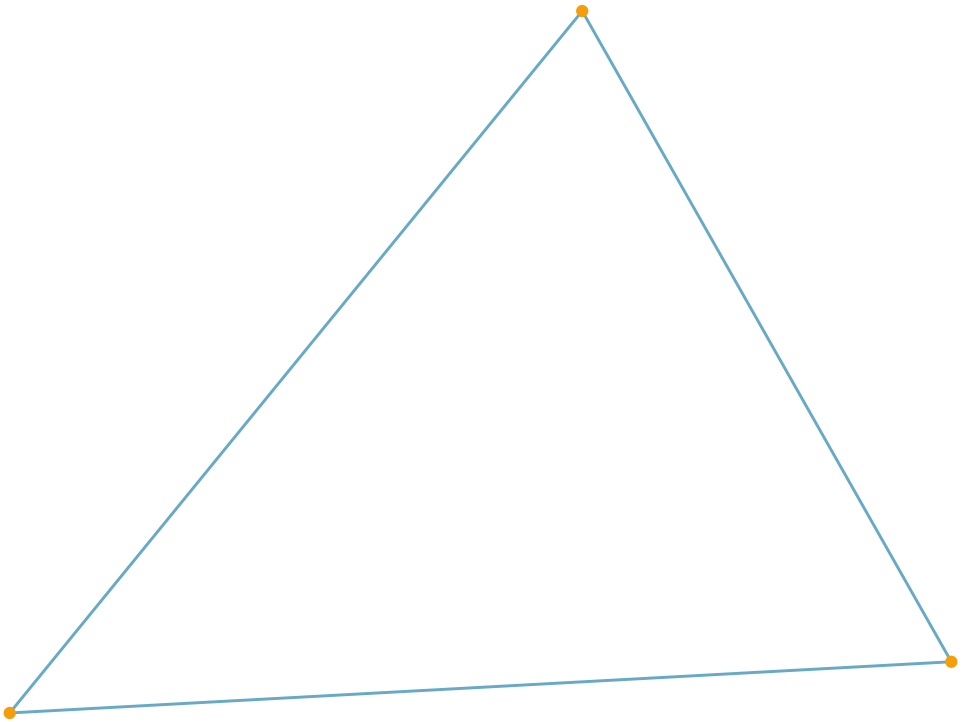
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

81/88 Triangulum Australe (TrA) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

81/88 Triangulum Australe (TrA) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 82273	252.166125	-69.027639	HIP 74946, HIP 77952
HIP 74946	229.727875	-68.679472	HIP 82273, HIP 77952
HIP 77952	238.786708	-63.429750	HIP 74946, HIP 82273

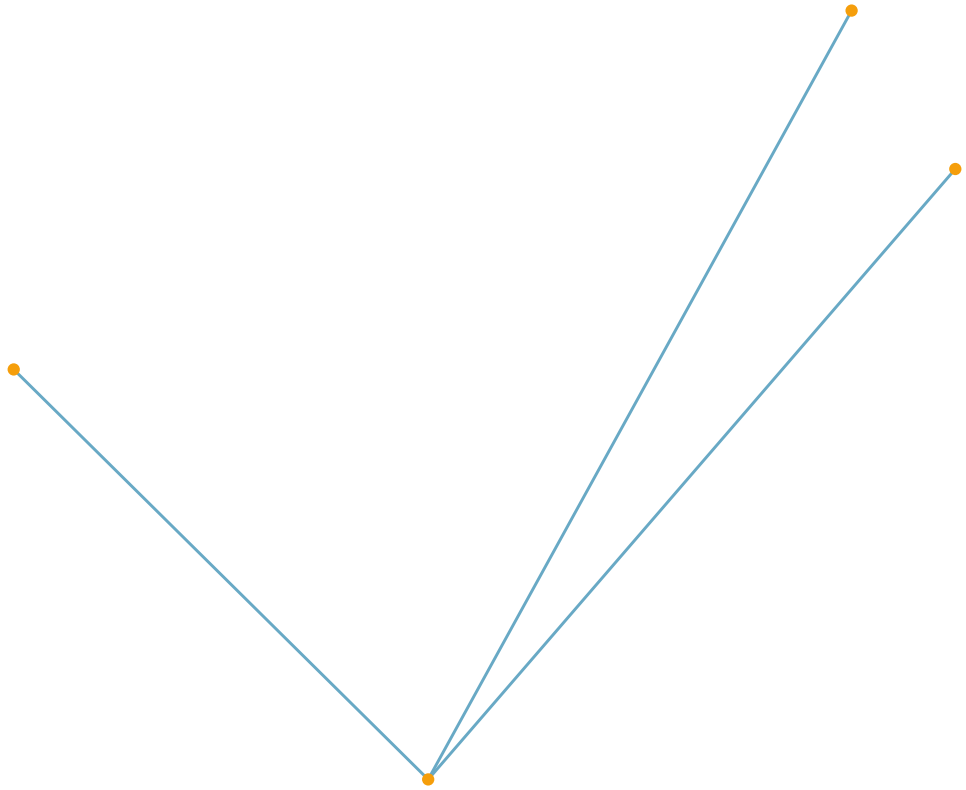
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

82/88 Tucana (Tuc) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

82/88 Tucana (Tuc) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 110130	334.625750	-60.259500	HIP 114996
HIP 114996	349.357542	-58.235917	HIP 110130, HIP 1599, HIP 2484
HIP 1599	5.007958	-64.877611	HIP 114996
HIP 2484	7.885667	-62.958083	HIP 114996

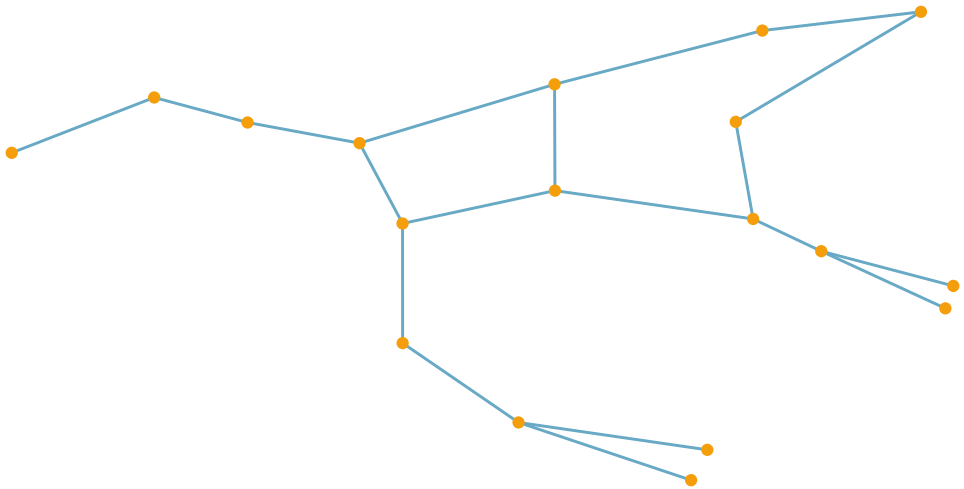
Effective release edges: 3 • unique plotted points: 4 • source: release_effective_edges.csv

83/88 Ursa Major (UMa) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

83/88 Ursa Major (UMa) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 67301	206.885625	+49.313306	HIP 65378
HIP 65378	200.980917	+54.925417	HIP 67301, HIP 62956
HIP 62956	193.506792	+55.959833	HIP 65378, HIP 59774
HIP 59774	183.856042	+57.032611	HIP 62956, HIP 54061, HIP 58001
HIP 54061	165.932667	+61.751111	HIP 59774, HIP 53910, HIP 46733
HIP 53910	165.459958	+56.382333	HIP 54061, HIP 58001, HIP 48402
HIP 58001	178.457250	+53.694722	HIP 53910, HIP 59774, HIP 57399
HIP 57399	176.513042	+47.779333	HIP 58001, HIP 54539
HIP 54539	167.416083	+44.498556	HIP 57399, HIP 50372, HIP 50801
HIP 50372	154.274708	+42.914472	HIP 54539
HIP 50801	155.582500	+41.499444	HIP 54539
HIP 48402	148.026500	+54.064278	HIP 53910, HIP 46853, HIP 48319
HIP 46853	143.218042	+51.678611	HIP 48402, HIP 44471, HIP 44127
HIP 44471	135.906500	+47.156667	HIP 46853
HIP 44127	134.803500	+48.042361	HIP 46853
HIP 48319	147.748708	+59.039111	HIP 48402, HIP 41704
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HIP 46733	142.881542	+63.061806	HIP 41704, HIP 54061

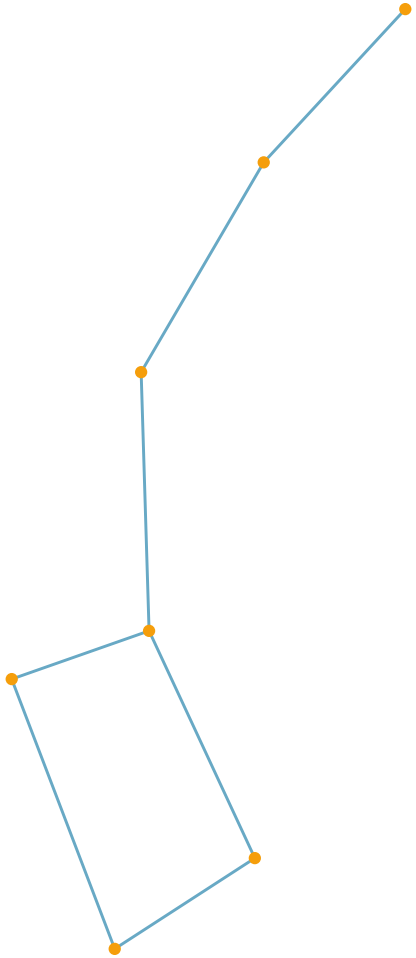
Effective release edges: 19 • unique plotted points: 18 • source: release_effective_edges.csv

84/88 Ursa Minor (UMi) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

84/88 Ursa Minor (UMi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 11767	37.946167	+89.264139	HIP 85822
HIP 85822	263.053750	+86.586333	HIP 11767, HIP 82080
HIP 82080	251.492333	+82.037250	HIP 85822, HIP 77055
HIP 77055	236.014417	+77.794500	HIP 82080, HIP 79822, HIP 72607
HIP 79822	244.377083	+75.754694	HIP 77055, HIP 75097
HIP 75097	230.182292	+71.833972	HIP 79822, HIP 72607
HIP 72607	222.676667	+74.155472	HIP 75097, HIP 77055

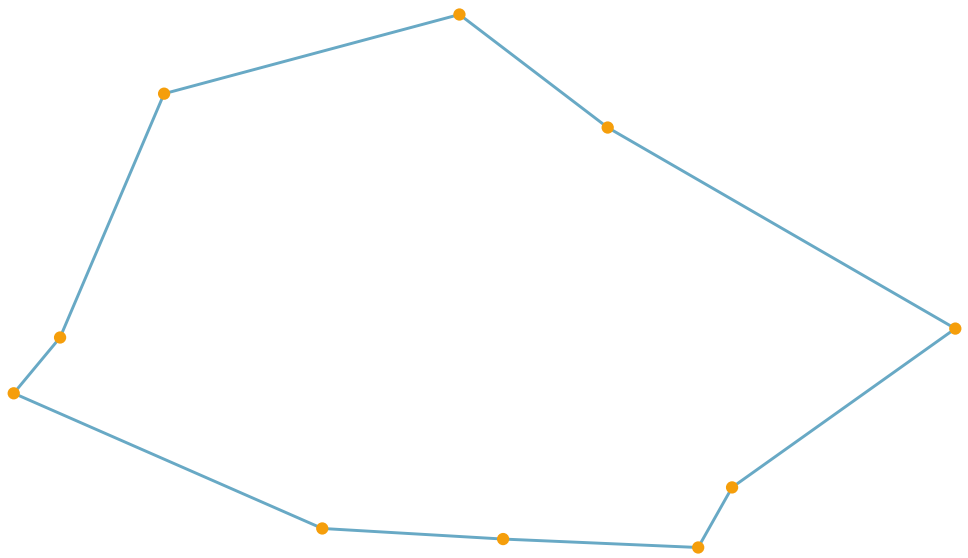
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

85/88 Vela (Vel) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

85/88 Vela (Vel) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 39953	122.383167	-47.336611	HIP 42536, HIP 44816
HIP 42536	130.073375	-52.921972	HIP 39953, HIP 42913
HIP 42913	131.175833	-54.708556	HIP 42536, HIP 45941
HIP 45941	140.528458	-55.010694	HIP 42913, HIP 48774
HIP 48774	149.215625	-54.567806	HIP 45941, HIP 52727
HIP 52727	161.692167	-49.420139	HIP 48774, HIP 51986
HIP 51986	159.326083	-48.225611	HIP 52727, HIP 50191
HIP 50191	153.684458	-42.122056	HIP 51986, HIP 46651
HIP 46651	142.675458	-40.466889	HIP 50191, HIP 44816
HIP 44816	136.999083	-43.432611	HIP 46651, HIP 39953

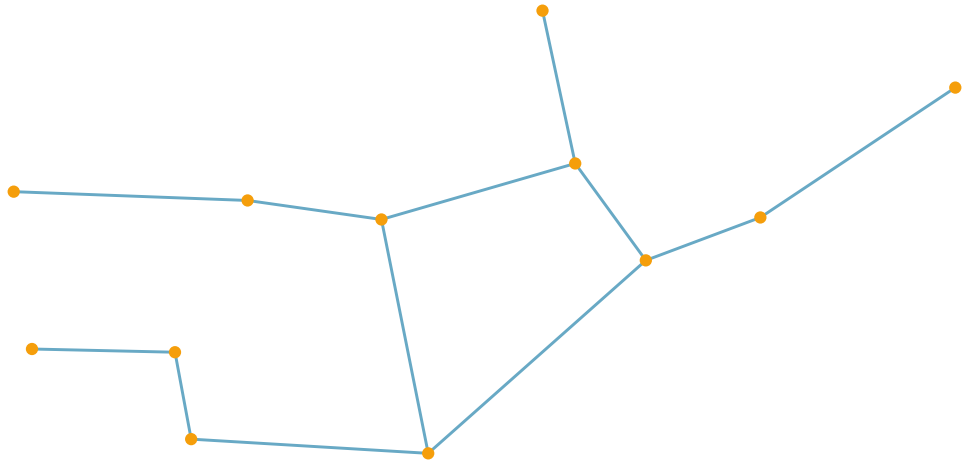
Effective release edges: 10 • unique plotted points: 10 • source: release_effective_edges.csv

86/88 Virgo (Vir) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

86/88 Virgo (Vir) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 57380	176.464875	+6.529806	HIP 60129
HIP 60129	184.976625	+0.666750	HIP 57380, HIP 61941
HIP 61941	190.416667	-1.449528	HIP 60129, HIP 65474, HIP 63090
HIP 65474	201.298333	-11.161250	HIP 61941, HIP 69427, HIP 66249
HIP 69427	213.223917	-10.274056	HIP 65474, HIP 69701
HIP 69701	214.003667	-5.999528	HIP 69427, HIP 71957
HIP 71957	220.764833	-5.657417	HIP 69701
HIP 66249	203.673958	+0.595944	HIP 65474, HIP 68520, HIP 63090
HIP 68520	210.411583	+1.544583	HIP 66249, HIP 72220
HIP 72220	221.562458	+1.892944	HIP 68520
HIP 63090	193.902000	+3.397611	HIP 66249, HIP 63608, HIP 61941
HIP 63608	195.544833	+10.959111	HIP 63090

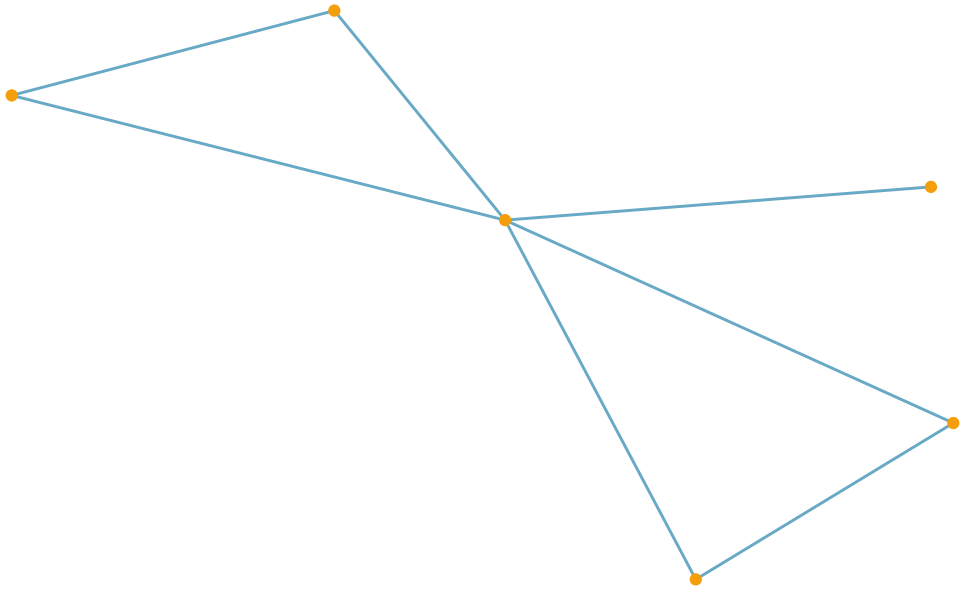
Effective release edges: 12 • unique plotted points: 12 • source: release_effective_edges.csv

87/88 Volans (Vol) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

87/88 Volans (Vol) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 37504	115.455000	-72.606139	HIP 34481, HIP 39794
HIP 34481	107.186750	-70.499194	HIP 37504, HIP 39794
HIP 39794	121.982667	-68.617139	HIP 34481, HIP 37504, HIP 35228, HIP 41312, HIP 44382
HIP 35228	109.207625	-67.957167	HIP 39794
HIP 41312	126.434375	-66.136528	HIP 39794, HIP 44382
HIP 44382	135.611667	-66.395833	HIP 41312, HIP 39794

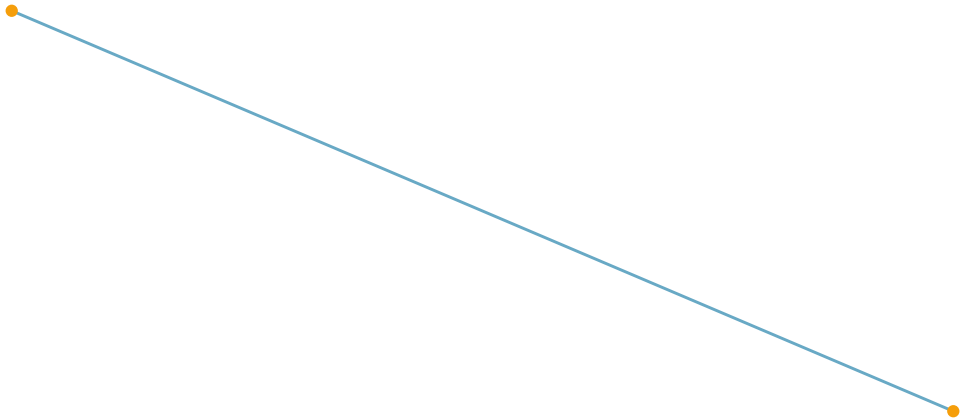
Effective release edges: 7 • unique plotted points: 6 • source: release_effective_edges.csv

88/88 Vulpecula (Vul) — wire-diagram comparison

Left: Galaxy Viewer release implementation • Right: upstream Hipparcos line dataset used by the overlay module

GALAXY VIEWER

Exact release JS edges + runtime source CSV; custom overrides applied



REFERENCE CHART / SOURCE WIRES

Official IAU / Sky & Telescope J2000 chart; fallback clearly marked if unavailable

OFFICIAL IAU CHART UNAVAILABLE

Chart could not be retrieved; no substitute diagram shown

Review: [ ] Correct [ ] Incorrect [ ] Needs review

Missing / extra connections: _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

88/88 Vulpecula (Vul) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

Point / HIP ID	RA (deg)	Dec (deg)	Connected to (star labels)
HIP 95771	292.176708	+24.665167	HIP 98543
HIP 98543	300.275042	+27.753556	HIP 95771

Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv